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(54) **FILTRATION APPARATUS CONTAINING
GRAPHENE OXIDE MEMBRANE**

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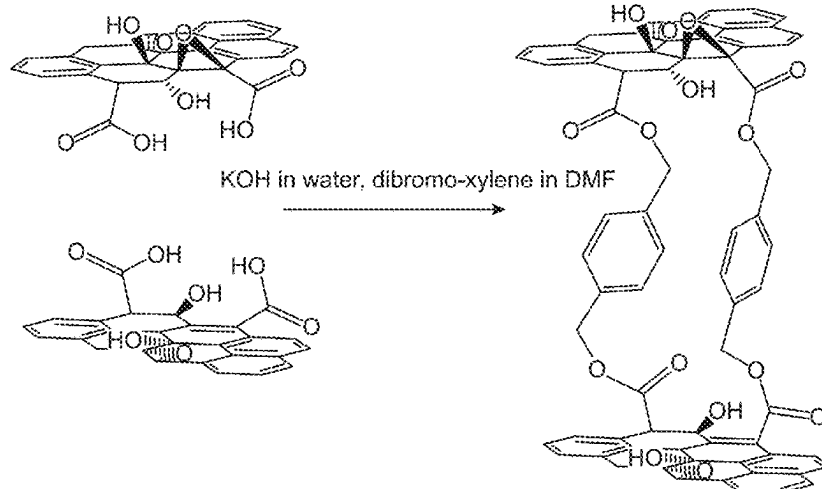
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(57) **ABSTRACT**

Embodiments described herein relate generally to graphene
oxide membranes for fluid filtration and more specifically to
graphene oxide membranes having tunable permeability,
rejection rate, and flux. Some embodiments of the graphene
oxide membranes disclosed herein are characterized as
having a flux of at least about 2.5×10^{-4} gallons per square
foot per day per psi with a 1 wt % lactose solution at room
temperature, and a lactose rejection rate of at least 50% with
a 1 wt % lactose solution.

10 Claims, 31 Drawing Sheets



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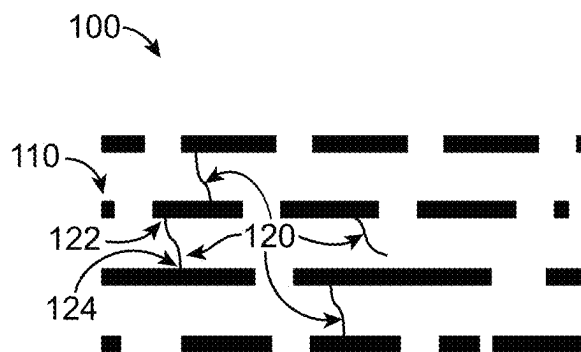


FIG. 1

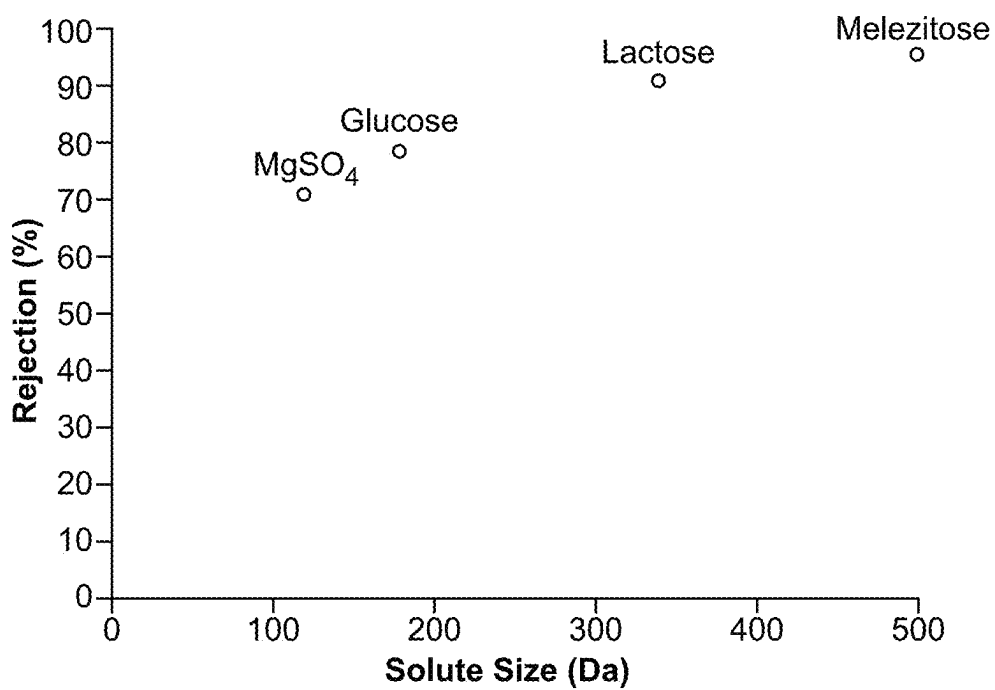


FIG. 2

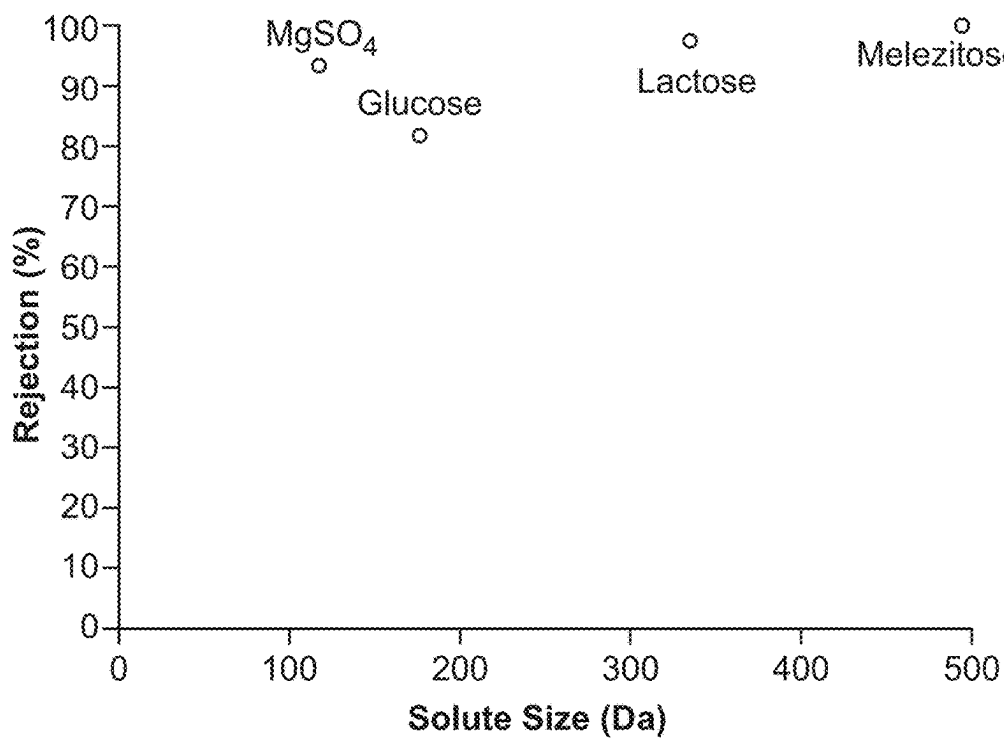


FIG. 3

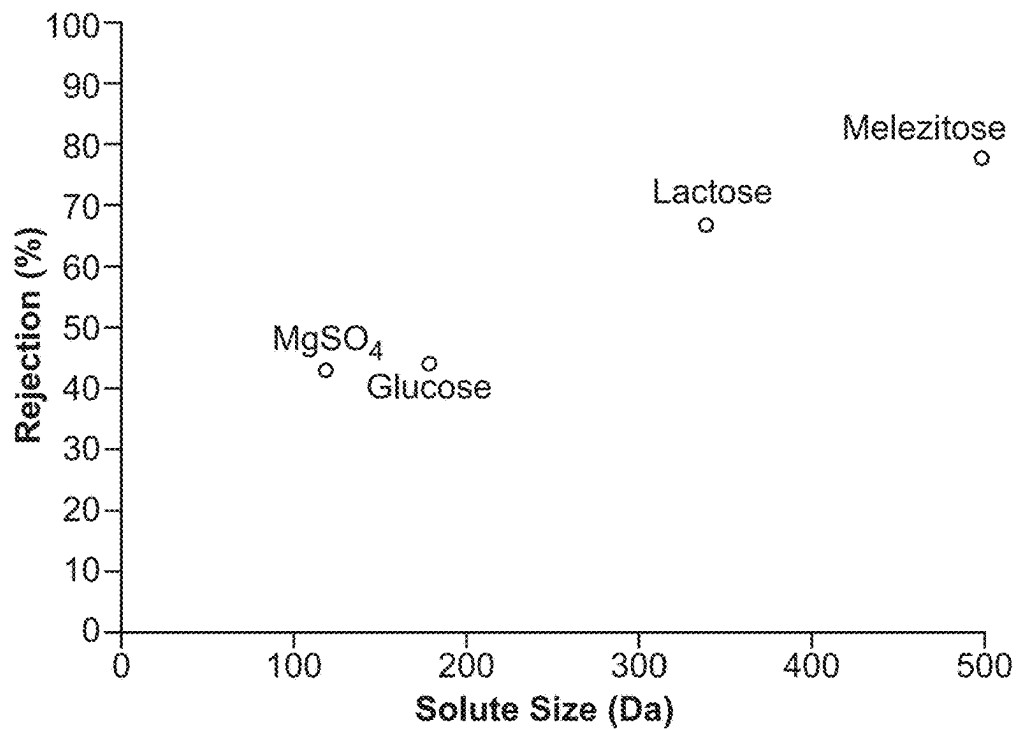


FIG. 4

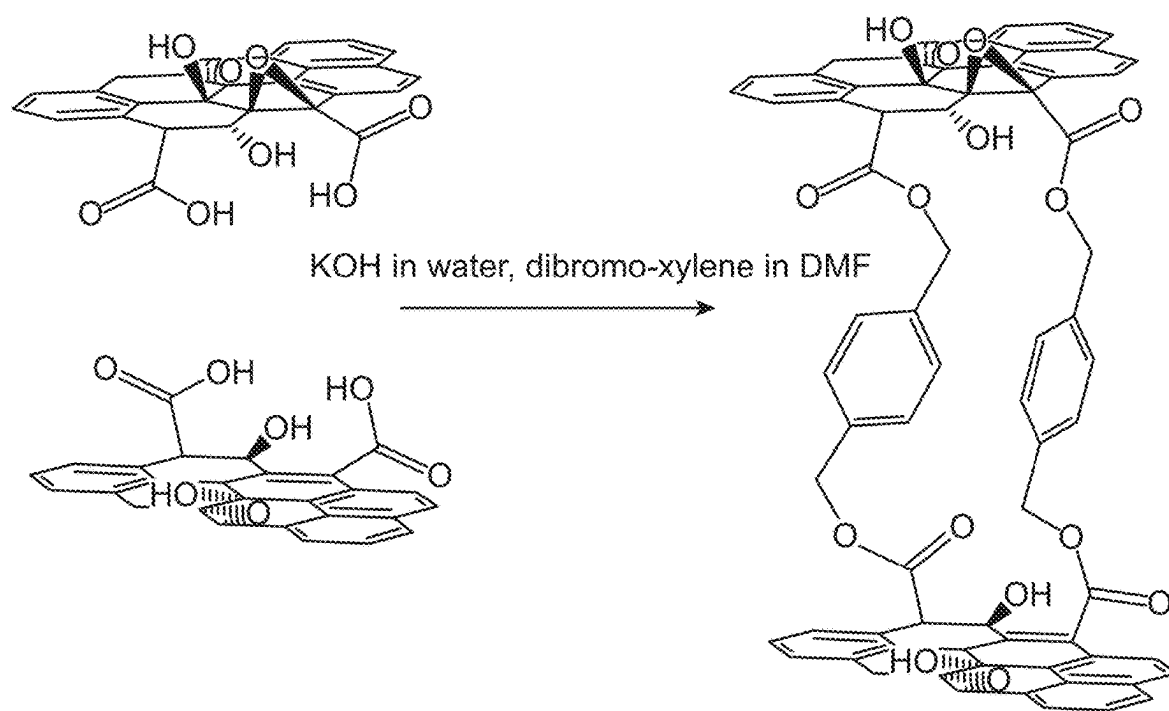


FIG. 5

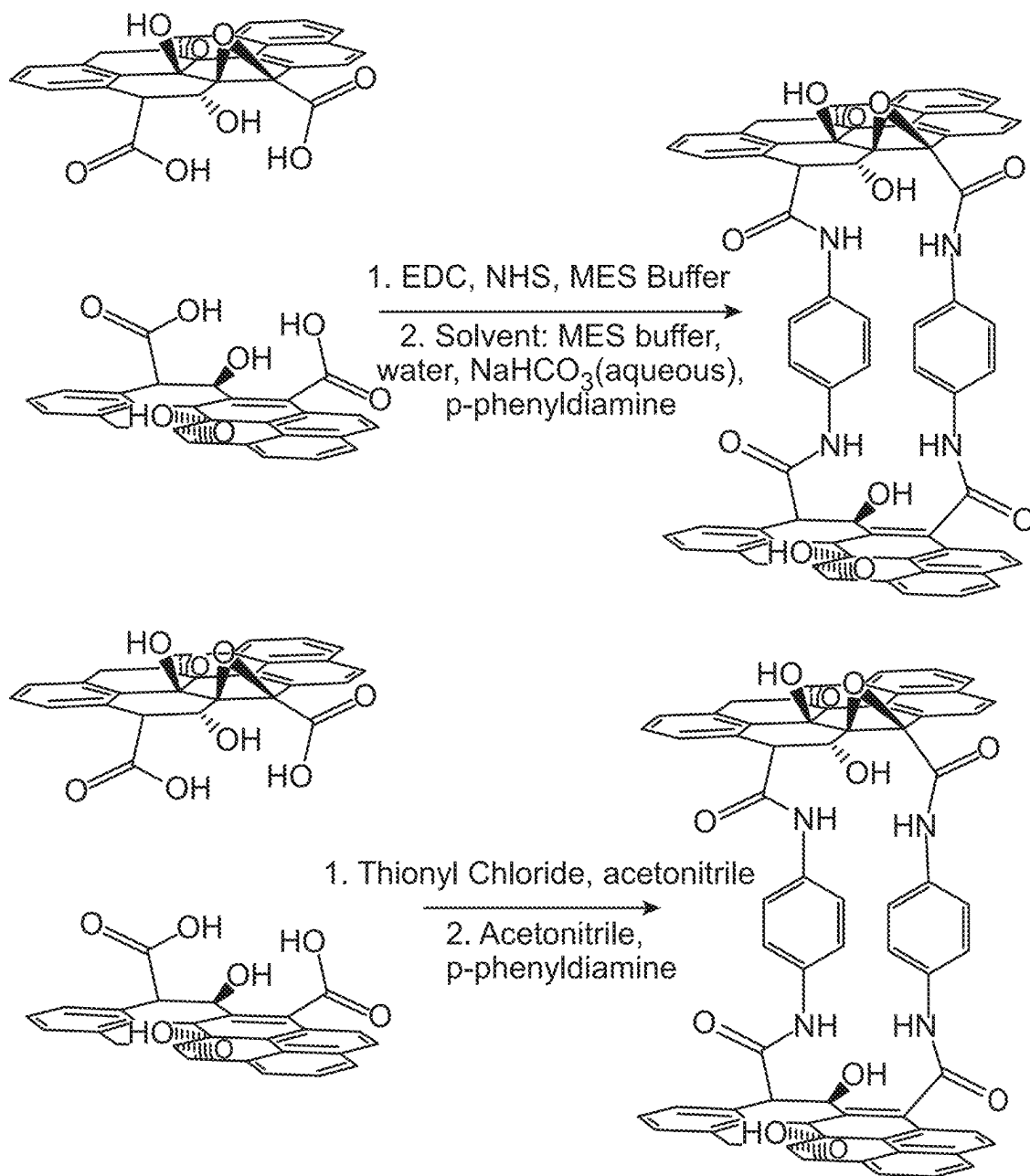


FIG. 6

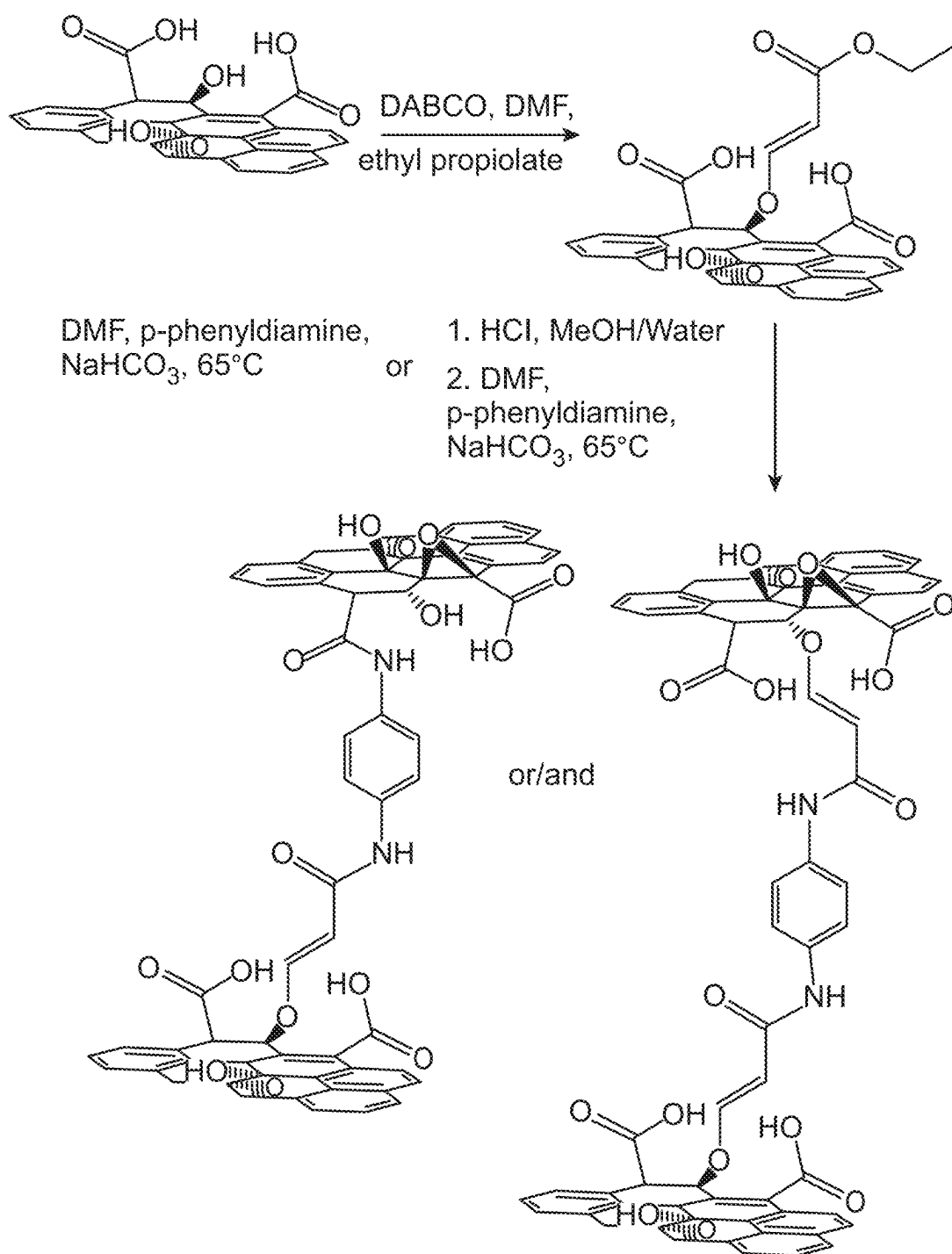


FIG. 7

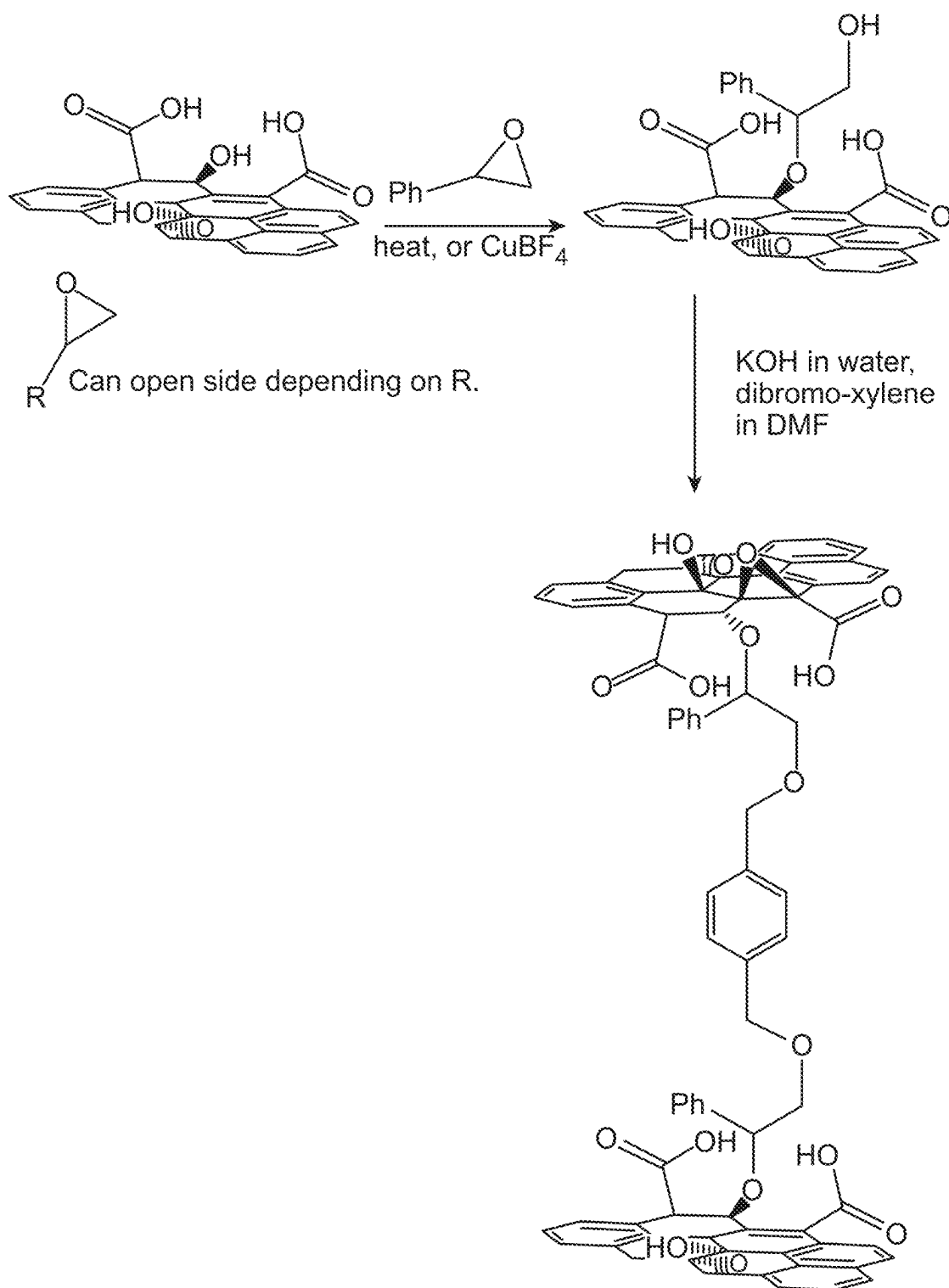


FIG. 8

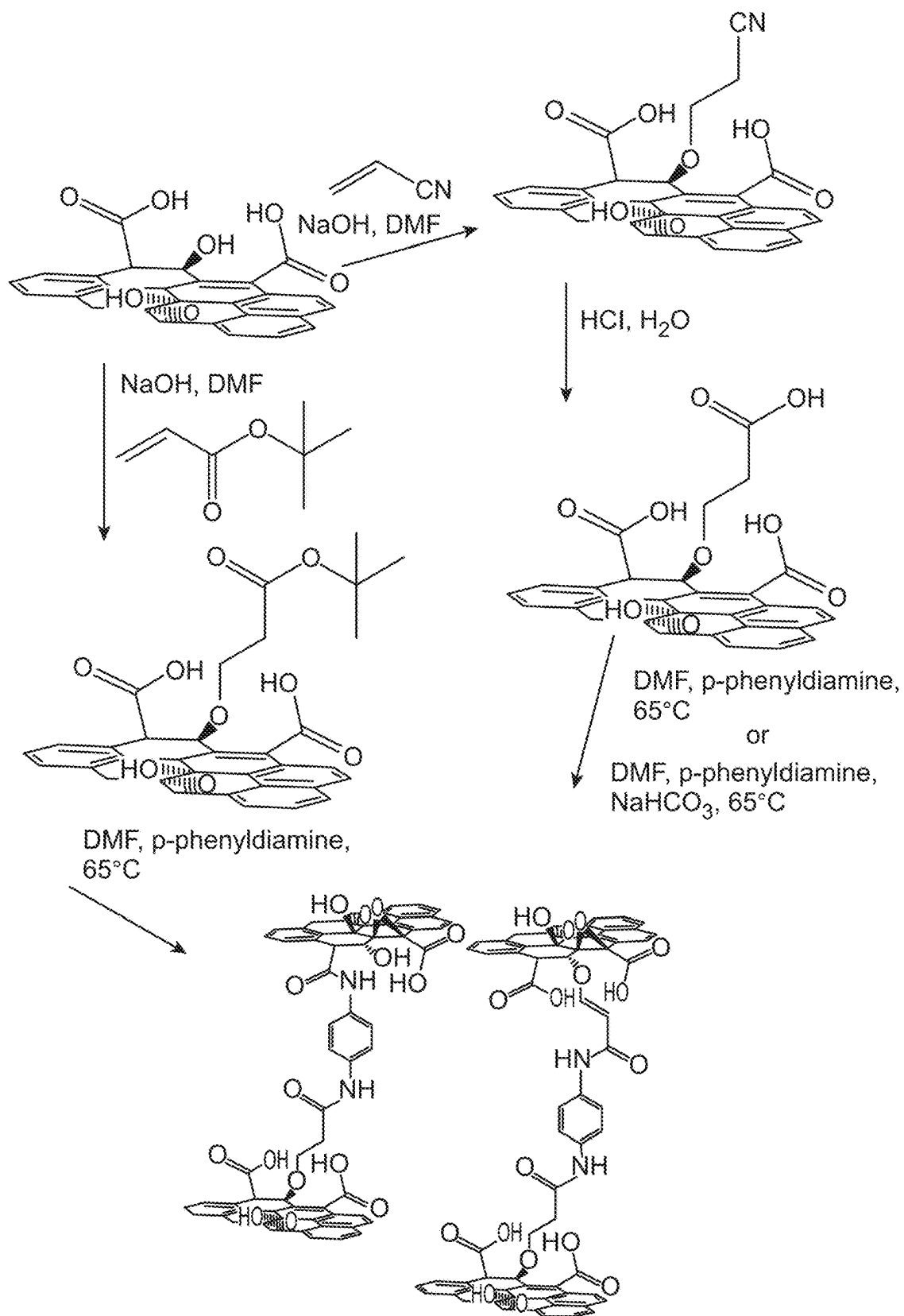


FIG. 9

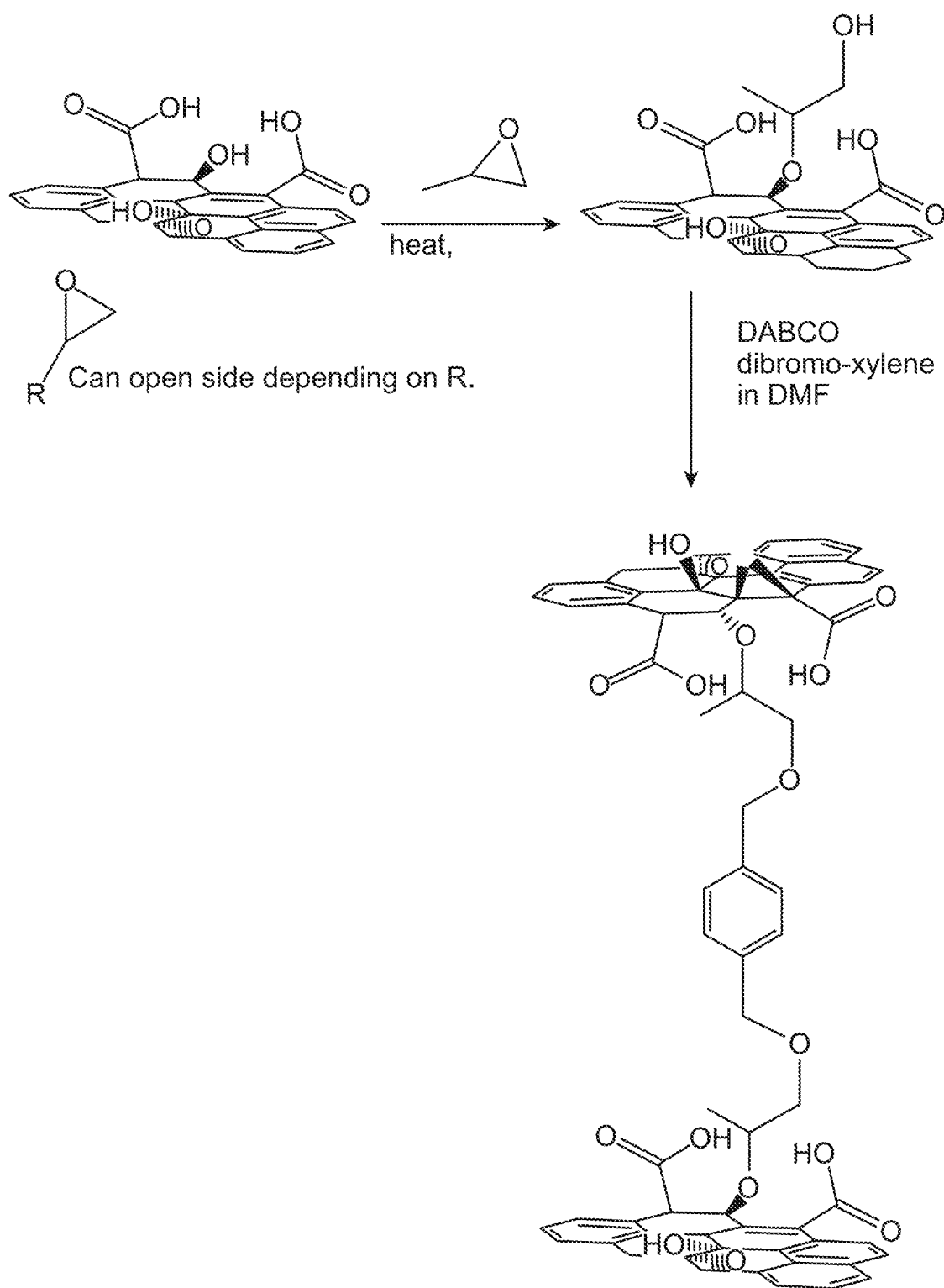


FIG. 10

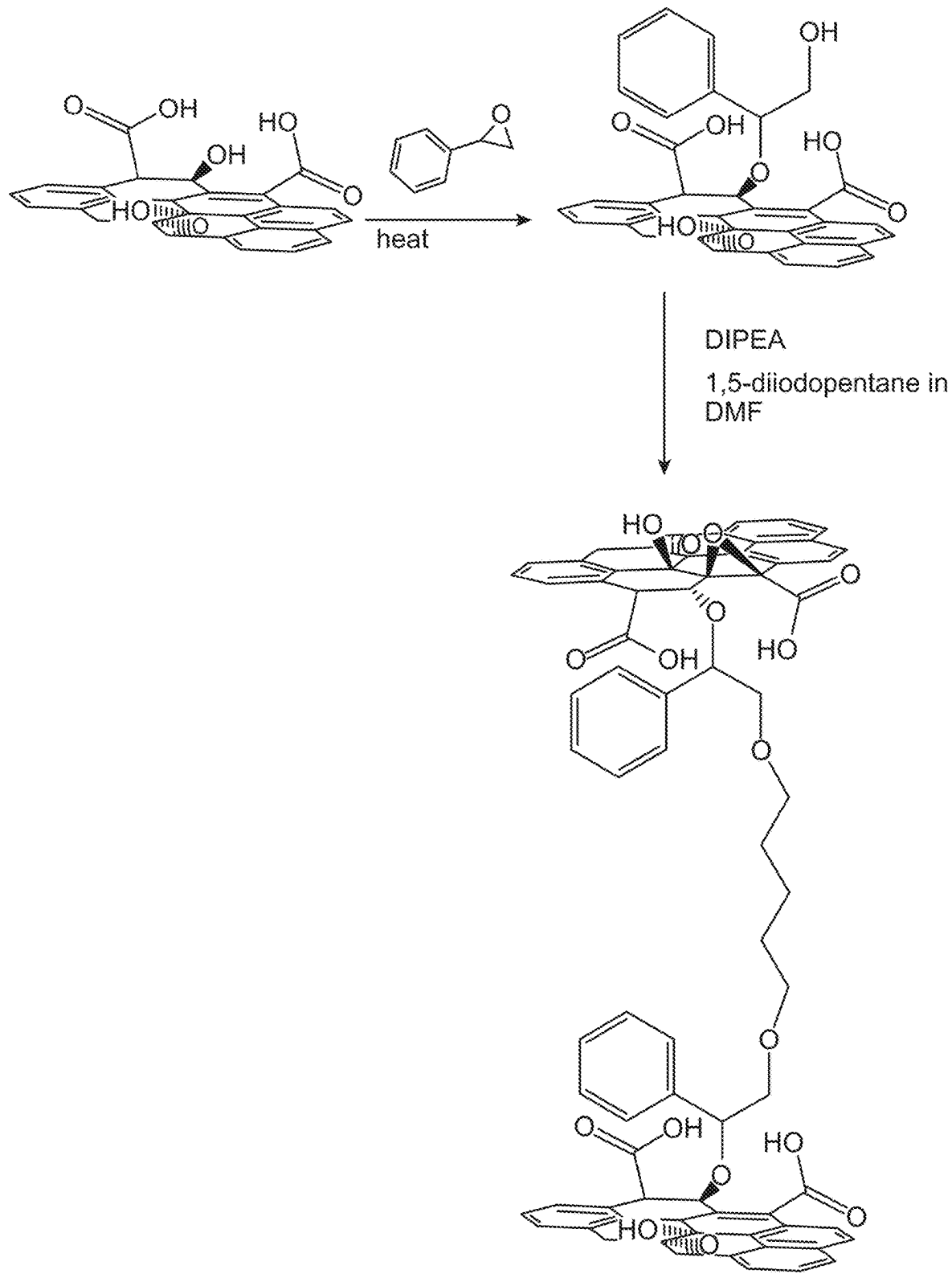


FIG. 11

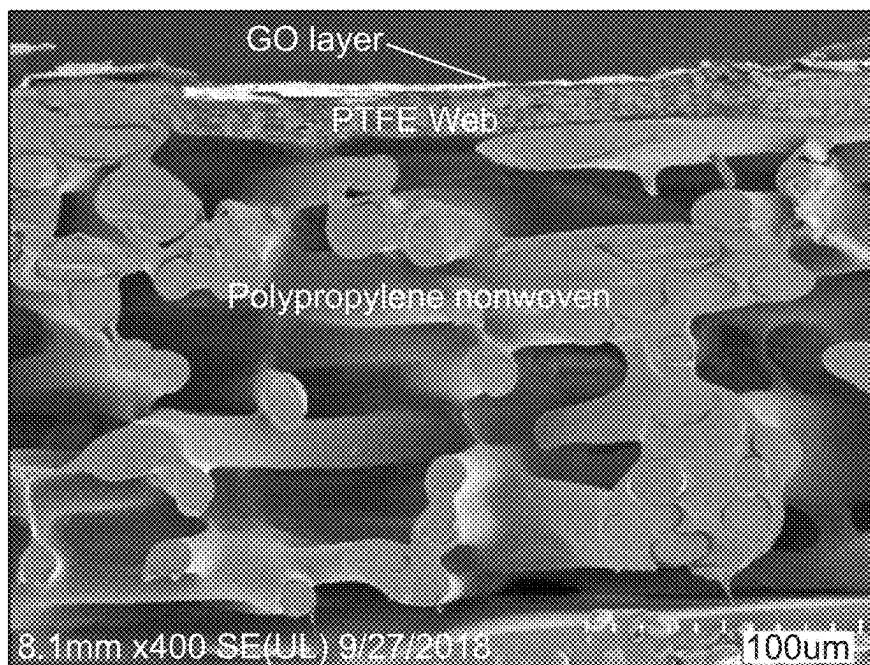


FIG. 12

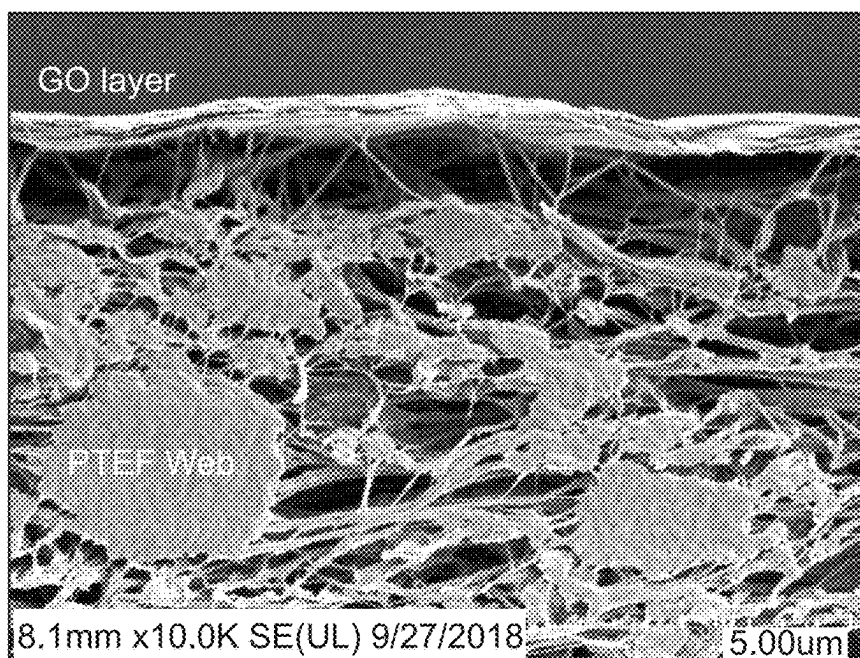


FIG. 13

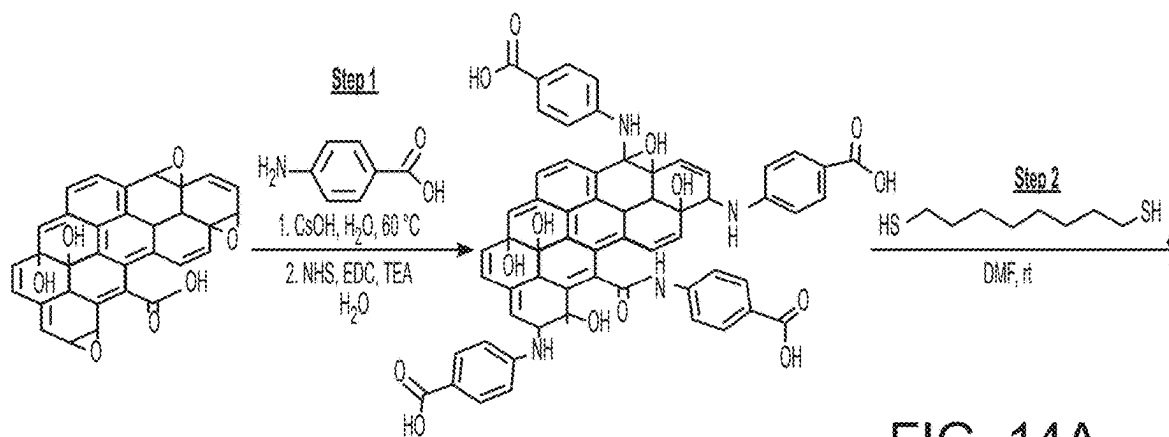


FIG. 14A

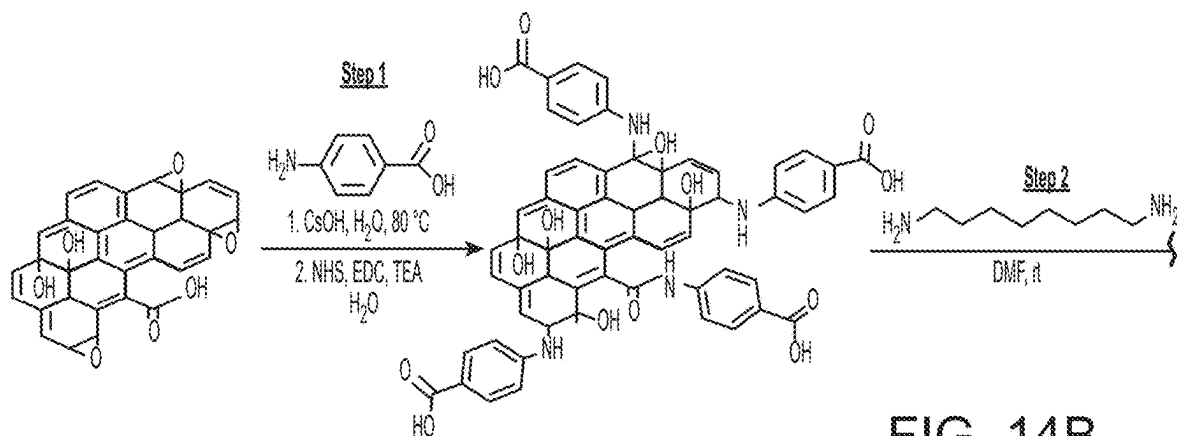


FIG. 14B

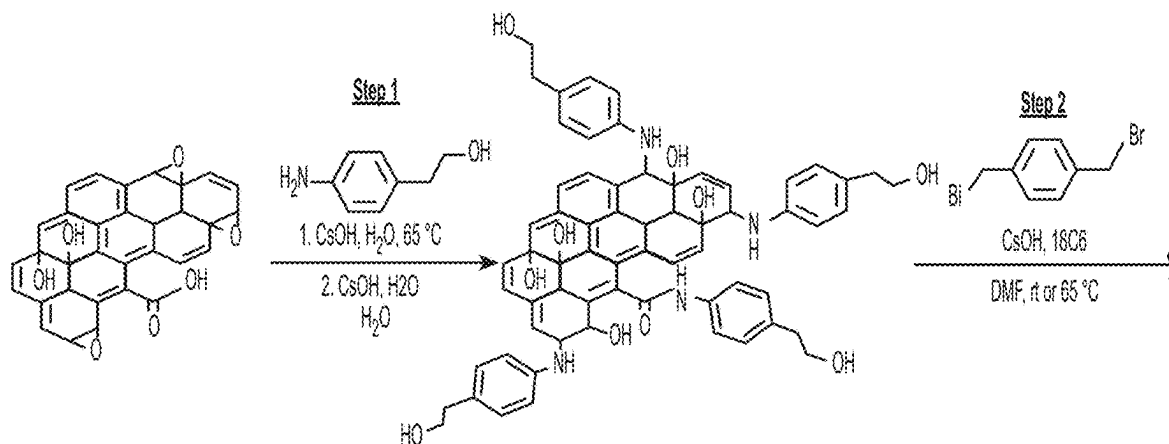
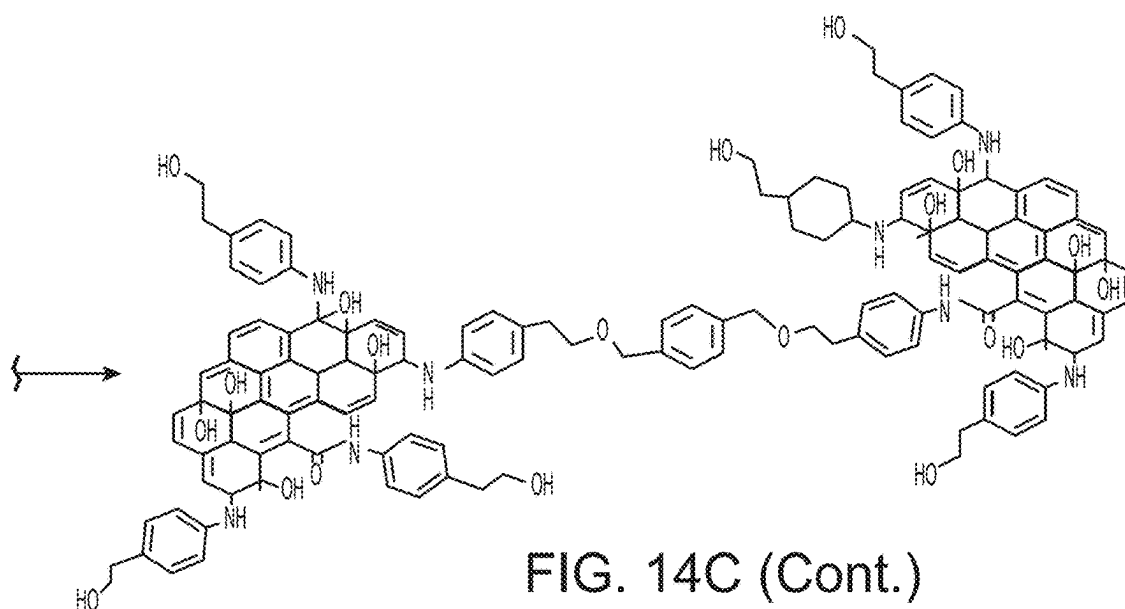
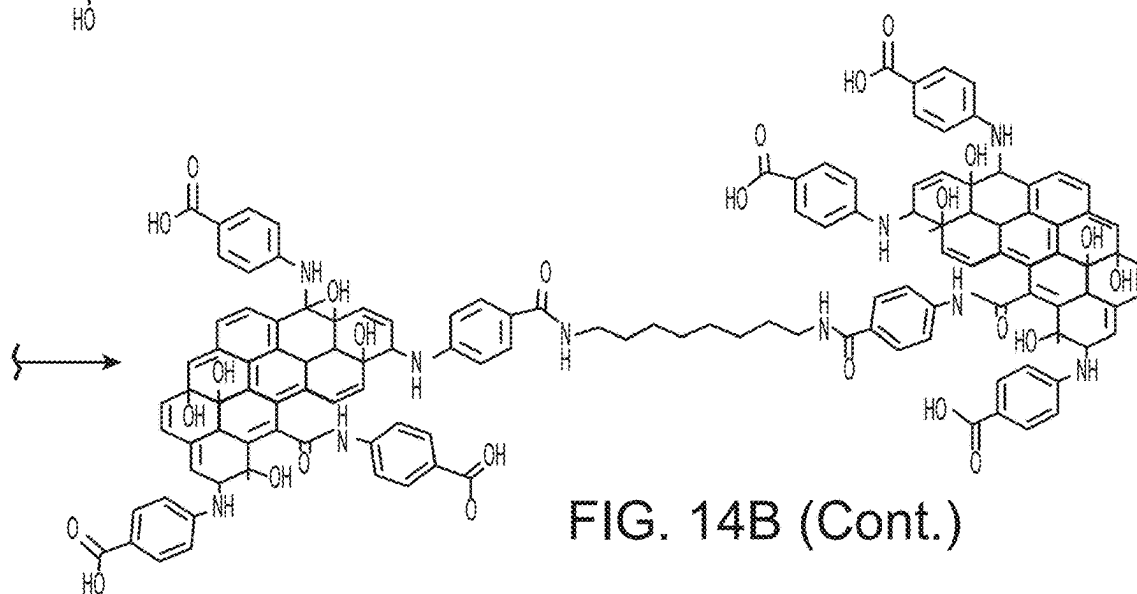
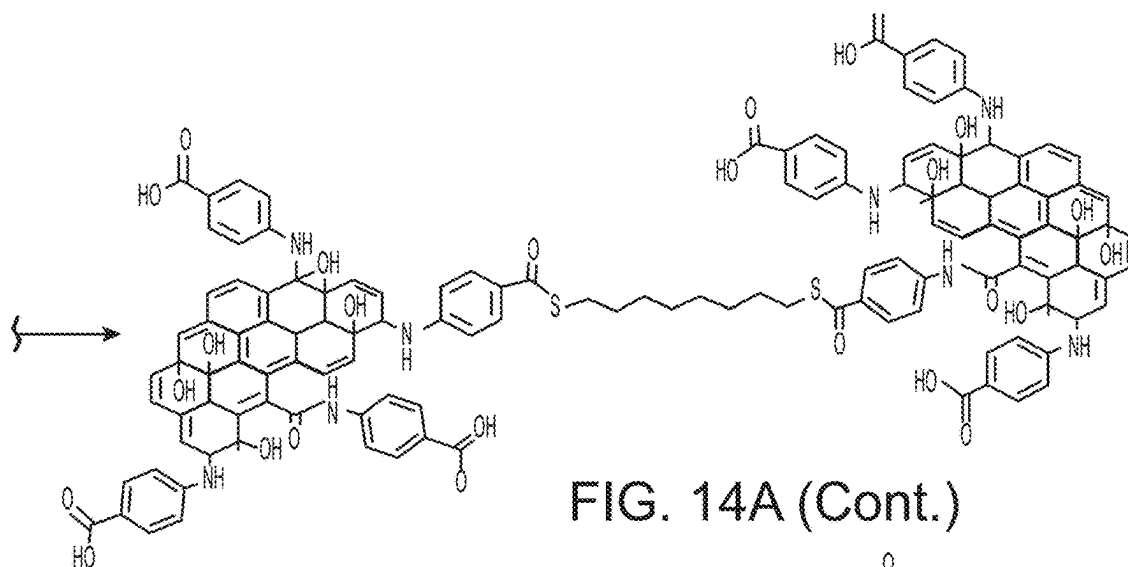


FIG. 14C



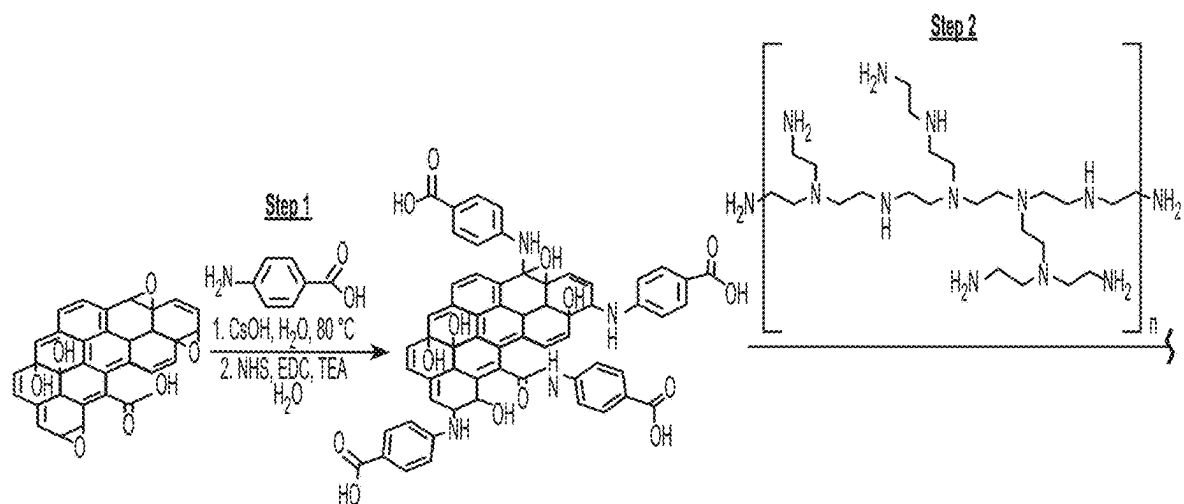


FIG. 14D

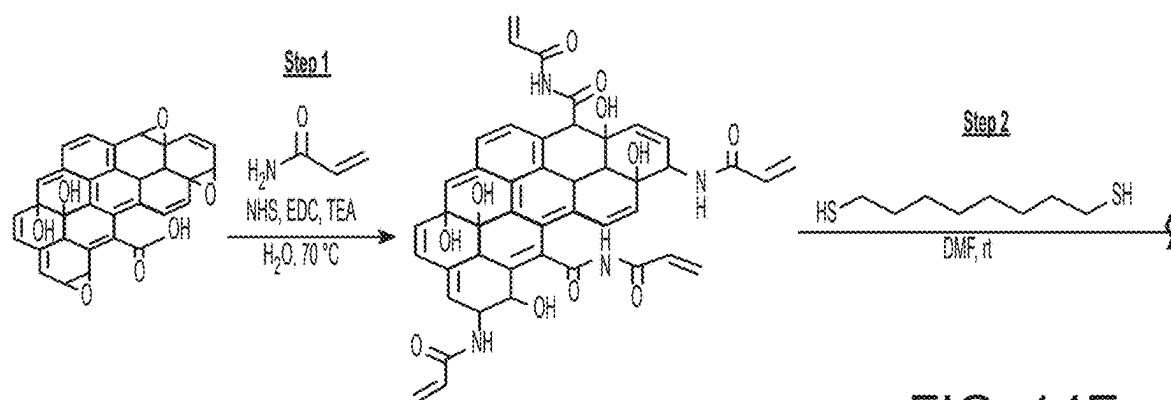


FIG. 14E

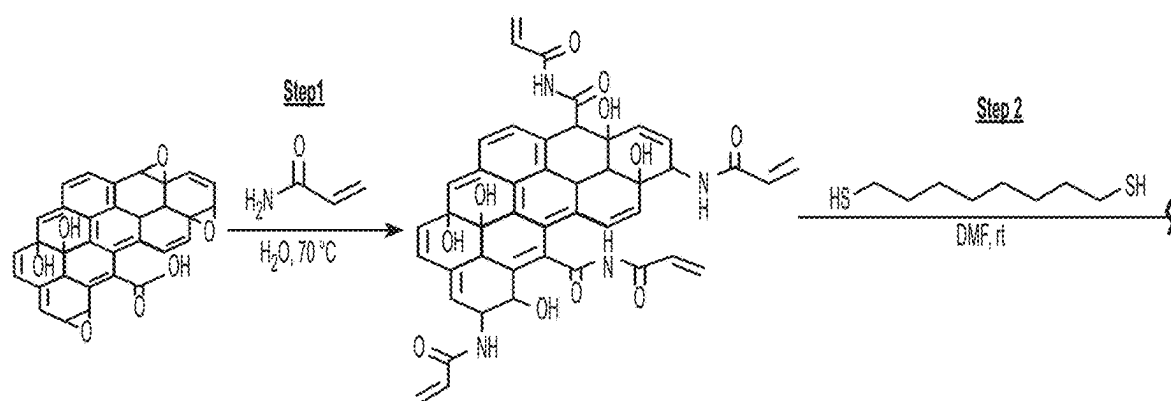


FIG. 14F

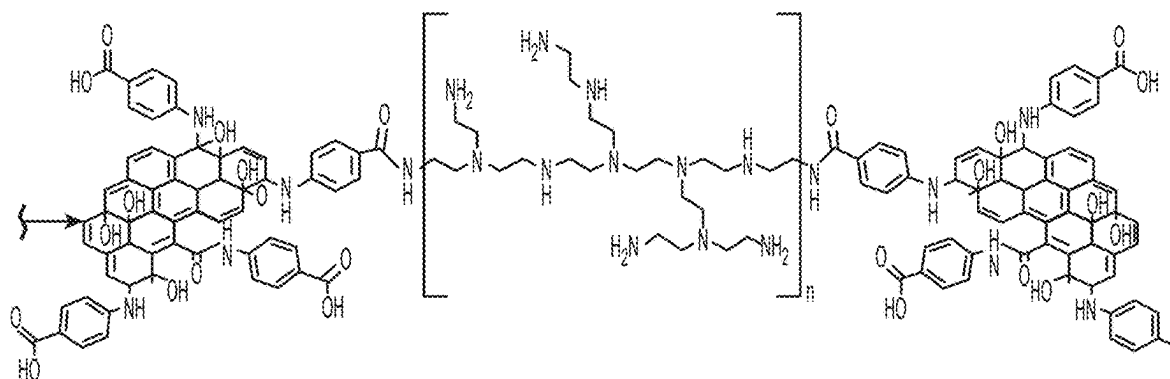


FIG. 14D (Cont.)

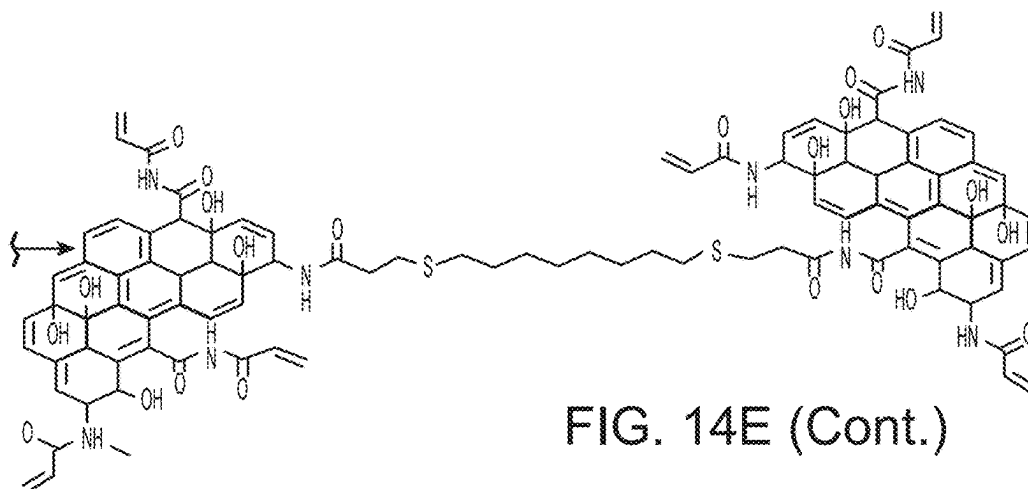


FIG. 14E (Cont.)

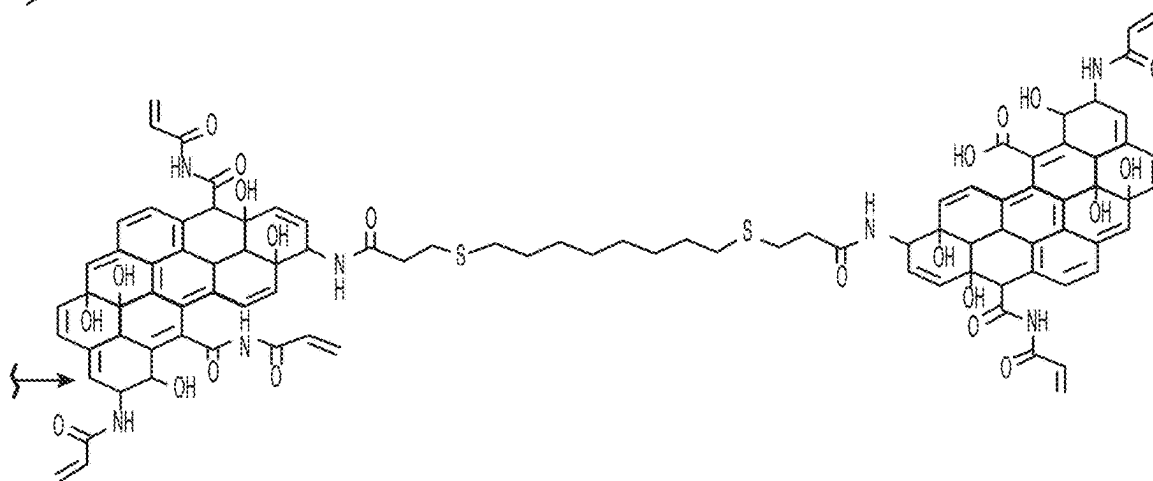


FIG. 14F (Cont.)

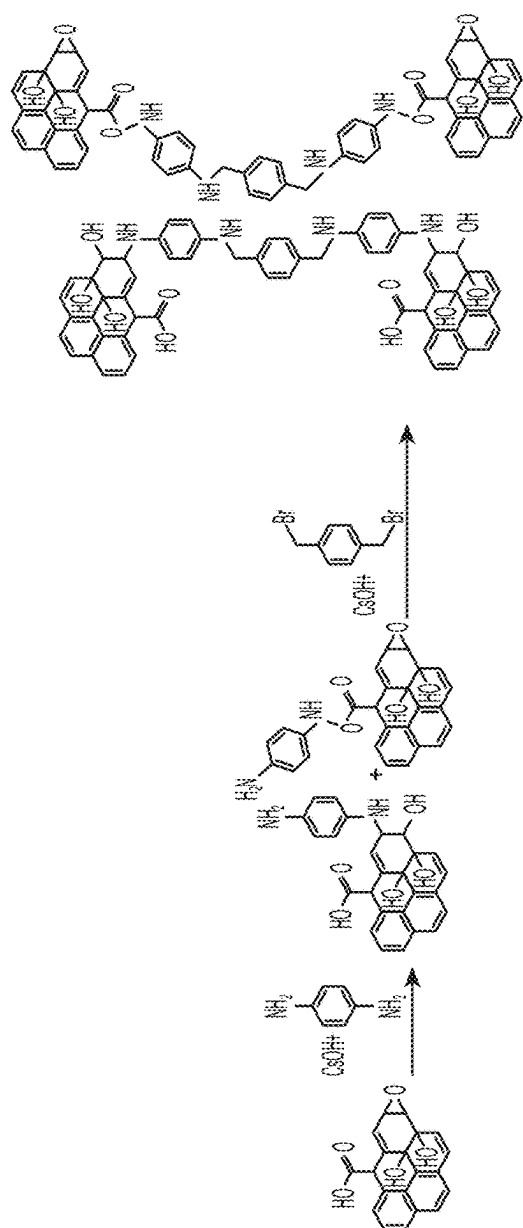


FIG. 14G

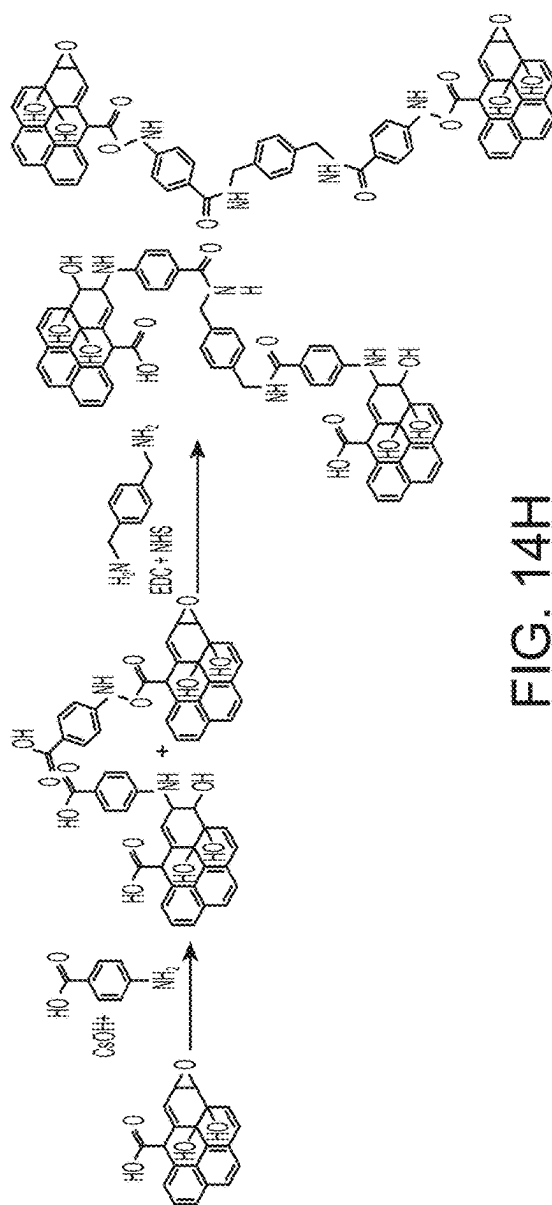


FIG. 14H

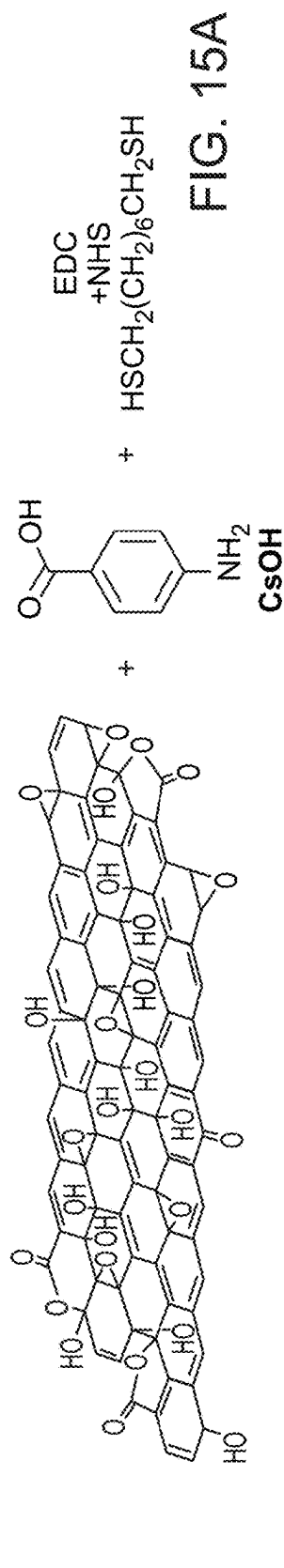


FIG. 15A

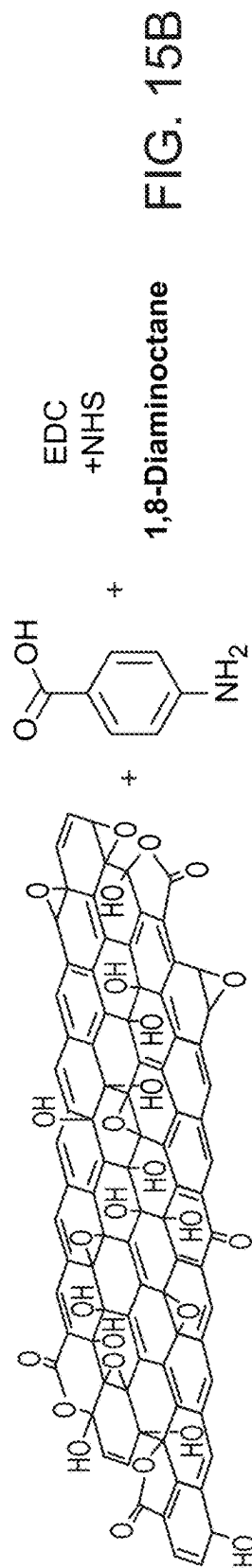


FIG. 15B

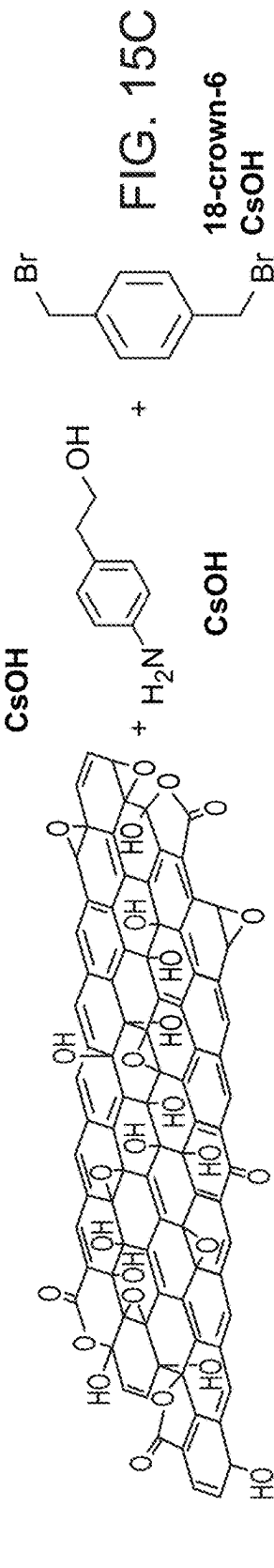


FIG. 15C

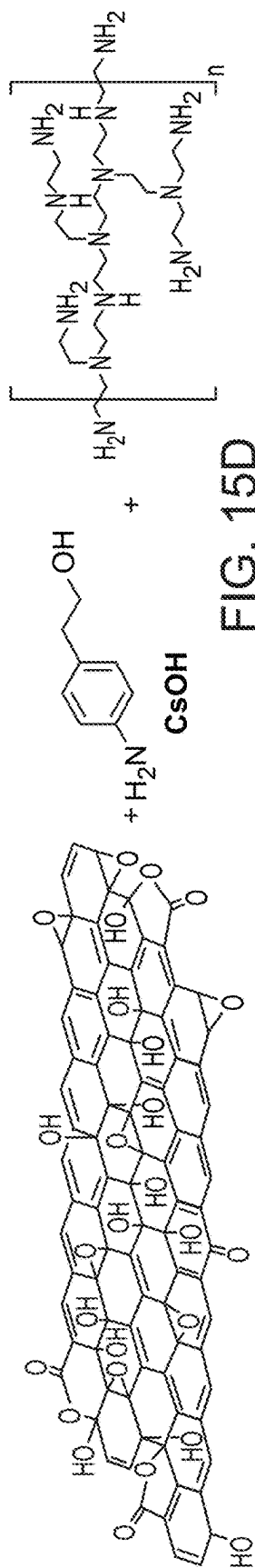


FIG. 15D

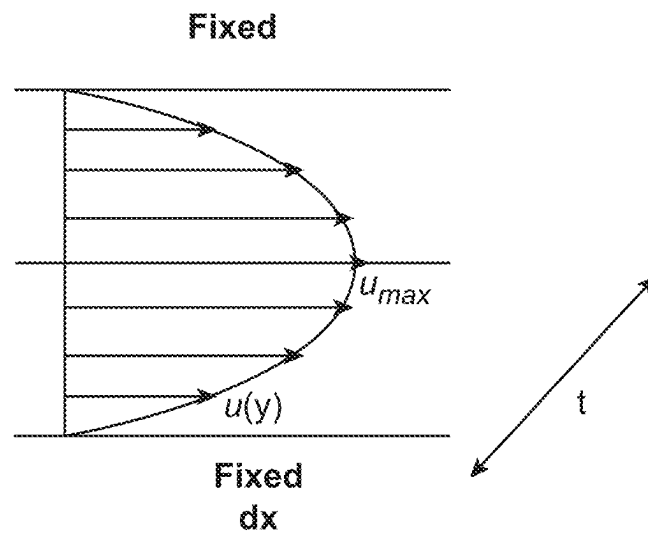


FIG. 16

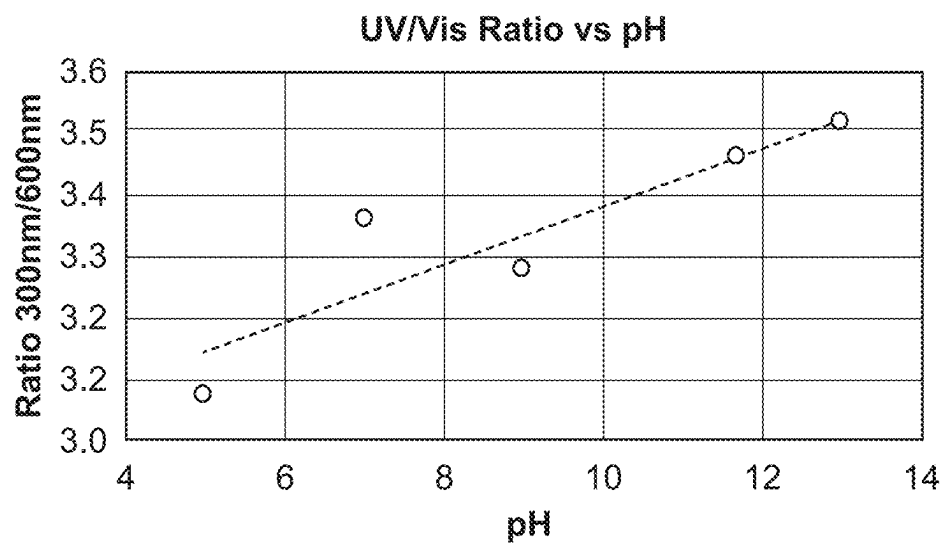


FIG. 17

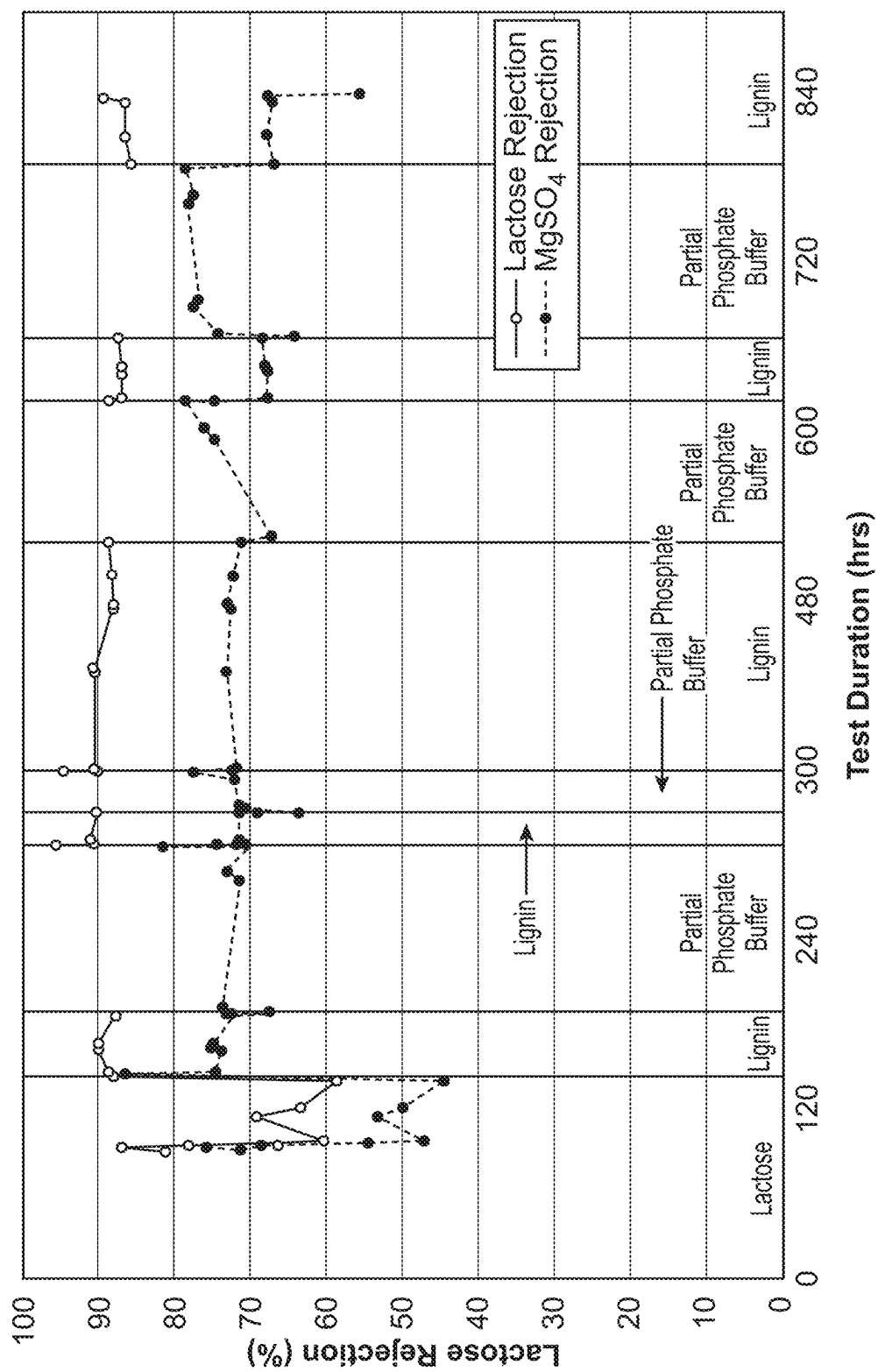


FIG. 18A

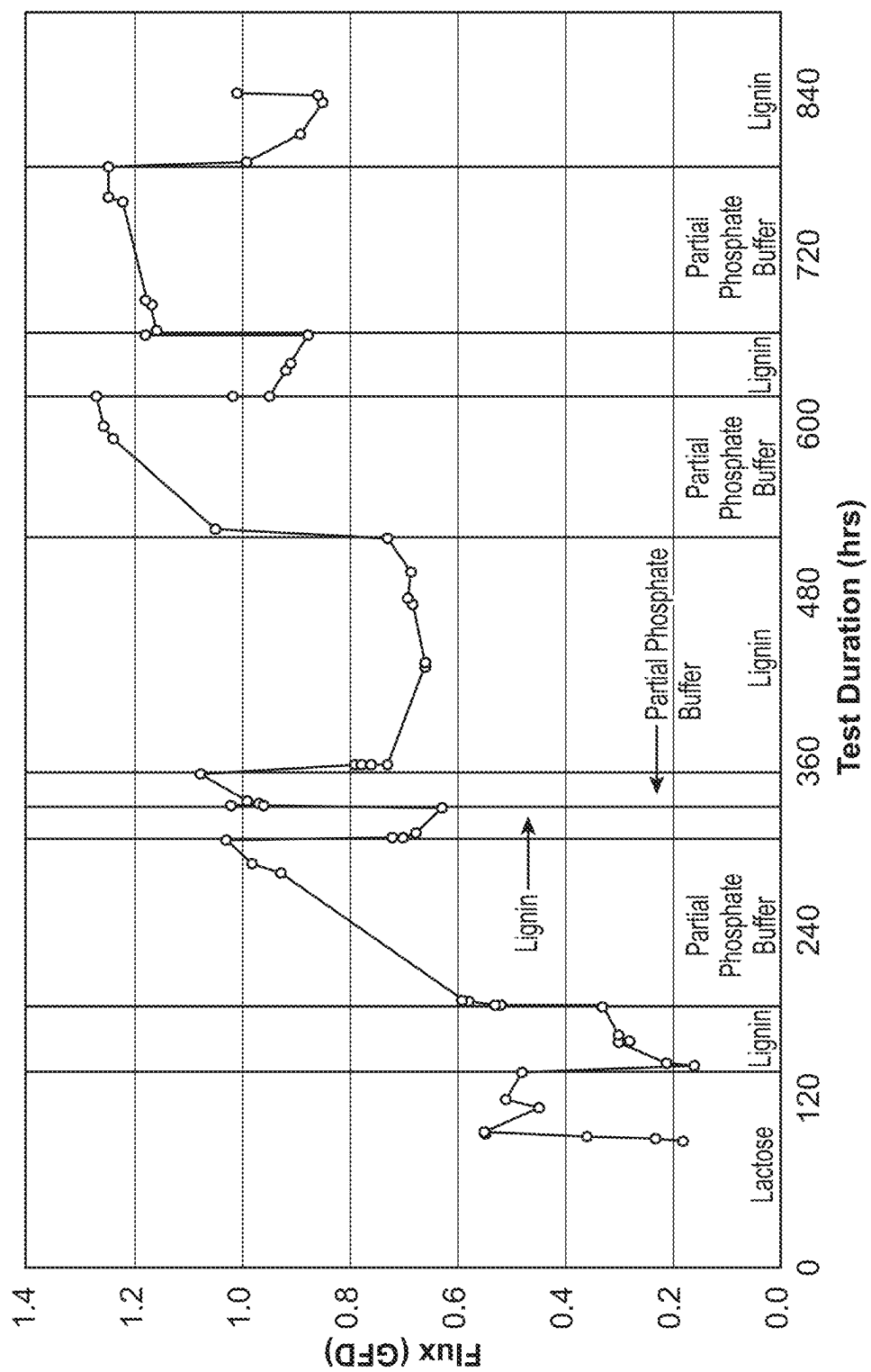


FIG. 18B

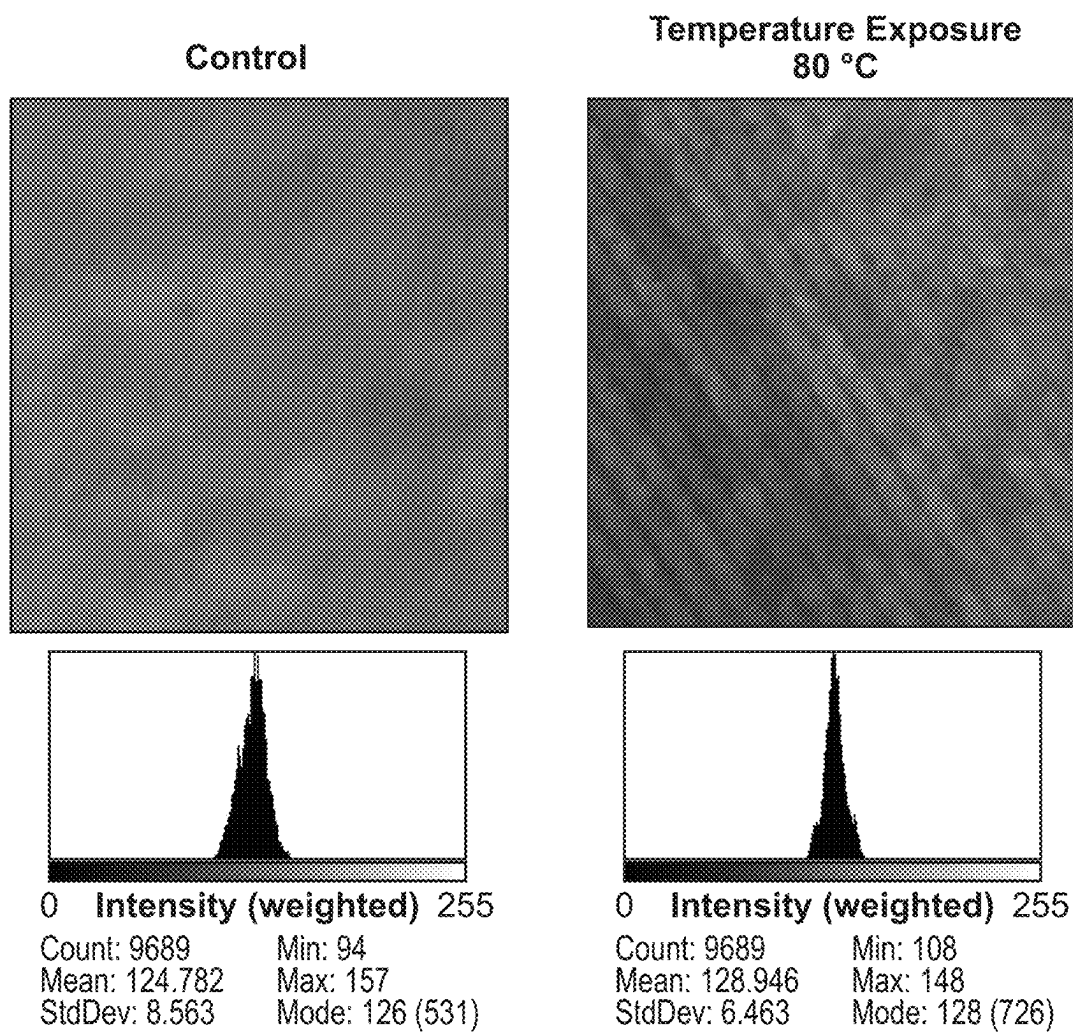
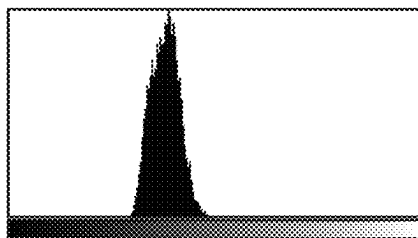
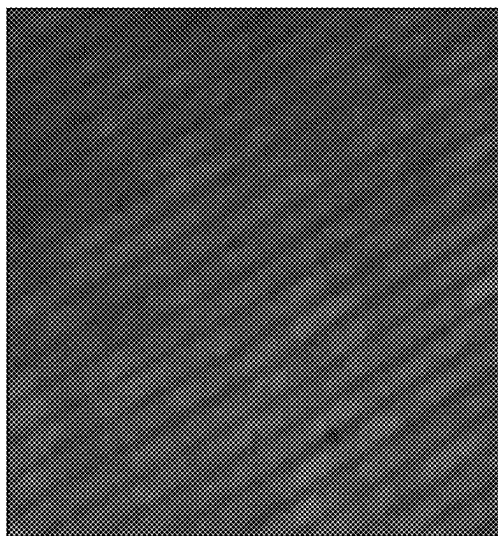


FIG. 19A

FIG. 19B

Temperature + pH Exposure
pH 11.5, 40 °C



0 Intensity (weighted) 255

Count: 13982

Min: 54

Mean: 95.948

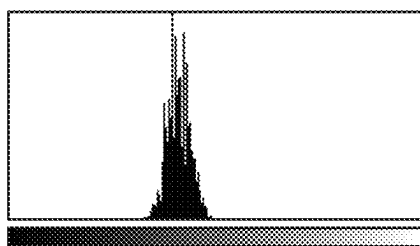
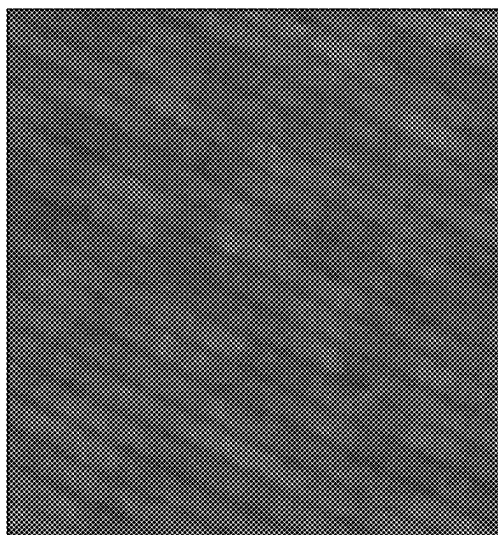
Max: 130

StdDev: 8.907

Mode: 98 (627)

FIG. 19C

pH Exposure
pH 11.5



0 Intensity (weighted) 255

Count: 7020	Min: 81
Mean: 103.930	Max: 128
StdDev: 7.113	Mode: 100 (555)

FIG. 19D

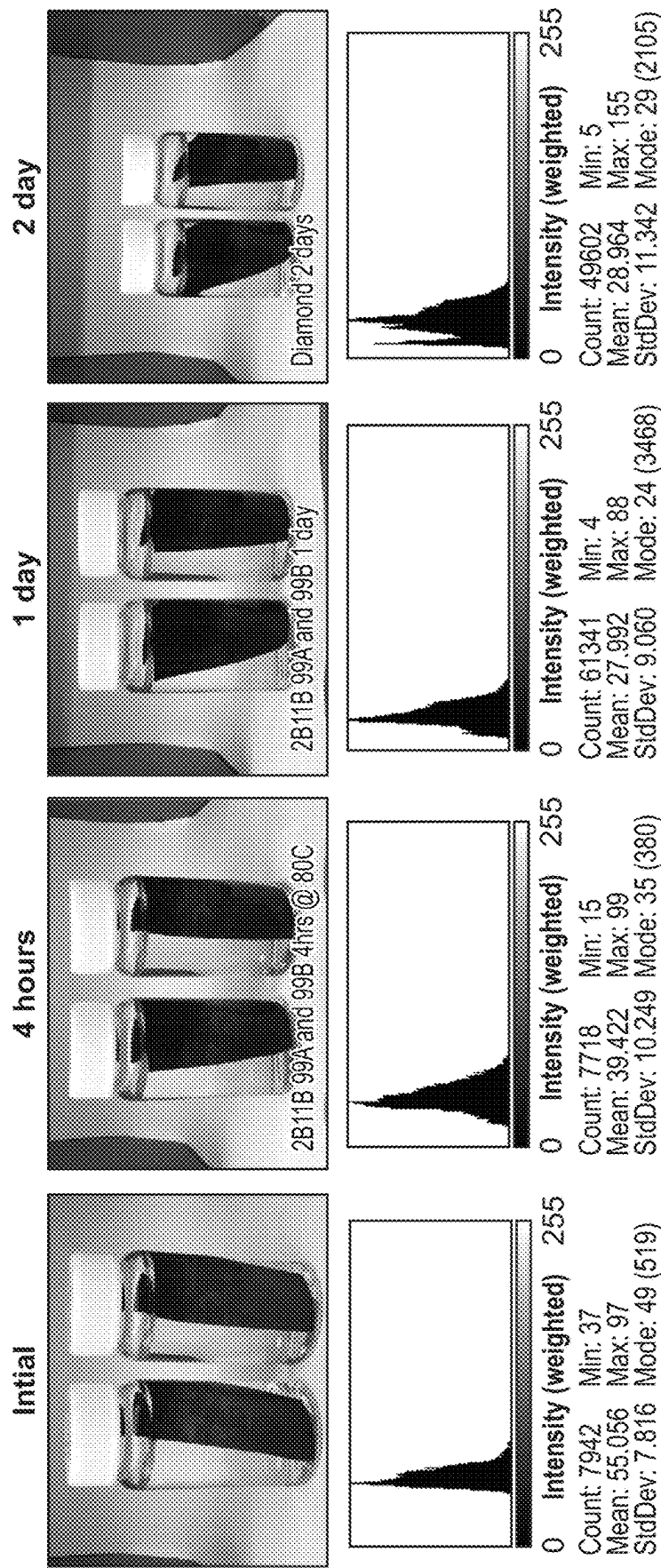
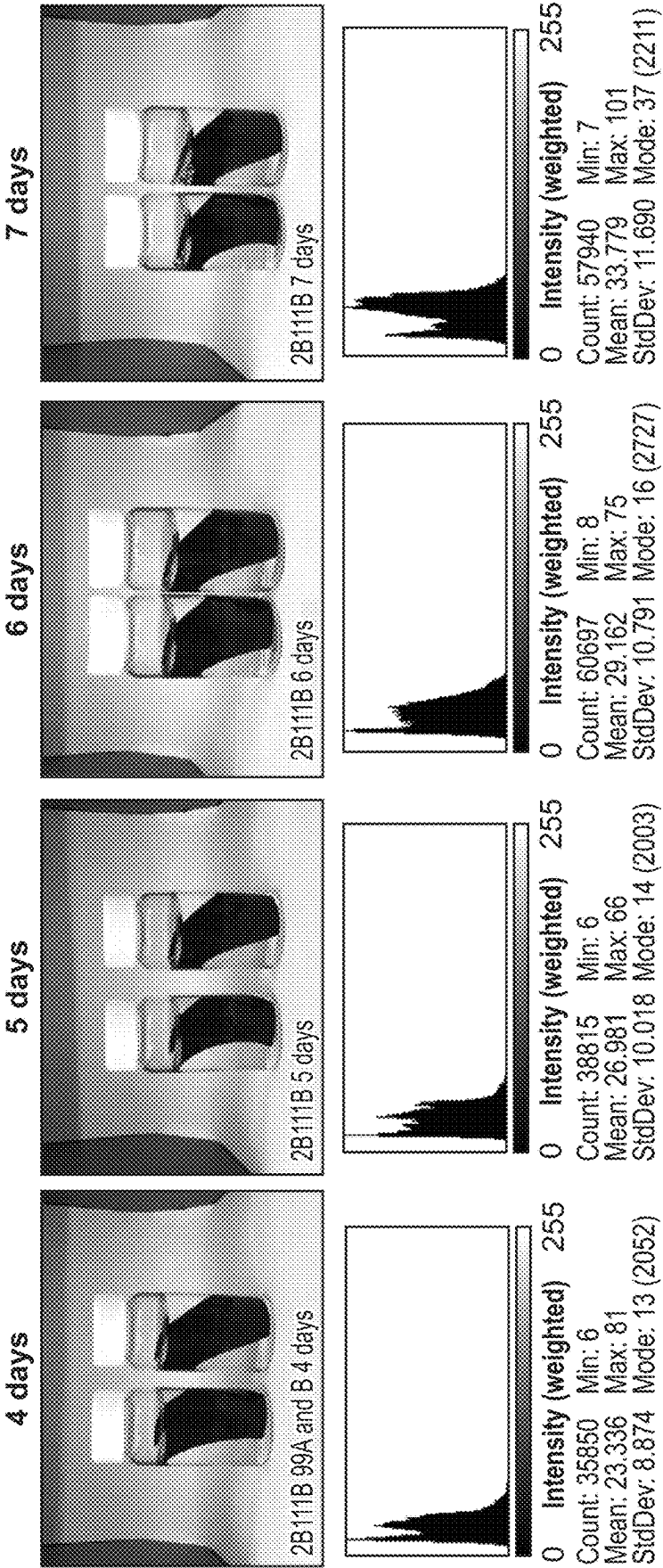


FIG. 20A

FIG. 20B

FIG. 20C

FIG. 20D



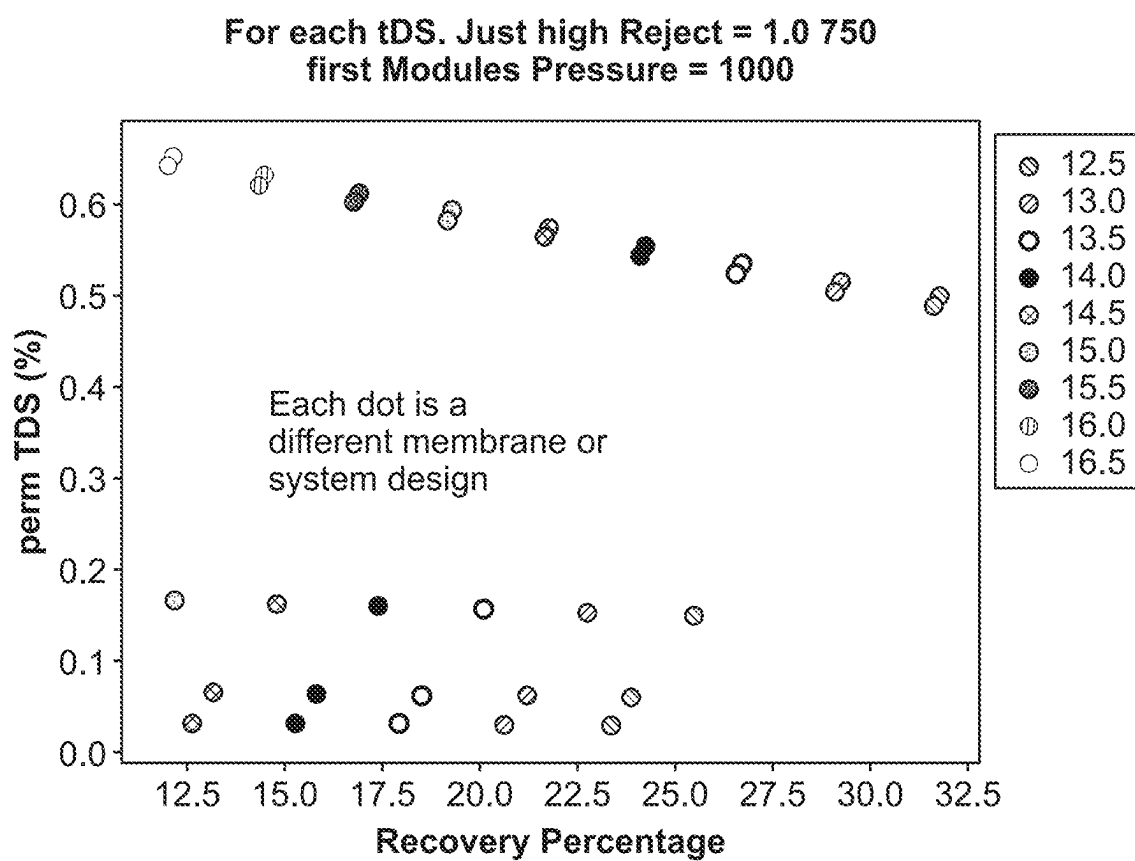
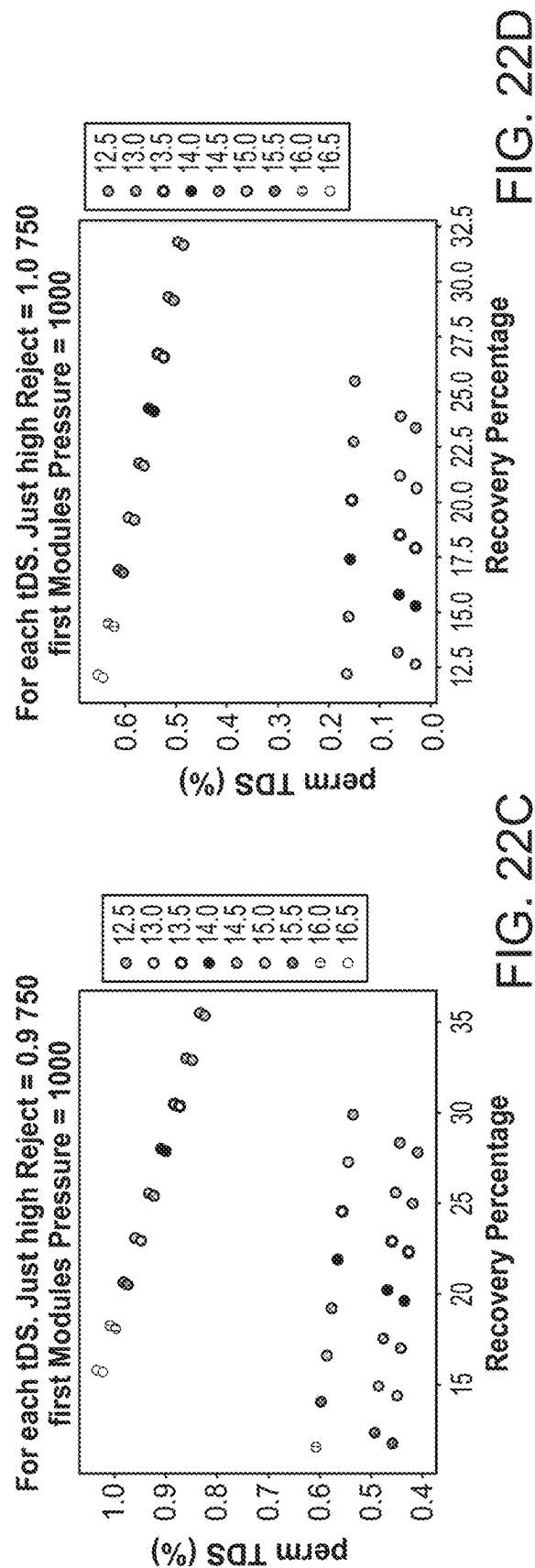
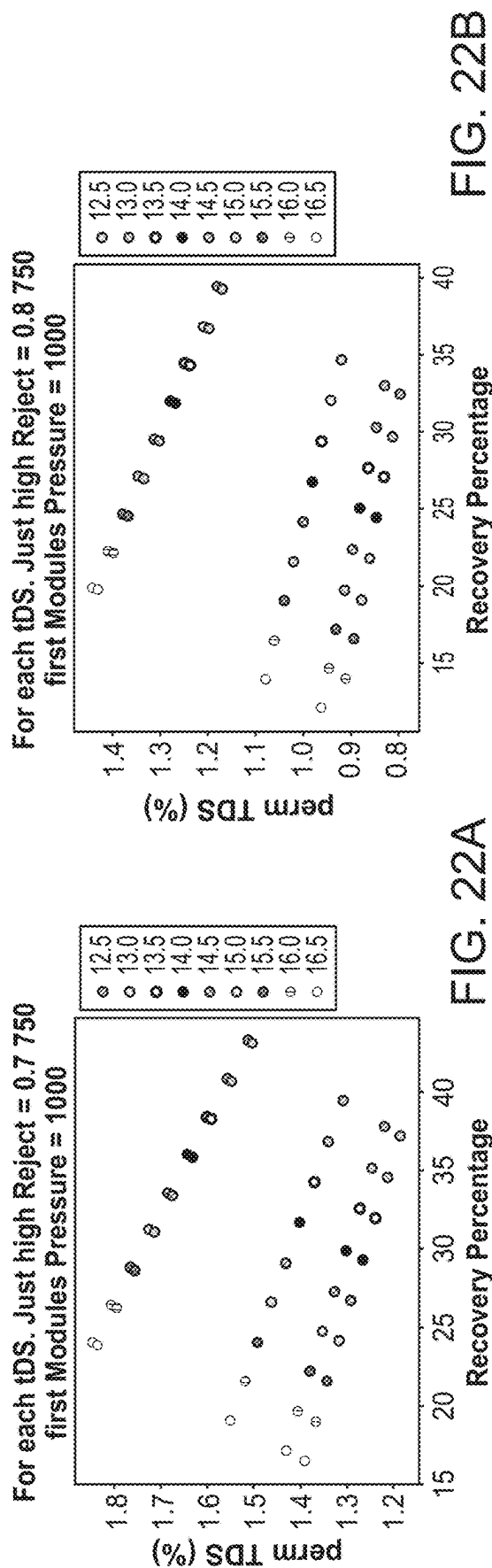
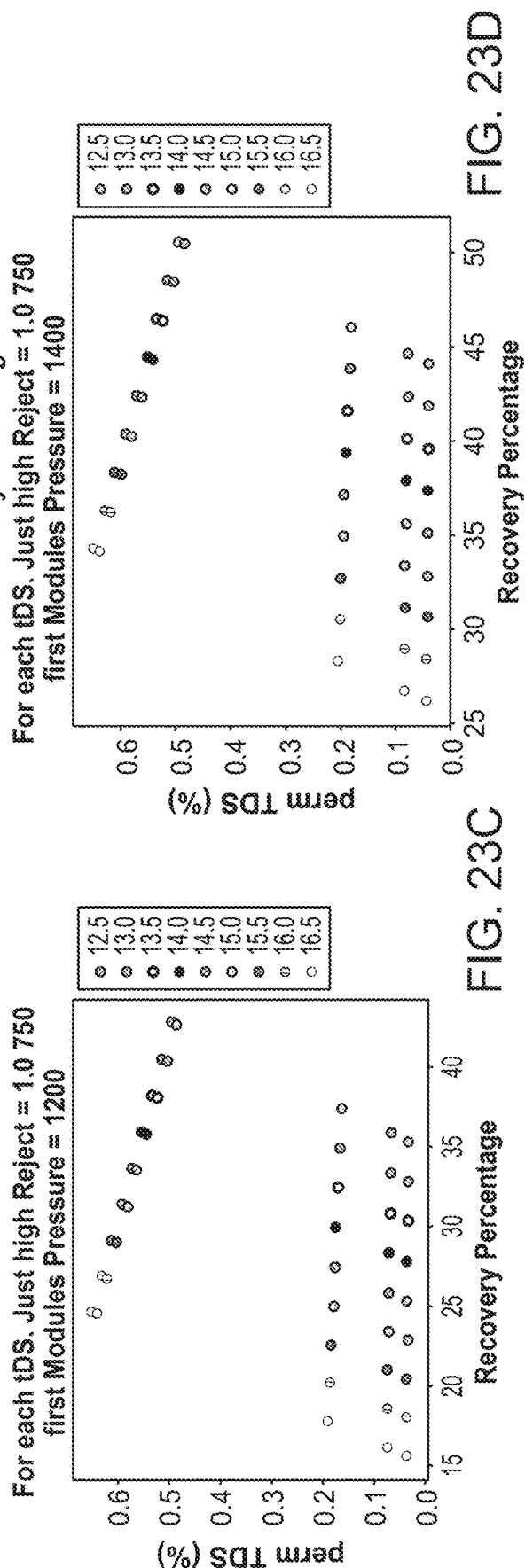
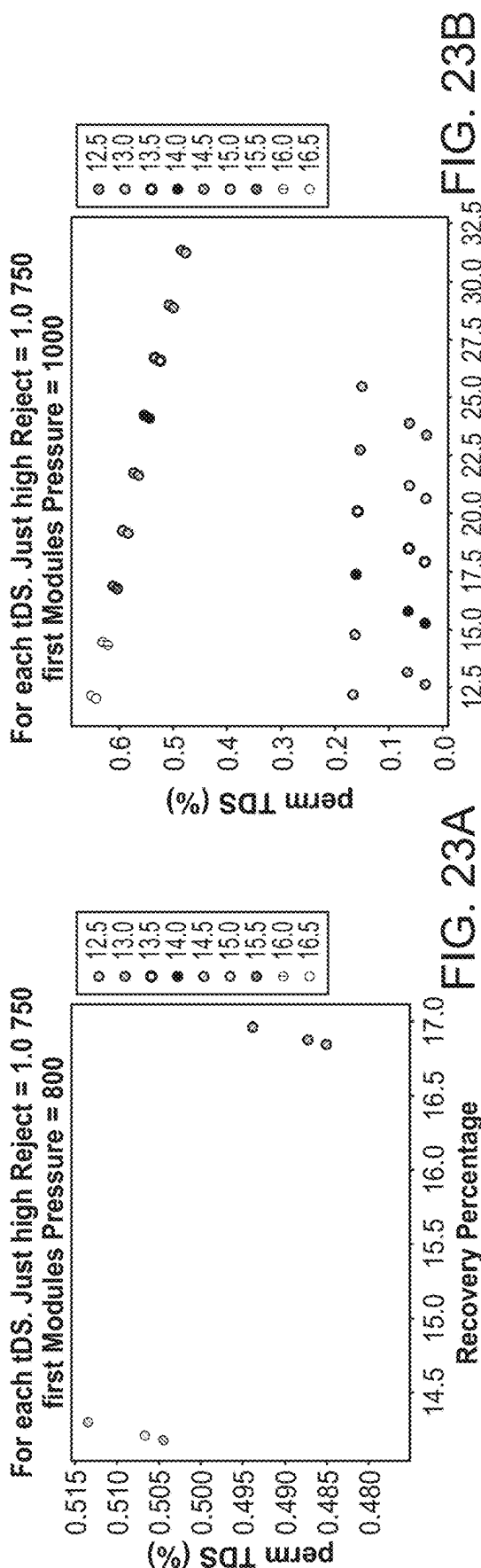
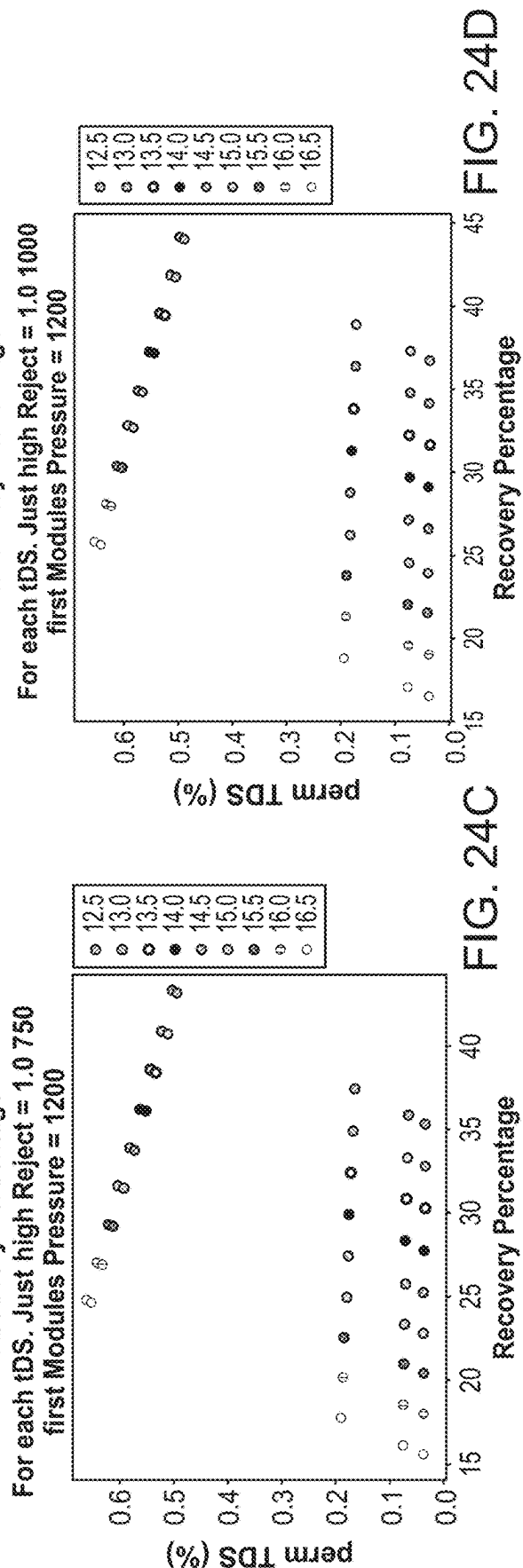
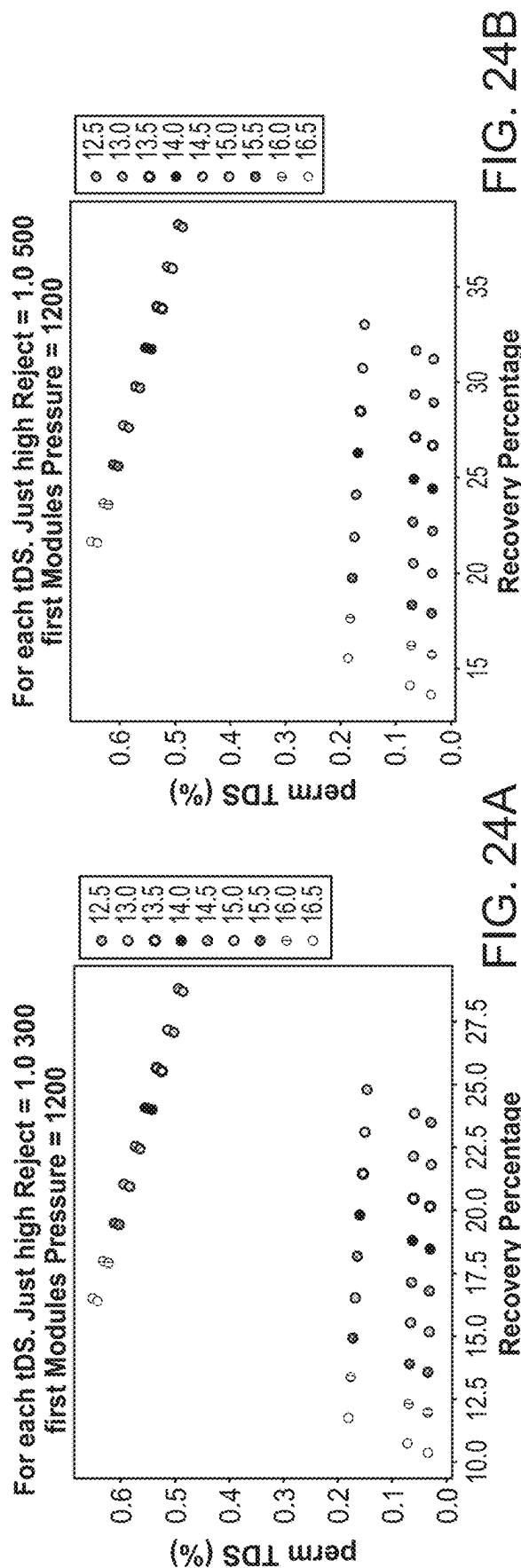
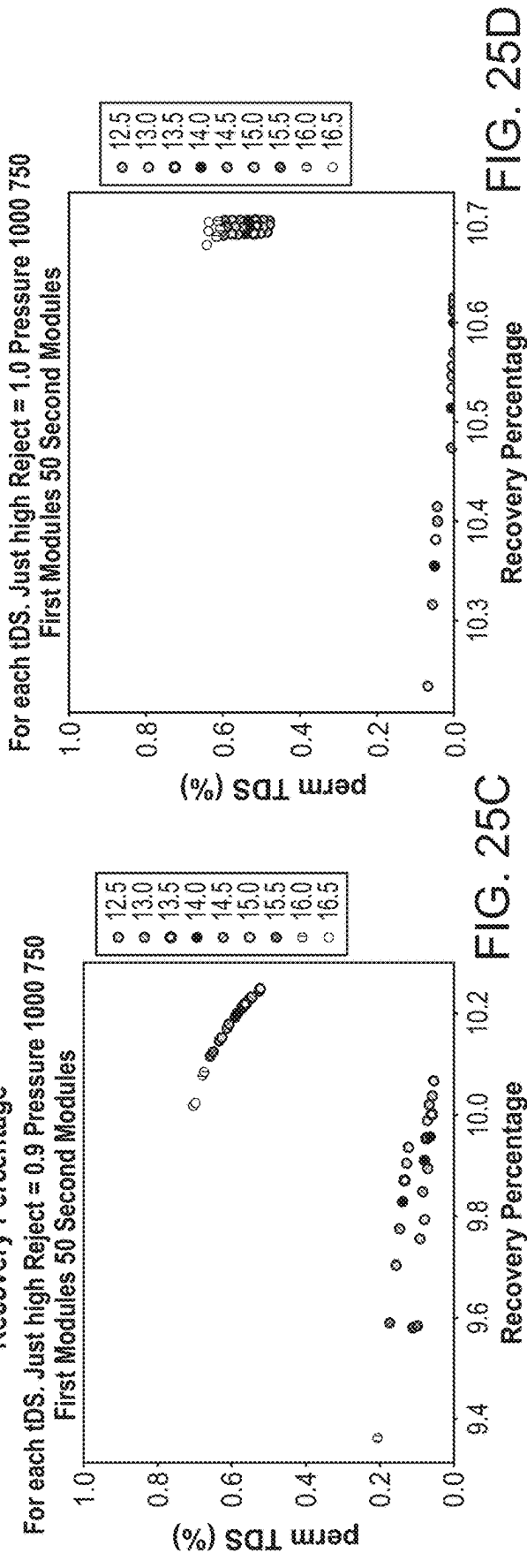
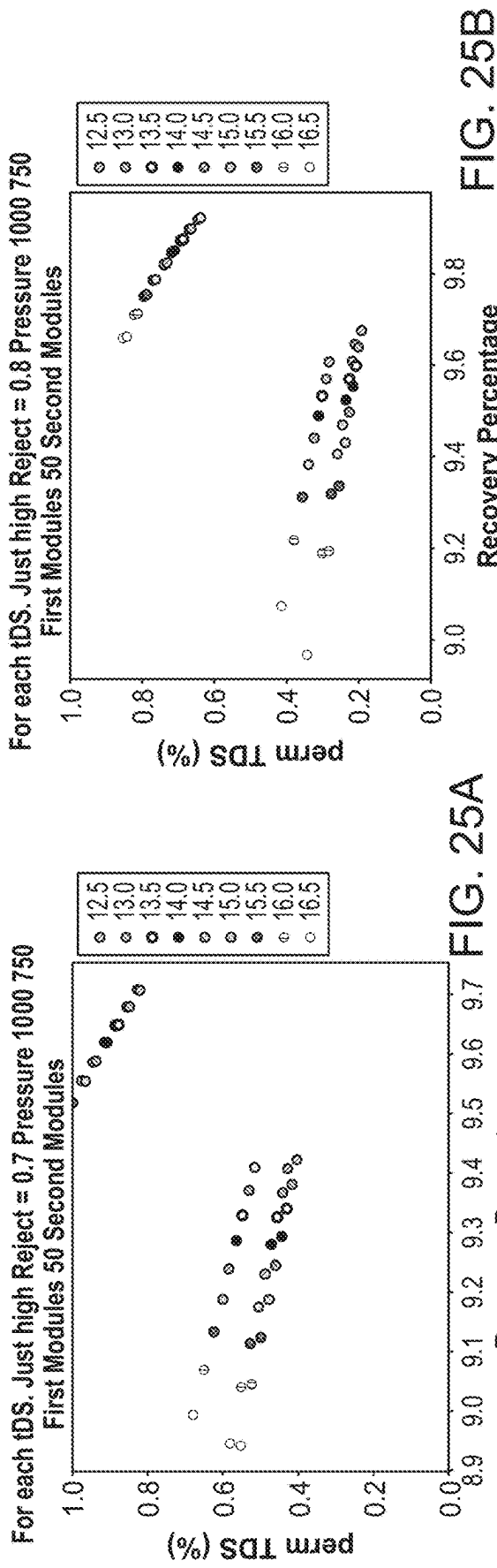


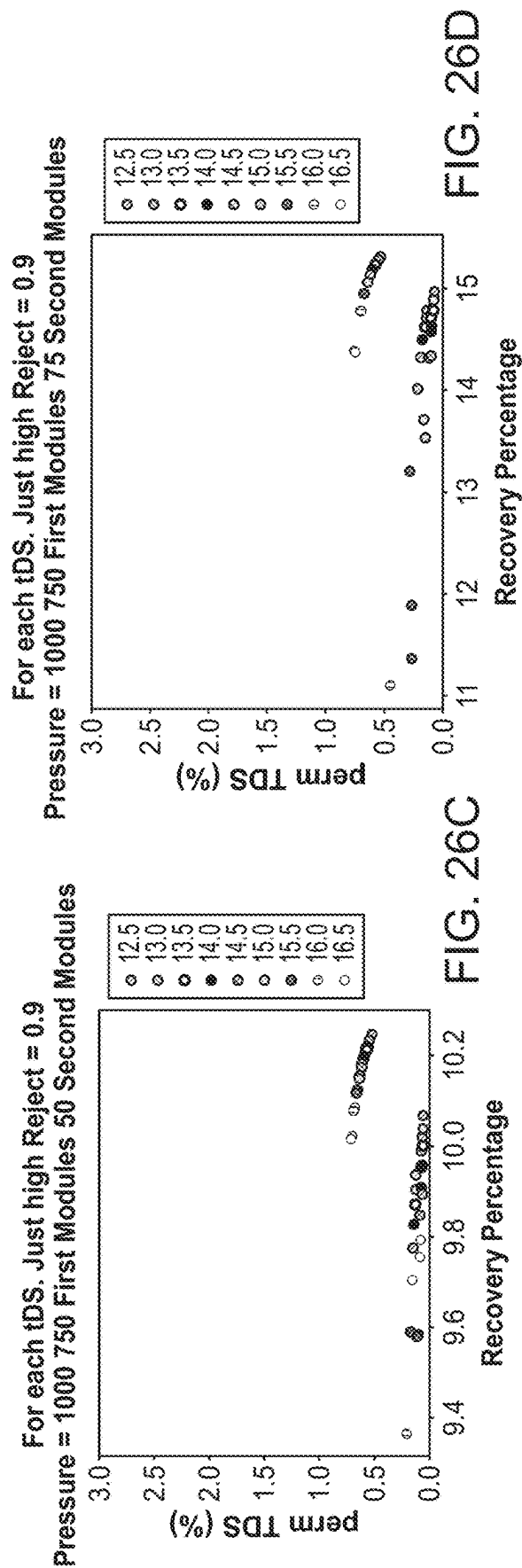
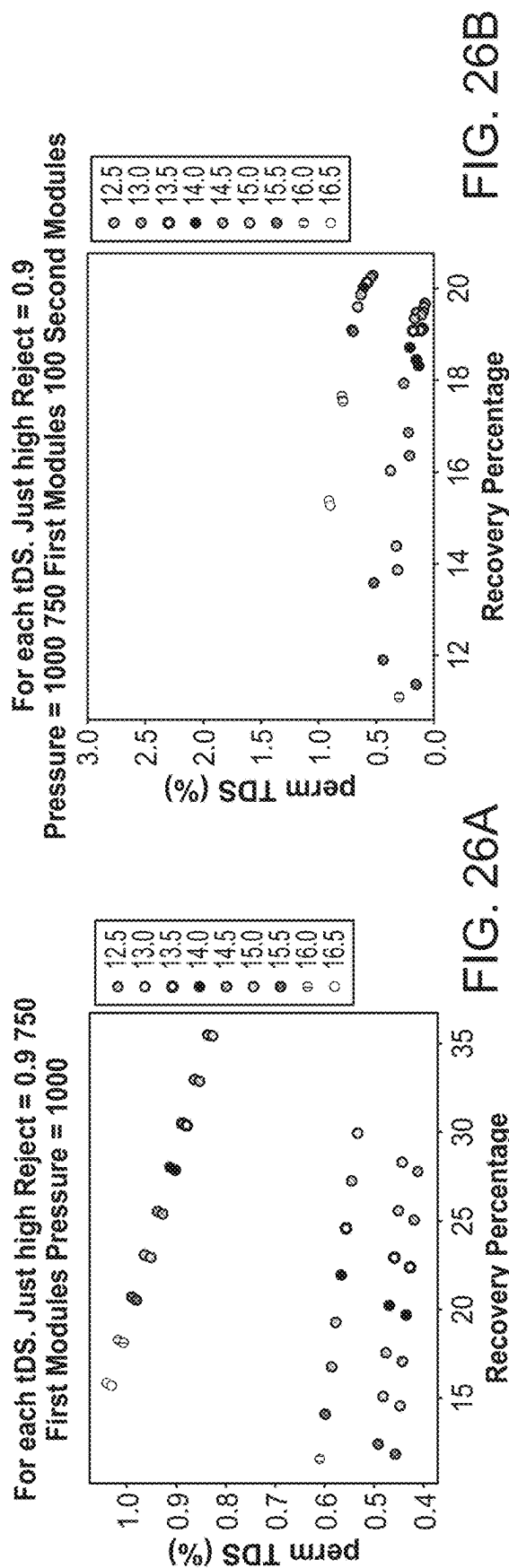
FIG. 21











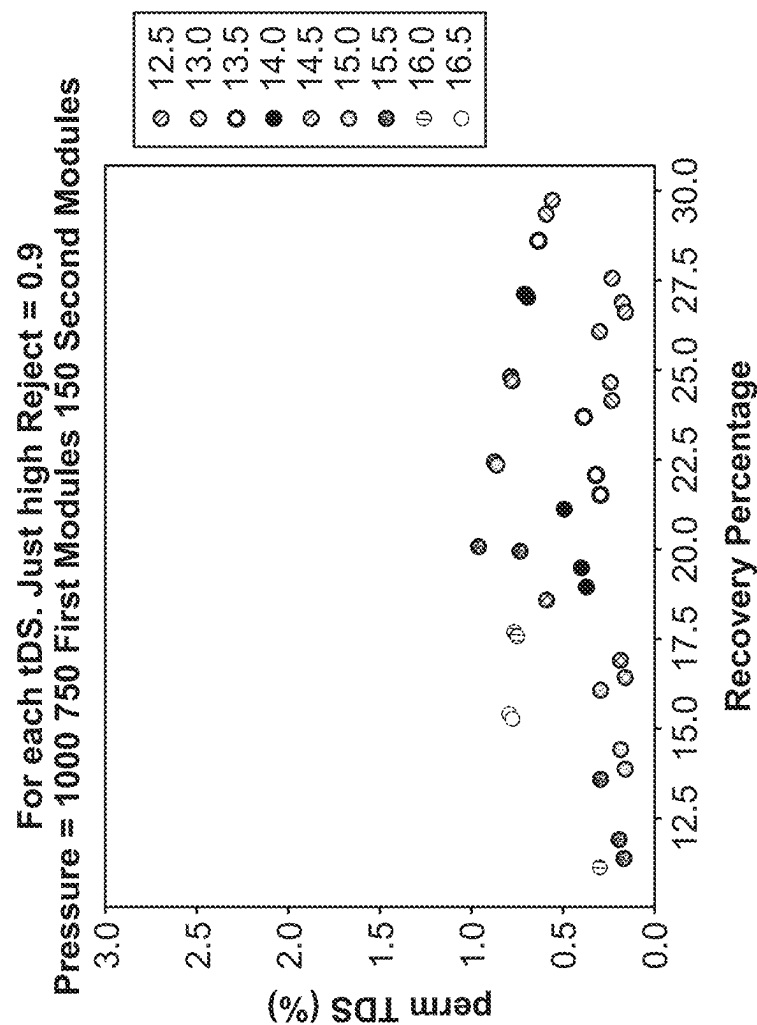


FIG. 26E

1

FILTRATION APPARATUS CONTAINING GRAPHENE OXIDE MEMBRANE

RELATED APPLICATIONS

This application is a continuation of U.S. patent application Ser. No. 16/885,864, filed May 28, 2020, and titled FILTRATION APPARATUS CONTAINING GRAPHENE OXIDE MEMBRANE, which is a continuation of International Patent Application No. PCT/US2020/033236, filed May 15, 2020, and titled FILTRATION APPARATUS CONTAINING GRAPHENE OXIDE MEMBRANE, which claims priority to and the benefit of U.S. Provisional Patent Application No. 62/848,014, filed on May 15, 2019, and U.S. Provisional Patent Application No. 62/910,789, filed on Oct. 4, 2019, the disclosures of which are hereby incorporated by reference in their entireties.

TECHNICAL FIELD

The present disclosure relates generally to graphene oxide membranes and their use in separation processes.

BACKGROUND

Membranes can be used to separate a mixture by passing some components (filtrate or permeate) and retaining others preferentially with a balance of the mixture (rejects) according to any of a variety of properties of the membrane and/or of the components of the material being filtered. For example, membranes can be configured to separate rejects from a filtrate based on size exclusion (i.e., a physical barrier such as pores that are smaller than the excluded particles). Other examples include membranes that are configured to separate rejects from a filtrate based on chemical, electrochemical, and/or physical binding with one or more components of the material being filtered.

Polymer membranes are a common type of membrane. They have been used commercially for water softening, desalination, and for the concentration, removal, and purification of different salts, small molecules, and macromolecules. However, in certain environments (e.g., oxidizing conditions, extreme pH, high temperatures, or in some solvents), polymer membranes can become damaged or fail due to swelling, oxidation reactions, degradation, or softening of the polymer. Accordingly, there is a need in the art for new membranes that address one or more deficiencies of polymer membranes.

SUMMARY

Embodiments described herein relate generally to graphene oxide membranes for fluid filtration. For example, the graphene oxide membranes can be used for concentration, removal, and purification of different salts.

One aspect of the present disclosure relates to a filtration apparatus comprising: (a) a support substrate, and (b) a graphene oxide membrane disposed on the support substrate. The graphene oxide membrane includes a plurality of graphene oxide sheets and each of the graphene oxide sheets are coupled to an adjacent graphene oxide sheet via a chemical linker. In some embodiments, the graphene oxide membrane has a flux of at least about 2.5×10^{-4} gallons per square foot per day per psi (GFD/psi) with a 1 wt % lactose solution at room temperature and a lactose rejection rate of at least 50% with a 1 wt % lactose solution.

2

In some embodiments, the rejection rate is measured at room temperature.

In some embodiments, the support substrate is a microporous polymer substrate.

5 In some embodiments, the support substrate has a root mean squared surface roughness of less than about 3.0 μm .

In some embodiments, the support substrate has an average pore size of 0.1 μm to 5 μm . In some embodiments, the support substrate has an average pore size of less than 1 μm .

10 In some embodiments, the support substrate has an average pore size of 0.2 μm to 1 μm . In some embodiments, the support substrate has an average pore size of about 0.45 μm .

In some embodiments, the support substrate includes a first layer and a second layer, wherein the graphene oxide membrane is disposed on the first layer, and wherein the first layer has a smaller average pore size than the second layer.

In some embodiments, the support substrate includes a material selected from polypropylene (PP), polystyrene, polyethylene, polyethylene oxide, polyethersulfone (PES), polytetrafluoroethylene (PTFE), polyvinylidene fluoride, polymethylmethacrylate, polydimethylsiloxane, polyester, cellulose, cellulose acetate, cellulose nitrate, polyacrylonitrile, glass fiber, quartz, alumina, silver, polycarbonate, nylon, Kevlar or other aramid, or polyether ether ketone.

25 In some embodiments, the support substrate is a polymer sheet.

In some embodiments, the support substrate is configured to be wound into a spiral filtration module.

In some embodiments, the support substrate is a hollow polymer tube.

In some embodiments, the graphene oxide membrane has a flux of at least about 2.5×10^{-3} GFD/psi with a 1 wt % lactose solution at room temperature.

35 In some embodiments, the graphene oxide membrane has a lactose rejection rate of at least 95% with a 1 wt % lactose solution.

In some embodiments, the graphene oxide membrane has a MgSO_4 rejection rate of at least 30% with a 0.1 wt % MgSO_4 solution.

40 In some embodiments, the filtration apparatus includes about 0.1 mg to 6 mg of the graphene oxide membrane per 5000 mm^2 . In some embodiments, the filtration apparatus includes about 2 mg of the graphene oxide membrane per 5000 mm^2 .

45 In some embodiments, the filtration apparatus further includes a housing.

One aspect of the present disclosure relates to a method of processing black liquor, wherein black liquor is flowed through any one of the aforementioned filtration apparatuses, wherein the black liquor comprises lignin, sodium sulfate, sodium carbonate, sodium hydrosulfide, sodium thiosulfate, and/or sodium hydroxide.

50 In some embodiments, at least a portion of the lignin is rejected by the graphene oxide membrane. In some embodiments, at least 50% of the lignin is rejected by the graphene oxide membrane. In some embodiments, at least 90% of the lignin is rejected by the graphene oxide membrane.

In some embodiments, at least a portion of the sodium sulfate is rejected by the graphene oxide membrane. In some embodiments, at least 50% of the sodium sulfate is rejected by the graphene oxide membrane. In some embodiments, at least 90% of the sodium sulfate is rejected by the graphene oxide membrane.

65 In some embodiments, at least a portion of the sodium carbonate is rejected by the graphene oxide membrane. In some embodiments, at least 50% of the sodium carbonate is rejected by the graphene oxide membrane. In some embodi-

ments, at least 90% of the sodium carbonate is rejected by the graphene oxide membrane.

In some embodiments, at least a portion of the sodium hydrosulfide is rejected by the graphene oxide membrane. In some embodiments, at least 50% of the sodium hydrosulfide is rejected by the graphene oxide membrane. In some embodiments, at least 90% of the sodium hydrosulfide is rejected by the graphene oxide membrane.

In some embodiments, at least a portion of the sodium thiosulfate is rejected by the graphene oxide membrane. In some embodiments, at least 50% of the sodium thiosulfate is rejected by the graphene oxide membrane. In some embodiments, at least 90% of the sodium thiosulfate is rejected by the graphene oxide membrane.

In some embodiments, at least a portion of the sodium hydroxide is rejected by the graphene oxide membrane. In some embodiments, at least 50% of the sodium hydroxide is rejected by the graphene oxide membrane. In some embodiments, at least 90% of the sodium hydroxide is rejected by the graphene oxide membrane.

One aspect of the present disclosure relates to a method of increasing a flux of the graphene oxide membrane described herein, the method comprising contacting the graphene oxide membrane with a solution that is at least about 50° C. for a period of time, wherein after the contacting step, the flux is increased by at least 50%.

Another aspect of the present disclosure relates to a method of increasing a flux of the graphene oxide membrane described herein, the method comprising contacting the graphene oxide membrane with a solution having a pH of at least about 10 for a period of time, wherein after the contacting step, the flux is increased by at least 50%.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1 is a schematic illustration of a graphene oxide membrane in accordance with some embodiments of the present disclosure.

FIG. 2 is a graph showing rejection rate as a function of molecule size for desalting, according to some embodiments of the present disclosure.

FIG. 3 is a graph showing rejection rate as a function of molecule size for concentration, according to some embodiments of the present disclosure.

FIG. 4 is a graph showing rejection rate as a function of molecule size for separation of larger and smaller molecules, according to some embodiments of the present disclosure.

FIG. 5 is a schematic illustration showing an esterification reaction to couple two graphene oxide sheets, according to some embodiments of the present disclosure.

FIG. 6 is a schematic illustration showing an amidation reaction to couple two graphene oxide sheets, according to some embodiments of the present disclosure.

FIG. 7 is a schematic illustration showing that the graphene oxide sheets are functionalized through etherification and then coupled through amidation, according to some embodiments of the present disclosure.

FIG. 8 is a schematic illustration showing that the graphene oxide sheets are functionalized through etherification and then coupled through etherification, according to some embodiments of the present disclosure.

FIG. 9 is a schematic illustration showing that the graphene oxide sheets are functionalized through etherification and then coupled through amidation, according to some embodiments of the present disclosure.

FIG. 10 is a schematic illustration showing that the graphene oxide sheets are functionalized through etherifi-

cation and then coupled through etherification, according to some embodiments of the present disclosure.

FIG. 11 is a schematic illustration showing that the graphene oxide sheets are functionalized through etherification and then coupled through etherification, according to some embodiments of the present disclosure.

FIG. 12 is a scanning electron microscope (SEM) image of the cross section of a graphene oxide membrane in accordance with some embodiments of the present disclosure. The graphene oxide membrane is disposed on a support substrate that includes polypropylene.

FIG. 13 is an SEM image of the cross section of a graphene oxide membrane in accordance with some embodiments of the present disclosure. The graphene oxide membrane is disposed on a support substrate that includes polytetrafluoroethylene.

FIGS. 14A-14H present examples of two-step reactions between graphene oxide sheets and chemical linkers.

FIGS. 15A-15D present examples of two-step reactions between graphene oxide sheets and chemical linkers.

FIG. 16 is a modeling diagram that models an arbitrarily sized membrane system using simple picture of permeability and rejection from dead end or crossflow. Every constituent species is tracked. Counter ions (e.g., sodium) given same rejection as pair ion (e.g., Na paired with lignin, Na paired with Cl, Na paired with SO₄, etc.). (Total or species flow rate in)*a*t=2*(flux through membrane)*dx*t+(flow rate out)*a*t.

FIG. 17 is a graph showing the UV/Vis ratio versus pH for a graphene oxide dispersion.

FIG. 18A is a graph showing a tangential flow test that has been running for >4 weeks at room temperature at 300 psi. Initially, performance is benchmarked against the 1 wt % lactose and 0.1 wt % MgSO₄ solution. Then, exposure was alternated between about 2 wt % lignin solution and a partial phosphate buffer (PPB) solution at pH 11.5. Lactose and lignin rejection, plotted on the y-axis, does not drop after repeated exposures to partial phosphate buffer at pH 11.5.

FIG. 18B is a graph showing the same tangential flow test as in FIG. 18A. The flux is shown to steadily improve from exposure to the PPB at room temperature and this improvement is retained during the alternating lignin steps showing the effect is persistent.

FIGS. 19A-19D are graphs showing the effect of pH on the color of the graphene oxide membrane, which can be indicative of the stability of the membrane. The membrane is functionalized by 2-(4-aminophenyl) ethanol prior to crosslinking. The color change may be quantified using image processing software by converting the red/green/blue (RGB) image to grayscale and comparing the grayscale mode value of the resulting signals. The histograms below each image show the distribution of gray values for each image with the x-axis representing possible gray values (0-255) and the y-axis representing the number of pixels found at each gray value. Grayscale mode values closer to zero correspond to darker images. Conversely, grayscale mode values closer to 255 correspond to lighter images. In, FIGS. 19A-19C, exposure at 80° C. has no effect on color (mode shifts from 126 in the control to 128 after exposure to 80° C. for 4 hours); after exposure to partial phosphate buffer (pH 11.5) for 4 hours at 40° C., the mode changes from 126 in the control to 98. In FIG. 19D, pH stability is shown.

FIGS. 20A-20H are a series of photographs and graphs showing the stability of graphene oxide membranes exposed to partial phosphate buffer at 80° C. and pH 11.5, where the membranes are crosslinked by p-phenylenediamine.

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FIG. 21 is a graph showing simulation results for a 1.0 rejection coefficient for divalent salts, a 750-module system size, and 1000 psi operating pressure. Each series of dots represents a different feed concentration. Some mills will have more or less concentrated feed liquor.

FIGS. 22A-22D, 23A-23D, 24A-24D, 25A-25D, and 26A-26E are graphs showing the recovery percentage of simulated permeate total dissolved solids (perm TDS) for different system sizes (as represented by recovery ratio). Each series of dots represents a different feed concentration. These figures show that sufficiently clean hot water streams can be produced by graphene oxide membranes. Multiple pass systems mean the permeate from the first membrane system passes through a second membrane system. The recovery is defined by the percentage of first system feed that becomes permeate from the second system in that case.

FIGS. 22A-22D are graphs showing simulation results for the effect of divalent rejection at 1000 psi in a single-pass system.

FIGS. 23A-23D are graphs showing simulation results for the effect of pressure at high rejection in a single-pass system.

FIGS. 24A-24D are graphs showing simulation results for the system size in a single-pass system.

FIGS. 25A-25D are graphs showing simulation results for the effect of rejection in a two-pass system.

FIGS. 26A-26E are graphs showing simulation results for multi-pass system sizing.

DETAILED DESCRIPTION

Graphite is a crystalline form of carbon with its atoms arranged in a hexagonal structure layered in a series of planes. Due to its abundance on earth, graphite is very cheap and is commonly used in pencils and lubricants. Graphene is a single, one atomic layer of carbon atoms (i.e., one of the layers of graphite) with several exceptional electrical, mechanical, optical, and electrochemical properties, earning it the nickname “the wonder material.” To name just a few, it is highly transparent, extremely light and flexible yet robust, and an excellent electrical and thermal conductor. Such extraordinary properties render graphene and related thinned graphite materials (e.g., few layer graphene) as promising candidates for a diverse set of applications. For example, graphene can be used in coatings to prevent steel and aluminum from oxidizing, and to filter salt, heavy metals, and oil from water.

Graphene oxide is an oxidized form of graphene having oxygen-containing pendant functional groups (e.g., epoxide, carboxylic acid, or hydroxyl) that exist in the form of single atom thick sheets. By oxidizing the graphene in graphite, graphene oxide sheets can be produced. For example, the graphene oxide sheets can be prepared from graphite using a modified Hummers method. Flake graphite is oxidized in a mixture of KMnO_4 , H_2SO_4 , and/or NaNO_3 , then the resulting pasty graphene oxide was diluted and washed through cycles of filtration, centrifugation, and resuspension. The washed graphene oxide suspension is subsequently ultrasonicated to exfoliate graphene oxide particles into graphene oxide sheets and centrifuged at high speed to remove unexfoliated graphite residues. The resulting yellowish/light brown solution is the final graphene oxide sheet suspension. This color indicated that the carbon lattice structure is distorted by the added oxygenated functional groups. The produced graphene oxide sheets are hydrophilic and can stay suspended in water for months without a sign of aggregation or deposition.

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Due in part to its low cost, high chemical stability, strong hydrophilicity, and compatibility with a variety of environments, graphene oxide has been explored for its use in membranes in filtration applications. For example, as compared to polymer membranes, which can be prone to oxidation, graphene oxide membranes can remain stable when exposed to oxidation conditions. However, existing graphene oxide membranes are plagued by issues of lower flux and lower rejection rate for small molecules (e.g., lactose) and salt (e.g., MgSO_4). For example, some existing graphene oxide membranes can achieve a high flux, but the rejection rate for lactose and/or MgSO_4 is low in these membranes, rendering them ineffective for filtration applications. In contrast, the present disclosure provides graphene oxide membranes having at least about 2.5×10^{-4} gallons per square foot per day per psi (GFD/psi) with a 1 wt % lactose solution at room temperature, and the graphene oxide membranes can achieve a rejection rate of greater than 50% with a 1 wt % lactose solution.

However, many filtration applications require a high flux and/or high rejection rate for small molecules and salt. Thus, the present disclosure provides graphene oxide membranes having one or more superior properties over existing graphene oxide membranes. At least by tuning the chemistries that couple adjacent graphene oxide sheets, the present disclosure also provides graphene oxide membranes having tunable rejection rate and flux.

As shown in FIG. 1, one aspect of the present disclosure provides a graphene oxide membrane 100 that includes a plurality of graphene oxide sheets 110 and a plurality of chemical linkers 120.

In some embodiments, the graphene oxide sheets 110 can be arranged and oriented generally parallel to each other and each of the graphene oxide sheets 110 can be coupled to an adjacent graphene oxide sheet 110 via at least one chemical linker 120.

The chemical linker 120 can be either linear or branched. In some embodiments, the chemical linkers 120 coupling adjacent graphene oxide sheets 110 can include a combination of linear and branched structures. In some embodiments, the length of the chemical linker 120 may be selected to impart desirable properties and/or control the spacing between the graphene oxide sheets 110. The spacing between the graphene oxide sheets 110 can be either interlayer spacing or intralayer spacing. The spacing between the graphene oxide sheets 110 can be engineered to control the molecular weight cutoff of the graphene oxide membrane 100.

The chemical linker 120 can have at least two ends that are coupled to adjacent graphene oxide sheets 110. For example, the chemical linker can include a first end 122 coupled to a first graphene oxide sheet and a second end 124 coupled to a second graphene oxide sheet. The first end 122 can be coupled to the first graphene oxide sheet through a covalent bond or a noncovalent interaction. The second end 124 can be coupled to the second graphene oxide sheet through a covalent bond or a noncovalent interaction. In some embodiments, an end of the chemical linker 120 (e.g., the first end 122, the second end 124, or another end) may be dangling, i.e., not coupled to anything.

In some embodiments, the chemical linker 120 can form a covalent bond with the oxygen-containing functional groups on the graphene oxide sheet 110. For example, the chemical linker 120 can form a covalent bond with the epoxide groups, carboxylic groups, or hydroxyl groups on graphene oxide. In some embodiments, the chemical linker 120 can also form a covalent bond with non-oxygen-con-

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taining groups (e.g., amine) on the graphene oxide sheet **110**. In some embodiments, the chemical linker **120** can also form a covalent bond with a carbon atom on the graphene oxide sheet **110**.

In some embodiments, the chemical linker **120** can form a noncovalent interaction with graphene through a variety of mechanisms. In some embodiments, the chemical linker **120** can be coupled to the graphene oxide sheet **110** through an ionic interaction. In some embodiments, the chemical linker **120** can be coupled to the graphene oxide sheet **110** through hydrogen bonding. In some embodiments, the chemical linker **120** can be coupled to the graphene oxide sheet **110** through one or more Van der Waals forces. In some embodiments, the chemical linker **120** can be coupled to the graphene oxide sheet **110** through one or more π -effects. In some embodiments, the chemical linker **120** can be coupled to the graphene oxide sheet **110** through the hydrophobic effect.

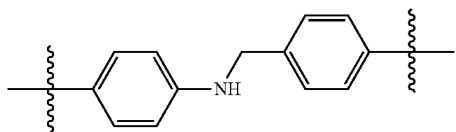
In some embodiments, the chemical linker **120** can have a structure in accordance with Formula I:



wherein: A is absent, aryl, heteroaryl, C_1 - C_{10} alkylenyl linker, C_2 - C_{10} alkenylenyl linker, or $(-CH_2-CH_2-O-)_p$ ($p=1$ to 5), each of which can be optionally substituted; and

R_1 and R_2 are independently selected from C_1 - C_{10} alkyl, C_1 - C_{10} alkenyl, C_1 - C_{10} hydroxyalkyl, $-C_0$ - C_6 alkyl- $C(O)-O-C_0$ - C_6 alkyl, $-C(O)-O-C_1$ - C_{10} alkyl, $-C_0$ - C_6 alkyl- $C(O)-S-C_0$ - C_6 alkyl, $-C(O)-S-C_1$ - C_{10} alkyl, $-C_0$ - C_6 alkyl- $O-C_0$ - C_6 alkyl, $-O-C_1$ - C_{10} alkyl, $-C_0$ - C_6 alkyl- $S-C_0$ - C_6 alkyl, $-S-C_1$ - C_{10} alkyl, $-C_0$ - C_6 alkyl- $NH-C_0$ - C_6 alkyl, $-NH-$, $-NH-(C_1-C_{10} \text{ alkyl})_2$, $-NH-C_1$ - C_{10} alkyl, $-C_0$ - C_6 alkyl- $NH-C(O)-C_0$ - C_6 alkyl, $-NH-C(O)-C_1$ - C_{10} alkyl, and $(-CH_2-CH_2-O-)_p$ ($p=1$ to 5), each of which can be optionally substituted, wherein one end of each of R_1 and R_2 can be optionally coupled to a graphene oxide sheet. In some embodiments, the alkyl, alkenyl, or hydroxyalkyl R_1 and/or R_2 can be optionally coupled to a graphene oxide sheet.

In some embodiments, A is phenyl, biphenyl, naphthyl, or



where $\sim$ denotes the point of coupling with R_1 or R_2 .

In some embodiments, A is a C_1 - C_6 alkylenyl linker or a C_2 - C_6 alkenylenyl linker, each of which can be optionally substituted.

In some embodiments, A is absent.

In some embodiments, R_1 and R_2 independently includes an ether, amine, amide, thioether, or a combination thereof.

In some embodiments, R_1 and R_2 are independently selected from $-(CH_2)_{1-10}O-$, $-(CH_2)_{1-10}OC(O)-$, $-(CH_2)_{0-6}NH-C(O)-(CH_2)_{0-6}-$, $-(CH_2)_{0-6}O-$, $(CH_2)_{0-6}-$, $-(CH_2)_{0-6}S-(CH_2)_{0-6}-$, or $-NH-$, each of which can be optionally substituted.

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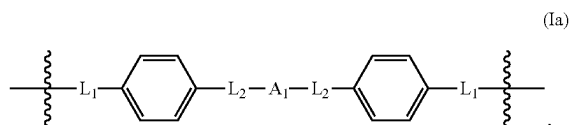
In some embodiments, R_1 and R_2 are independently C_1 - C_{10} hydroxyalkyl, which can be optionally substituted, and the hydroxyalkyl can be optionally coupled to a graphene oxide sheet.

In some embodiments, R_1 and R_2 are independently $-NH-$, $-NH-C(O)-$, $-NH-C(O)-(CH_2)_2-O-$, $-CH_2-NH$ -phenyl- $NH-C(O)-$, $-CH_2-S-(CH_2)_2-NH-C(O)-$, or $-CH_2-O-C(O)-$.

In some embodiments, R_1 and R_2 are independently $-C_1$ - C_6 alkyl- $O-C_1$ - C_6 alkyl, which can be optionally substituted, and the alkyl can be optionally coupled to a graphene oxide sheet.

In some embodiments, R_1 and R_2 are independently $-NH-C(O)-C_1$ - C_{10} alkyl, which can be optionally substituted, and the alkyl can be optionally coupled to a graphene oxide sheet. For example, R_1 and R_2 can be independently $-NH-C(O)-(CH_2)_q-O-$ ($q=1$ to 10).

In some embodiments, the chemical linker **120** can have a structure in accordance with Formula Ia:



wherein:

L_1 is selected from $-NH-$, $-C(=O)-NH-$, or absent;

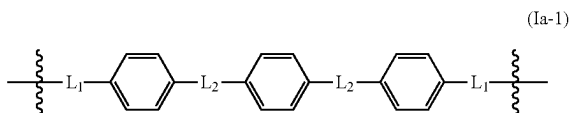
L_2 is selected from $-C(=O)-NH-(CH_2)_n-$, $-(CH_2)_2-O-(CH_2)_n-$, or $-NH-(CH_2)_n-$;

A_1 is selected from aryl, heteroaryl, C_4 - C_{10} heterocycloalkyl, C_4 - C_{10} cycloalkyl, or C_4 - C_{10} alkyl, wherein the aryl, heteroaryl, heterocycloalkyl, cycloalkyl, and alkyl can each be optionally substituted by one or more substituents selected from halo, C_1 - C_4 alkoxy, or C_1 - C_4 alkyl;

n is 0-4; and

$\sim$ denotes the point of coupling with a carbon atom on a graphene oxide sheet.

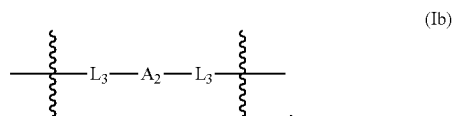
In some embodiments, A_1 is phenyl. For example, the chemical linker **120** can have a structure in accordance with Formula Ia-1:



In some embodiments, A_1 is linear C_5 alkyl. In some embodiments, A_1 is linear C_6 alkyl.

In some embodiments, n is 0. In some embodiments, n is 1. In some embodiments, n is 2. In some embodiments, n is 3. In some embodiments, n is 4.

In some embodiments, the chemical linker **120** can have a structure in accordance with Formula Ib:



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wherein:

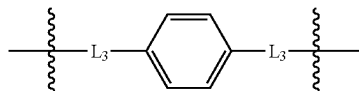
L_3 is selected from $-\text{C}(=\text{O})-\text{NH}-(\text{CH}_2)_m-$,
 $-\text{C}(=\text{O})-\text{NH}-\text{C}(=\text{O})-(\text{CH}_3)_2-\text{S}-$
 $(\text{CH}_2)_m-$, or $-\text{NH}-\text{C}(=\text{O})-(\text{CH}_3)_2-\text{S}-$
 $(\text{CH}_2)_m-$;

A_2 is selected from aryl, heteroaryl, C_4 - C_{10} heterocycloalkyl, C_4 - C_{10} cycloalkyl, or C_4 - C_{10} alkyl, wherein the aryl, heteroaryl, heterocycloalkyl, cycloalkyl, and alkyl can each be optionally substituted by one or more substituents selected from halo, C_1 - C_4 alkoxy, or C_1 - C_4 alkyl;

m is 0-4; and

 denotes the point of coupling with a carbon atom on a graphene oxide sheet.

In some embodiments, A_2 is phenyl. For example, the chemical linker **120** can have a structure in accordance with Formula Ib-1:

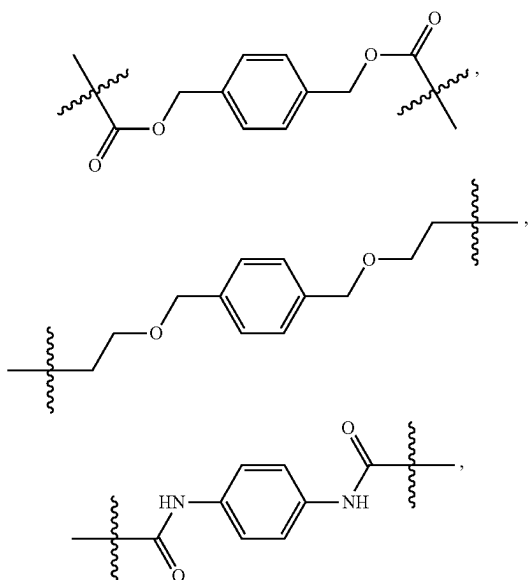


(Ib-1)

In some embodiments, A_2 is linear C_5 alkyl. In some embodiments, A_2 is linear C_6 alkyl.

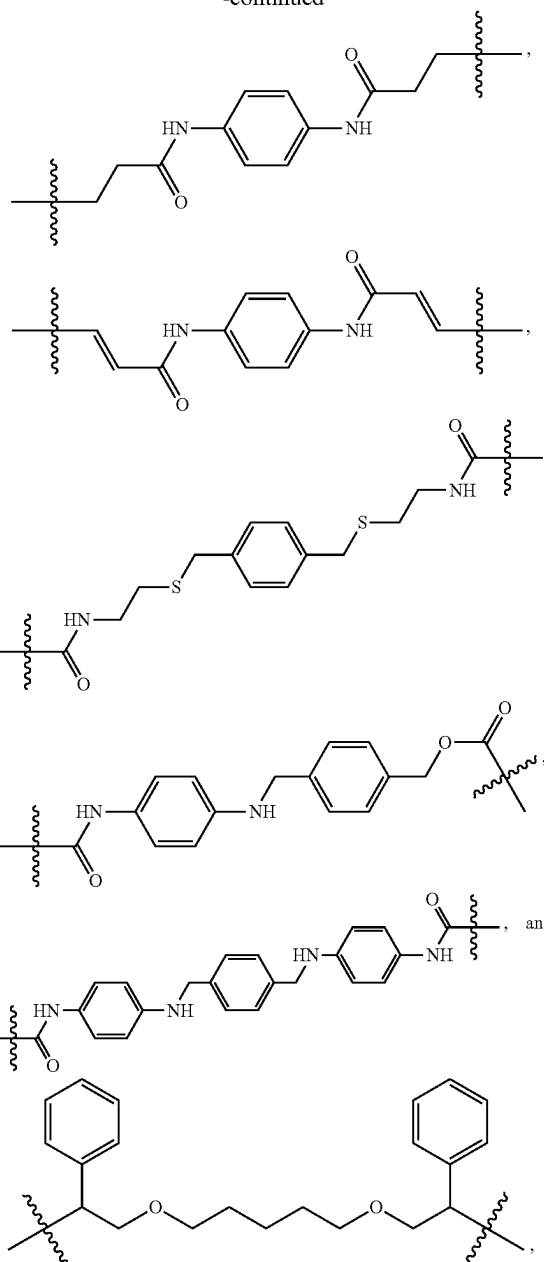
In some embodiments, m is 0. In some embodiments, m is 1. In some embodiments, m is 2. In some embodiments, m is 3. In some embodiments, m is 4.

In some embodiments, the chemical linker **120** can have one of the following structures:



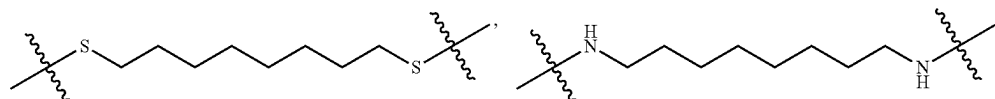
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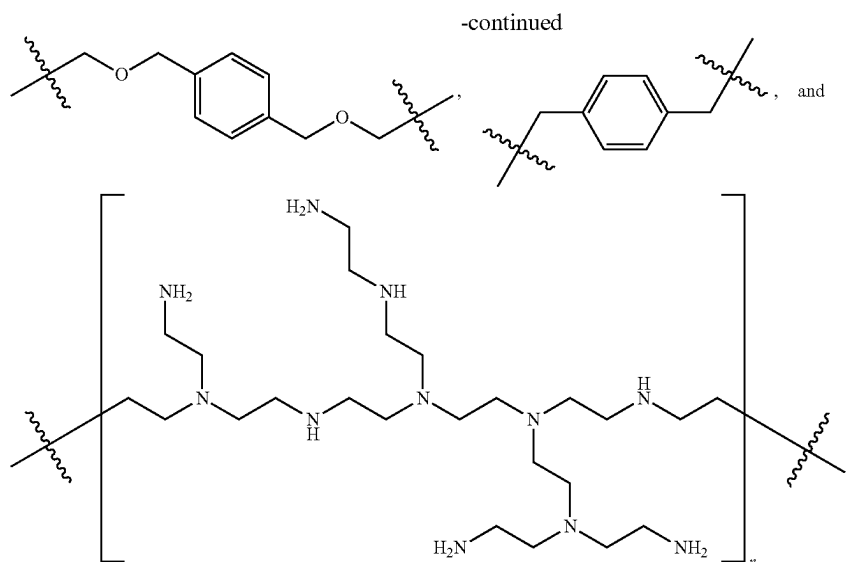
where  denotes the point of coupling with a graphene oxide sheet. Each of these chemical linkers can be optionally substituted.

In some embodiments, the chemical linker **120** can have one of the following structures:



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wherein: n is 1 to 5; $\sim$ denotes the point of coupling to $-L_4-NH-$ or $-L_5-C(=O)-NH-$, the nitrogen being coupled to a carbon atom on a graphene oxide sheet; L_4 is aryl; and L_5 is C_1-C_6 alkylene or C_2-C_6 alkenylene.

Swelling of membranes can be problematic because it can adversely affect the structural integrity of the membrane, change the molecular weight cutoff, etc. Without being bound by any particular theory, it is believed that the interaction (e.g., van der Waals interactions) between the graphene oxide sheets are relatively weak and certain solvents and/or solvents at certain temperatures enter into the region between the sheets and disrupt some of these interactions resulting in swelling and/or destabilization. The chemical linkers **120** may serve to stabilize the graphene oxide membrane **100** from destabilization in solvents and/or at elevated temperatures. In some embodiments, the chemical linker **120** may have a length and/or density that substantially reduces swelling of the graphene oxide membrane **100** in certain environments (e.g., solvents, elevated temperatures, etc.) and/or prevents destabilization of the graphene oxide membrane **100**.

In some embodiments, the weight ratio of graphene oxide to chemical linker in the finished membrane can be less than about 1,000, less than about 500, less than about 400, less than about 300, less than about 200, less than about 100, less than about 50, less than about 25, less than about 15, less than about 10, or less than about 5, inclusive of all values and ranges therebetween. In some embodiments, the weight ratio of graphene oxide to chemical linker in the finished membrane can be at least about 5, at least about 10, at least about 15, at least about 20, at least about 25, at least about 30, at least about 35, at least about 40, at least about 45, or at least about 50, inclusive of all values and ranges therebetween.

Combinations of the above-referenced ranges for the weight ratio are also possible (e.g., at least about 5 and less than about 1000, or at least about 10 and less than about 200).

In some embodiments, the at % content of carbon present on the surface of the graphene oxide membrane measured by X-ray photoelectron spectroscopy can be less than about 75%, less than about 72%, or less than about 70%, inclusive

of all values and ranges therebetween. In some embodiments, the at % content of carbon present on the surface of the graphene oxide membrane measured by X-ray photoelectron spectroscopy can be at least about 60%, at least about 62%, or at least about 65%, inclusive of all values and ranges therebetween.

Combinations of the above referenced ranges for the at % content of carbon are also possible (e.g., at least about 60% and less than about 75%, or at least about 65% and less than about 72%). In contrast, existing graphene oxide membranes that are deliberately or unintentionally reduced often have at % content of carbon of greater than 80% or even greater than 95%.

In some embodiments, the graphene oxide membrane can include at least about 100 layers, at least about 125 layers, at least about 150 layers, at least about 200 layers, at least about 225 layers, or at least about 250 layers of graphene oxide sheets, inclusive of all values and ranges therebetween. In some embodiments, the graphene oxide membrane **100** can include no more than about 600 layers, no more than about 550 layers, no more than about 500 layers, no more than about 450 layers, no more than about 400 layers, no more than about 350 layers, or no more than about 300 layers of graphene oxide sheets.

Combinations of the above-referenced ranges for the number of layers are also possible (e.g., at least about 100 and no more than about 600, or at least about 300 and no more than about 600).

In some embodiments, the graphene oxide membrane can include about 100 to about 600 layers of graphene oxide sheets, e.g., 200-500 layers, 200-400 layers, 200-300 layers, 200-250 layers, 300-600 layers, 300-500 layers, or 300-400 layers.

In some embodiments, the graphene oxide membrane can have a thickness greater than or equal to about 25 nm, greater than or equal to about 50 nm, greater than or equal to about 0.1 microns, greater than or equal to about 0.15 microns, greater than or equal to about 0.2 microns, greater than or equal to about 0.3 microns, greater than or equal to about 0.4 microns, greater than or equal to about 0.5 microns, greater than or equal to about 0.75 microns, greater than or equal to about 1 micron, or greater than or equal to about 2 microns. In some embodiments, the thickness of the

graphene oxide membrane may be less than or equal to about 5 microns, less than or equal to about 1 micron, less than or equal to about 0.5 microns, or less than or equal to about 0.5 microns.

Combinations of the above-referenced ranges for the thickness of the graphene oxide membrane are also possible (e.g., greater than or equal to about 25 nm and less than or equal to about 5 microns, greater than or equal to about 0.15 microns and less than or equal to about 0.5 microns).

In some embodiments, the graphene oxide membrane can have a thickness of about 25 nm, about 50 nm, about 0.1 microns, about 0.15 microns, about 0.2 microns, about 0.25 microns, about 0.3 microns, about 0.35 microns, about 0.4 microns, about 0.45 microns, about 0.5 microns, about 0.55 microns, about 0.6 microns, about 0.65 microns, about 0.7 microns, about 0.75 microns, about 0.8 microns, about 0.85 microns, about 0.9 microns, about 0.95 microns, about 1.0 micron, about 1.5 microns, or about 2 microns.

In some embodiments, the aspect ratio (on the plane of the graphene oxide sheets) of the graphene oxide membrane can be less than about 5,000,000, less than about 1,000,000, less than about 500,000, less than about 250,000, less than about 100,000, less than about 50,000, less than about 25,000, less than about 10,000, less than about 5,000, or less than about 1,000, inclusive of all values and ranges therebetween.

In some embodiments, the size of the space between graphene oxide sheets is the d-spacing, which can be measured by X-ray diffraction such as grazing incidence X-ray diffraction (GIXRD). In some embodiments, the d-spacing for dried graphene oxide sheets can be less than about 20 Å, less than about 15 Å, or less than about 10 Å, inclusive of all values and ranges therebetween. In some embodiments, the d-spacing for dried graphene oxide sheets can be in the range of about 5 Å to about 20 Å, about 5 Å to about 15 Å, about 8 Å to about 20 Å, about 8 Å to about 15 Å, inclusive of all values and ranges therebetween. In some embodiments, the d-spacing for dried graphene oxide sheets can be about 17 Å, about 16 Å, about 15 Å, about 14 Å, about 13 Å, about 12 Å, about 11 Å, about 10 Å, about 9 Å, about 8 Å, or about 7 Å. The length of the chemical linker is an important factor in controlling the d-spacing.

In some embodiments, the graphene oxide membrane can have a flux of at least about 2.5×10^{-4} GFD/psi, at least about 5.0×10^{-4} GFD/psi, at least about 7.5×10^{-4} GFD/psi, at least about 1.0×10^{-3} GFD/psi, at least about 1.25×10^{-3} GFD/psi, at least about 1.5×10^{-3} GFD/psi, at least about 1.75×10^{-3} GFD/psi, at least about 2.0×10^{-3} GFD/psi, at least about 2.25×10^{-3} GFD/psi, at least about 2.5×10^{-3} GFD/psi, at least about 5.0×10^{-3} GFD/psi, at least about 10.0×10^{-3} GFD/psi, at least about 15.0×10^{-3} GFD/psi, or at least about 20.0×10^{-3} GFD/psi, inclusive of all values and ranges therebetween, measured with a 1 wt % lactose solution at room temperature.

In some embodiments, the graphene oxide membrane can have a flux of no more than about 40.0×10^{-3} GFD/psi, no more than about 35.0×10^{-3} GFD/psi, no more than about 30.0×10^{-3} GFD/psi, no more than about 20.0×10^{-3} GFD/psi, no more than about 15.0×10^{-3} GFD/psi, no more than about 10.0×10^{-3} GFD/psi, inclusive of all values and ranges therebetween, measured with a 1 wt % lactose solution at room temperature.

Combinations of the above-referenced ranges for the flux are also possible (e.g., at least about 2.5×10^{-4} GFD/psi and no more than about 40.0×10^{-3} GFD/psi, or at least about 5.0×10^{-3} GFD/psi and no more than about 30.0×10^{-3} GFD/psi).

In some embodiments, the flux is measured at 50 psi to 1000 psi, such as about 50 psi, about 75 psi, about 100 psi, about 125 psi, about 150 psi, about 175 psi, about 200 psi, about 225 psi, about 250 psi, about 275 psi, about 300 psi, about 325 psi, about 350 psi, about 375 psi, about 400 psi, about 425 psi, about 450 psi, about 475 psi, about 500 psi, about 525 psi, about 550 psi, about 575 psi, about 600 psi, about 625 psi, about 650 psi, about 675 psi, about 700 psi, about 725 psi, about 750 psi, about 775 psi, about 800 psi, about 825 psi, about 850 psi, about 875 psi, about 900 psi, about 925 psi, about 950 psi, about 975 psi, or about 1000 psi.

In some embodiments, the graphene oxide membrane can have a flux of 2.5×10^{-4} to 3.75×10^{-2} GFP/psi, 2.5×10^{-4} to 2.5×10^{-2} GFP/psi, 2.5×10^{-3} to 2.5×10^{-2} GFP/psi, or 1.25×10^{-2} to 2.5×10^{-2} GFP/psi, as measured with a 1 wt % lactose solution at room temperature.

In some embodiments, the graphene oxide membrane can have a lactose rejection rate of at least 50%, at least 60%, at least 65%, at least 70%, at least 75%, at least 80%, at least 85%, at least 90%, at least 95%, at least 98%, or at least 99%, with a 1 wt % lactose solution. The lactose rejection rate can be measured at room temperature.

In some embodiments, the graphene oxide membrane can have a lactose rejection rate of 50% to 99.5% with a 1 wt % lactose solution. In some embodiments, the graphene oxide membrane can have a lactose rejection rate of 60% to 99.5% with a 1 wt % lactose solution. In some embodiments, the graphene oxide membrane can have a lactose rejection rate of 70% to 99.5% with a 1 wt % lactose solution. In some embodiments, the graphene oxide membrane can have a lactose rejection rate of 80% to 99.5% with a 1 wt % lactose solution. In some embodiments, the graphene oxide membrane can have a lactose rejection rate of 90% to 99.5% with a 1 wt % lactose solution. In some embodiments, the graphene oxide membrane can have a lactose rejection rate of 95% to 99.5% with a 1 wt % lactose solution.

In some embodiments, the graphene oxide membrane can have a MgSO_4 rejection rate of at least 30%, at least 40%, at least 50%, at least 60%, at least 65%, at least 70%, at least 75%, at least 80%, at least 85%, at least 90%, at least 95%, at least 98%, or at least 99%, with a 0.1 wt % MgSO_4 solution. The MgSO_4 rejection rate can be measured at room temperature.

In some embodiments, the graphene oxide membrane can have a MgSO_4 rejection rate of 30% to 99.5% with a 0.1 wt % MgSO_4 solution. In some embodiments, the graphene oxide membrane can have a MgSO_4 rejection rate of 40% to 99.5% with a 0.1 wt % MgSO_4 solution. In some embodiments, the graphene oxide membrane can have a MgSO_4 rejection rate of 50% to 99.5% with a 0.1 wt % MgSO_4 solution. In some embodiments, the graphene oxide membrane can have a MgSO_4 rejection rate of 60% to 99.5% with a 0.1 wt % MgSO_4 solution. In some embodiments, the graphene oxide membrane can have a MgSO_4 rejection rate of 70% to 99.5% with a 0.1 wt % MgSO_4 solution. In some embodiments, the graphene oxide membrane can have a MgSO_4 rejection rate of 80% to 99.5% with a 0.1 wt % MgSO_4 solution. In some embodiments, the graphene oxide membrane can have a MgSO_4 rejection rate of 90% to 99.5% with a 0.1 wt % MgSO_4 solution. In some embodiments, the graphene oxide membrane can have a MgSO_4 rejection rate of 95% to 99.5% with a 0.1 wt % MgSO_4 solution.

The procedure for characterizing rejection and permeability of a graphene oxide membrane is shown below: (1) cut a 47 to 50 mm disc from the graphene oxide membrane using a razor blade or laser cutter; (2) load the disc with the

graphene oxide side up onto a porous stainless steel frit, which is then mounted into a Sterlitech HP4750 filtration cell; (3) add 60 to 100 mL of 1 wt % lactose solution; (4) place the setup on a stir plate at approximately 750 rpm; (4) close the feed chamber and pressurize it to 50 to 100 psi. Under this procedure, at least 15 mL of permeate is collected across three samples to ensure that the performance measurement was steady.

In some embodiments, the molecular weight cutoff for the graphene oxide membrane is about 100 Da. In some embodiments, the molecular weight cutoff for the graphene oxide membrane is about 150 Da. In some embodiments, the molecular weight cutoff for the graphene oxide membrane is about 200 Da. In some embodiments, the molecular weight cutoff for the graphene oxide membrane is about 250 Da. In some embodiments, the molecular weight cutoff for the graphene oxide membrane is about 300 Da. In some embodiments, the molecular weight cutoff for the graphene oxide membrane is about 350 Da.

In some embodiments, the color of the graphene oxide membrane can be used to assess the membrane's stability under elevated temperatures and/or basic pH levels. The color of the graphene oxide membrane can be characterized by recording images of the graphene oxide membrane, and calculating the grayscale mode value with the aid of using image processing software. The range of grayscale mode value that an image can assume is zero to 255, with values closer to zero corresponding to darker images, and values closer to 255 corresponding to lighter images.

In some embodiments, the graphene oxide membrane can display a grayscale mode value of less than about 180, less than about 160, less than about 140, less than about 120, less than about 100, less than about 80, less than about 60, or less than about 40, inclusive of all values and ranges therebetween, where the image of the graphene oxide membrane is collected in a lightbox with dimensions 9.4×9.1×8.7" and two rows of 20 white LEDs on the top front and rear edge of the lightbox. In some embodiments, the graphene oxide membrane can display a grayscale mode value of at least about 20, at least about 25, at least about 30, at least about 45, at least about 60, at least about 75, inclusive of all values and ranges therebetween, where the image of the graphene oxide membrane is collected in a lightbox with dimensions 9.4×9.1×8.7" and two rows of 20 white LEDs on the top front and rear edge of the lightbox.

Combinations of the above-referenced ranges for the grayscale mode value are also possible (e.g., at least about 20 and less than about 180, or at least 90 and less than about 120).

Other ways to quantify the color of the graphene oxide membrane can also be used. For example, distribution shape, center of a fit, and/or standard deviation can be used.

The lighting level to obtain an image of the graphene oxide membrane can have an effect on the grayscale mode value. For comparison purposes, the same or substantially the same lighting level should be used to obtain two or more images of the same membrane at different time points or different membranes.

Another aspect of the present disclosure relates to graphene oxide membranes having temperature, pH, and/or pressure durability.

The temperature and/or pH durability of the graphene oxide membrane can be assessed by contacting the graphene oxide membrane with a test solution that has a high temperature and/or non-neutral pH, and comparing the solute rejection rates before and after the contact.

In some embodiments, the graphene oxide membrane can have a first solute rejection rate with a first solution having the solute at room temperature before exposure to a test solution with a high temperature and/or non-neutral pH. In some embodiments, the solute can include lactose, lignin, sodium sulfate, sodium carbonate, sodium hydrosulfide, sodium thiosulfate, sodium hydroxide, hydrochloric acid, nitric acid, phosphoric acid, sodium chloride, potassium chloride, and/or magnesium sulfate. The solute is dissolved in a solvent. In some embodiments, the solvent can include water, ethanol, methanol, acetone, or any other suitable solvent. In some embodiments, the graphene oxide membrane can have a second solute rejection rate with a second solution having the solute at room temperature after exposure to the test solution for a contact period.

In some embodiments, the first solution and/or the second solution can include about 0.1 wt % solute, about 0.2 wt % solute, about 0.3 wt % solute, about 0.4 wt % solute, about 0.5 wt % solute, about 0.6 wt % solute, about 0.7 wt % solute, about 0.8 wt % solute, about 0.9 wt % solute, about 1 wt % solute, about 1.5 wt % solute, about 2 wt % solute, about 2.5 wt % solute, about 3 wt % solute, about 3.5 wt % solute, about 4 wt % solute, about 4.5 wt % solute, about 5 wt % solute, about 5.5 wt % solute, about 6 wt % solute, about 6.5 wt % solute, about 7 wt % solute, about 7.5 wt % solute, about 8 wt % solute, about 8.5 wt % solute, about 9 wt % solute, about 9.5 wt % solute, or about 10 wt % solute, inclusive of all values and ranges therebetween.

In some embodiments, the test solution can have a high temperature, a non-neutral pH, or any combination thereof. In some embodiments, the high temperature can be greater than about 50° C., greater than about 55° C., greater than about 60° C., greater than about 65° C., greater than about 70° C., greater than about 75° C., greater than about 80° C., greater than about 85° C., greater than about 90° C., or greater than about 95° C., inclusive of all values and ranges therebetween. In some embodiments, the high temperature can be no more than about 100° C., no more than about 95° C., no more than about 90° C., no more than about 85° C., or no more than about 80° C., inclusive of all values and ranges therebetween.

Combinations of the above-referenced ranges for the temperature are also possible (e.g., greater than about 50° C. and no more than about 100° C., greater than about 60° C. and no more than about 100° C., or greater than about 80° C. and no more than about 100° C.).

In some embodiments, the non-neutral pH can be at least about 8, at least about 8.5, at least about 9, at least about 9.5, at least about 10, at least about 10.5, at least about 11, at least about 11.5, at least about 12, at least about 12.5, at least about 13, at least about 13.5, or at least about 14. In some embodiments, the non-neutral pH can be less than about 6, less than about 5.5, less than about 5, less than about 4.5, less than about 4, less than about 3.5, less than about 3, less than about 2.5, less than about 2, less than about 1.5, less than about 1, or less than about 0.5, inclusive of all values and ranges therebetween. In some embodiments, the non-neutral pH can be about 0, about 0.5, about 1, about 1.5, about 2, about 2.5, about 3, about 3.5, about 4, about 4.5, about 5, about 5.5, about 6, about 8, about 8.5, about 9, about 9.5, about 10, about 10.5, about 11, about 11.5, about 12, about 12.5, about 13, about 13.5, or about 14, inclusive of all values and ranges therebetween.

In some embodiments, the first solute rejection rate can be at least about 50%, at least about 55%, at least about 60%, at least about 65%, at least about 70%, at least about 75%, at least about 80%, at least about 85%, at least about 90%,

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at least about 91%, at least about 92%, at least about 93%, at least about 94%, at least about 95%, at least about 96%, at least about 97%, at least about 98%, at least about 99%, at least about 99.5%, or about 100%, inclusive of all values and ranges therebetween.

In some embodiments, the second solute rejection rate can be at least about 50%, at least about 55%, at least about 60%, at least about 65%, at least about 70%, at least about 75%, at least about 80%, at least about 85%, at least about 90%, at least about 91%, at least about 92%, at least about 93%, at least about 94%, at least about 95%, at least about 96%, at least about 97%, at least about 98%, at least about 99%, at least about 99.5%, or about 100%, inclusive of all values and ranges therebetween.

In some embodiments, the first lactose rejection rate with a 1 wt % lactose solution can be at least about 50%, at least about 55%, at least about 60%, at least about 65%, at least about 70%, at least about 75%, at least about 80%, at least about 85%, at least about 90%, at least about 91%, at least about 92%, at least about 93%, at least about 94%, at least about 95%, at least about 96%, at least about 97%, at least about 98%, at least about 99%, at least about 99.5%, or about 100%, inclusive of all values and ranges therebetween.

In some embodiments, the second lactose rejection rate with a 1 wt % lactose solution can be at least about 50%, at least about 55%, at least about 60%, at least about 65%, at least about 70%, at least about 75%, at least about 80%, at least about 85%, at least about 90%, at least about 91%, at least about 92%, at least about 93%, at least about 94%, at least about 95%, at least about 96%, at least about 97%, at least about 98%, at least about 99%, at least about 99.5%, or about 100%, inclusive of all values and ranges therebetween.

In some embodiments, when the membrane is temperature and/or pH durable, the second solute rejection rate can be substantially the same as the first solute rejection rate. In some embodiments, when the membrane is temperature and/or pH durable, the second solute rejection rate can be greater than the first solute rejection rate. For example, the second solute rejection rate can be at least about 5%, at least about 10%, at least about 15%, or at least about 20% greater than the first solute rejection rate.

In some embodiments, the test solution can be a buffer. In some embodiments, the test solution can include a phosphate buffer. In some embodiments, the test solution can include Na_2HPO_4 . In some embodiments, the test solution can have a concentration of about 0.1 M, about 0.2 M, about 0.3 M, about 0.4 M, about 0.5 M, about 0.6 M, about 0.7 M, about 0.8 M, about 0.9 M, about 1 M, about 2 M, about 3 M, about 4 M, about 5 M, or about 6 M, inclusive of all values and ranges therebetween.

In some embodiments, the contact period can be greater than about 10 minutes, greater than about 20 minutes, greater than about 30 minutes, greater than about 40 minutes, greater than about 50 minutes, greater than about 1 hour, greater than about 1.5 hours, greater than about 2 hours, greater than about 2.5 hours, greater than about 3 hours, greater than about 3.5 hours, greater than about 4 hours, greater than about 4.5 hours, greater than about 5 hours, greater than about 5.5 hours, greater than about 6 hours, greater than about 6.5 hours, greater than about 7 hours, greater than about 7.5 hours, greater than about 8 hours, greater than about 8.5 hours, greater than about 9 hours, greater than about 9.5 hours, or greater than about 10 hours, inclusive of all values and ranges therebetween.

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In some embodiments, the contact period can be less than about 10 hours, less than about 9.5 hours, less than about 9 hours, less than about 8.5 hours, less than about 8 hours, less than about 7.5 hours, less than about 7 hours, less than about 6.5 hours, less than about 6 hours, less than about 5.5 hours, less than about 5 hours, less than about 4.5 hours, less than about 4 hours, less than about 3.5 hours, less than about 3 hours, less than about 2.5 hours, less than about 2 hours, less than about 1.5 hours, less than about 1 hour, less than about 50 minutes, less than about 40 minutes, less than about 30 minutes, or less than about 20 minutes, inclusive of all values and ranges therebetween.

In some embodiments, the contact period can be about 1 hour to about 10 hours, about 1 hour to about 9 hours, about 1 hour to about 8 hours, about 1 hour to about 7 hours, about 1 hour to about 6 hours, about 1 hour to about 5 hours, about 1 hour to about 4 hours, about 1 hour to about 3 hours, about 1 hour to about 2 hours, about 2 hours to about 10 hours, about 2 hours to about 9 hours, about 2 hours to about 8 hours, about 2 hours to about 7 hours, about 2 hours to about 6 hours, about 2 hours to about 5 hours, about 2 hours to about 4 hours, about 2 hours to about 3 hours, about 3 hours to about 10 hours, about 3 hours to about 9 hours, about 3 hours to about 8 hours, about 3 hours to about 7 hours, about 3 hours to about 6 hours, about 3 hours to about 5 hours, or about 3 hours to about 4 hours.

The temperature and/or pH durability of the graphene oxide membrane can also be assessed by contacting the graphene oxide membrane with a test solution that has a high temperature and/or non-neutral pH for a period of time, and comparing the grayscale mode values before and after the contact. In some embodiments, when the membrane is temperature and/or pH durable, the grayscale mode value changes by no more than about 30%, no more than about 25%, no more than about 20%, no more than about 15%, no more than about 10% after the contact. In some embodiments, when the membrane is temperature and/or pH durable, the grayscale mode value changes by no more than about 50, no more than about 40, no more than about 30, or no more than about 20 after the contact. The contact period can be about a few hours (e.g., about 2 hours, about 4 hours, about 6 hours, about 8 hours, about 10 hours, about 12 hours, about 14 hours, about 16 hours, about 18 hours, about 20 hours, or about 22 hours), about one day, or about 2 days.

In one aspect, the flux of the graphene oxide membrane can be increased by contacting the membrane with a pretreatment solution that is at least about 50° C. for a period of time. In some embodiments, the pretreatment solution flows through the membrane during the contacting step.

In some embodiments, the membrane is contacted with the pretreatment solution at no more than about 100° C., no more than about 90° C., no more than about 80° C., no more than about 70° C., or no more than about 60° C., inclusive of all values and ranges therebetween. In some embodiments, the membrane is contacted with the pretreatment solution that is at least about 50° C., at least about 55° C., at least about 60° C., at least about 65° C., or at least about 70° C., inclusive of all values and ranges therebetween.

Combinations of the above-referenced ranges for the temperature of the test solution are also possible (e.g., a temperature of at least about 50° C. and no more than about 100° C., or at least about 60° C. and no more than about 90° C.).

In another aspect, the flux of the graphene oxide membrane can be increased by contacting the membrane with a basic pretreatment solution at room temperature or at a temperature greater than room temperature. In some

embodiments, the pretreatment solution flows through the membrane during the contacting step.

In some embodiments, the basic pretreatment solution has a pH of about 14, less than about 13.5, less than about 13, less than about 12.5, less than about 12, less than about 11.5, less than about 11, less than about 10.5, less than about 10, less than about 9.5, less than about 9, inclusive of all values and ranges therebetween. In some embodiments, the basic pretreatment solution has a pH of at least about 8.0, at least about 8.5, at least about 9.0, at least about 9.5, at least about 10, or at least about 10.5, inclusive of all values and ranges therebetween.

Combinations of the above referenced ranges for the pH of the basic pretreatment solution are also possible (e.g., at least about 8.0 and less than about 13, or at least about 9.0 and less than about 12).

In some embodiments, the period of time that the membrane is in contact with the pretreatment solution to increase the flux can be less than about 400 hours, less than about 300 hours, less than about 200 hours, less than about 100 hours, less than about 40 hours, less than about 20 hours, less than about 10 hours, less than about 8 hours, less than about 6 hours, less than about 4 hours, or less than about 2 hours, inclusive of all values and ranges therebetween. In some embodiments, the period of time that the membrane is in contact with a pretreatment solution to increase the flux can be at least about 0.5 hours, at least about 1 hour, at least about 1.5 hours, at least about 2 hours, at least about 3 hours, at least about 4 hours, or at least about 8 hours, inclusive of all values and ranges therebetween. Generally, when elevated temperature is used, a shorter period of time is needed to increase the flux than that when room temperature or lower temperature is used.

Combinations of the above referenced ranges for the period of time are also possible (e.g., at least about 0.5 hours and less than about 400 hours, at least about 2 hours and less than about 100 hours, or about 2 hours and less than about 10 hours).

In some embodiments, the flux increases after the contacting step can be less than about 1,000%, less than about 900%, less than about 800%, less than about 700%, less than about 600%, less than about 500%, less than about 400%, less than about 300%, less than about 200%, or less than about 100%, inclusive of all values and ranges therebetween. In some embodiments, the flux increase after the contacting step can be at least about 40%, at least about 50%, at least about 60%, at least about 90%, at least about 150%, or at least about 200%, inclusive of all values and ranges therebetween.

Combinations of the above-referenced ranges for the increase in flux are also possible (e.g., at least about 40% and less than about 1,000%, or at least about 60% and less than about 600%).

Filtration Apparatus

In another aspect, the present disclosure provides a filtration apparatus that includes a support substrate, a graphene oxide membrane of the present disclosure, and optionally a housing. The graphene oxide membrane can be disposed on the support substrate. The housing can enclose the support substrate and graphene oxide membrane.

The support substrate can include a non-woven fiber or polymer. In some embodiments, the support substrate can include a material selected from polypropylene (PP), polystyrene, polyethylene, polyethylene oxide, polyethersulfone (PES), polytetrafluoroethylene (PTFE), polyvinylidene fluoride, polymethylmethacrylate, polydimethylsiloxane, polyester, polyolefin, cellulose, cellulose acetate, cellulose

nitrate, polyacrylonitrile, glass fiber, quartz, alumina, silver, polycarbonate, nylon, Kevlar or other aramid, or polyether ether ketone.

In some embodiments, the support substrate is a microporous substrate. The support substrate can have an average pore size of 0.1 μm to 10 μm , e.g., 0.1 μm to 8 μm , 0.1 μm to 5 μm , 0.2 μm to 5 μm , 0.2 μm to 2 μm , or 0.2 μm to 1 μm . In some embodiments, the support substrate can have an average pore size less than 1 μm , such as about 0.2 μm , about 0.3 μm , about 0.4 μm , about 0.45 μm , about 0.5 μm , about 0.55 μm , about 0.6 μm , about 0.65 μm , about 0.7 μm , or about 0.75 μm .

In some embodiments, the graphene oxide membrane **100** and the support substrate can have a combined thickness of about 100 microns, about 150 microns, about 200 microns, about 250 microns, about 300 microns, about 350 microns, about 400 microns, about 450 microns, about 500 microns, inclusive of all values and ranges therebetween.

In some embodiments, the support substrate can include two or more layers. For example, the support substrate can include a first layer and a second layer, the first layer is disposed on the second layer, wherein the first layer and the second layer have different average pore sizes. In some embodiments, the graphene oxide membrane is disposed on the first layer, and the first layer has a smaller average pore size than the second layer.

It was discovered that the roughness of the support substrate can have an impact on the flux of the graphene oxide membrane. Specifically, a smooth support substrate can improve the flux and/or rejection rate of the graphene oxide membrane as compared to a rough support substrate. Accordingly, in some embodiments, the support substrate can be smooth. For example, the support substrate has a root mean squared surface roughness of less than about 3 μm , less than about 2.5 μm , less than about 2 μm , less than about 1.5 μm , or less than about 1 μm . In some embodiments, the support substrate of the graphene oxide membrane **100** can have a root mean squared surface roughness of at least about 1 μm , at least about 1.2 μm , at least about 1.4 μm , at least about 1.5 μm , inclusive of all values and ranges therebetween. In some embodiments, the surface roughness is measured by a Dektak 6M Contact Profilometer.

Combinations of the above-referenced ranges for the root mean squared surface roughness are also possible (e.g., at least about 1 μm and less than 2.5 μm , or at least 1.4 μm and less than about 3 μm). In some embodiments, the support substrate has a root mean squared surface roughness of about 3 μm , about 2.5 μm , about 2 μm , about 1.5 μm , or about 1 μm .

In some embodiments, the filtration apparatus includes about 0.1 mg to 6 mg of the graphene oxide membrane per 5000 mm^2 . In some embodiments, the filtration apparatus includes about 0.1 mg to 5 mg, about 0.1 mg to 4 mg, about 0.1 mg to 3 mg, about 0.5 mg to 5 mg, about 0.5 mg to 4 mg, about 0.5 mg to 3 mg, about 1 mg to 4 mg, or about 1 mg to 3 mg of the graphene oxide membrane per 5000 mm^2 . For example, the filtration apparatus can include about 1 mg, about 1.5 mg, about 2 mg, about 2.5 mg, or about 3 mg of the graphene oxide membrane per 5000 mm^2 .

In some embodiments, the support substrate can comprise a hollow polymer tube. The hollow polymer tube can have a surface area greater than or equal to about 100 cm^2 .

In some embodiments, the graphene oxide membrane can comprise a plurality of flat polymer sheets combined to form a spiral filtration module. For example, in some embodiments, a spiral filtration module can comprise a plurality of flat polymer sheets stacked atop one another, and the plu-

ality of stacked flat polymer sheets may be rolled around a core tube. In some embodiments, prior to being rolled around the core tube, adjacent flat polymer sheets may be separated by a sheet of feed channel spacer to form a leaf, and each leaf may be separated by a sheet of permeate spacer. When the flat polymer sheets, the one or more feed channel spacers, and the one or more permeate spacers are rolled around the core tube, each permeate spacer may form a permeate channel.

To improve the membrane's durability under high-pressure operations, e.g., about 500 psi to 1600 psi or greater, in some embodiments, the support substrate can have a Young's modulus of no more than about 3.0 GPa, no more than about 2.5 GPa, no more than about 2.0 GPa, no more than about 1.8 GPa, no more than about 1.6 GPa, no more than about 1.4 GPa, or no more than about 1.2 GPa, inclusive of all values and ranges therebetween. In some embodiments, the support substrate can have a Young's modulus of at least about 0.5 GPa, at least about 0.6 GPa, at least about 0.7 GPa, at least about 0.8 GPa, at least about 0.9 GPa, at least about 1.0 GPa, at least about 1.1 GPa, at least about 1.3 GPa, at least about 1.5 GPa, or at least about 2.0 GPa, inclusive of all values and ranges therebetween.

Combinations of the above referenced ranges for the Young's modulus of the support substrate are also possible (e.g., a Young's modulus of at least about 0.5 GPa and no more than about 3.0 GPa, or at least about 1.5 GPa and no more than about 3.0 GPa). In some embodiments, the high pressure operation is about 900 psi, about 1000 psi, about 1100 psi, about 1200 psi, about 1300 psi, about 1400 psi, about 1500 psi, or about 1600 psi.

In some embodiments, the support substrate can have a thickness of no more than about 750 μm , no more than about 700 μm , no more than about 650 μm , no more than about 550 μm , no more than about 500 μm , no more than about 450 μm , or no more than about 400 μm , inclusive of all values and ranges therebetween. In some embodiments, the support substrate can have a thickness of at least about 200 μm , at least about 220 μm , or at least about 240 μm , inclusive of all values and ranges therebetween.

Combinations of the above referenced ranges for the thickness of the support substrate are also possible (e.g., a thickness of at least about 200 μm and no more than about 750 μm , at least about 240 μm and no more than about 500 μm).

In some embodiments, the support substrate suitable for high-pressure durability can comprise PES, PTFE, PP, PAN, polyolefin, nylon, or a combination thereof.

The support substrate can have one layer, two layers, three layers, or more. In some embodiments, the support substrate can comprise a first layer in contact with the membrane **100**, and a second layer disposed on the first layer, the second layer configured to provide further mechanical support. The first layer can comprise the same material as the second layer. For example, the first layer can comprise PES, and the second layer can comprise PES. The first layer can comprise a different material from the second layer.

In some embodiments, the support substrate can comprise a first layer in contact with the membrane **100**, a second layer disposed on the first layer, and a third layer disposed on the second layer. For example, the first layer can comprise PTFE; the second layer can comprise PP; and the third layer can comprise PES.

In some embodiments, the support substrate can comprise a first layer in contact with the membrane **100**, a second layer disposed on the first layer, a third layer disposed on the second layer, and a fourth layer disposed on the third layer.

For example, the first layer can comprise PTFE; the second layer can comprise PP; the third layer can comprise PTFE; and the fourth layer can comprise PP.

To test whether a graphene oxide membrane on a support substrate can endure high pressure operations, a graphene oxide membrane on a support substrate having a first lactose rejection rate of at least 50% with a first 1 wt % lactose solution at room temperature is subjected to a pressure of about 1000 psi for a period of time. Afterwards, if the graphene oxide membrane has a second lactose rejection rate of at least 50% with a second 1 wt % lactose solution at room temperature, then the membrane is deemed to be able to endure high pressure operations. In some embodiments, the period of time is about 1 hour to 24 hours, e.g., about 1 hour, about 2 hours, about 3 hours, about 4 hours, about 5 hours, about 6 hours, about 7 hours, about 8 hours, about 9 hours, about 10 hours, about 11 hours, about 12 hours, about 13 hours, about 14 hours, about 15 hours, about 16 hours, about 17 hours, about 18 hours, about 19 hours, about 20 hours, about 21 hours, about 22 hours, about 23 hours, or about 24 hours. In some embodiments, for testing purposes, the support substrate is mounted on a standard tricot permeate carrier (e.g. 0.012 inches thick, ~5 oz/sq yard, and 50 wales and 48 courses per inch).

Manufacture of the Graphene Oxide Membrane

The fabrication of the membrane includes dispersing the graphene oxide sheets in a solvent to produce a stable dispersion. In some embodiments, the solvent can be water. In some embodiments, the solvent can be organic. The dispersion may exhibit certain physical and chemical characteristics in order to produce continuous and uniform coatings substantially free of structural defects such as pinholes. For example, the hydrophilicity of the dispersion should be adequately matched to the support substrate to ensure wetting of the support substrate surface. This can be tested through contact angle measurements.

The stability of the dispersion can be inferred from the pH of the dispersion. For example, dispersions that exhibit acidic pH values (e.g., $\text{pH} \leq 5$) can develop visible aggregates. Fabricating coatings with such dispersions leads to poor coverage, coating non-uniformity, and poor membrane performance. In contrast, dispersions that have basic pH are stable. Moreover, addition of basic additives to the dispersion can increase the magnitude of the zeta potential on the graphene oxide sheets, which in turn leads to greater Coulombic stabilization.

The stability of the dispersion can be indirectly observed through UV-Vis spectroscopy measurements, owing to the absorption band at around 300 nm, attributed to n-to-p* transitions. At longer wavelengths (>500 nm) the graphene oxide sheets **110** absorb very weakly, and consequently, any signal in this region can be attributed to scattering, rather than absorption, due to the formation of aggregates. The ratio of UV-Vis signal at 300 nm (due to absorption) and that observed at 600 nm (due to aggregate scattering) can be used to characterize the dispersion in the solution. Generally, the higher this ratio is, the better the graphene oxide sheets **110** are dispersed.

In some embodiments, the ratio of UV-Vis signal at 300 nm and that observed at 600 nm can be less than about 4.4, less than about 4.2, less than about 4.0, less than about 3.8, less than about 3.6, less than about 3.4, less than about 3.2, or less than about 3.0, inclusive of all values and ranges therebetween. In some embodiments, the ratio of UV-Vis signal at 300 nm and that observed at 600 nm can be at least

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about 3.0, at least about 3.1, at least about 3.2, at least about 3.3, or at least about 3.4, inclusive of all values and ranges therebetween.

Combinations of the above referenced ranges for the ratio are also possible (e.g., a ratio of at least about 3.0 and less than about 4.4, at least about 3.2 and less than about 4.0).

In some embodiments, the dispersion can further include viscosity modifiers and/or surfactants. In some embodiments, the viscosity modifier is hydroxypropyl methylcellulose. For example, the dispersion can include about 0.01 wt % viscosity modifier. In some embodiments, the surfactant is sodium dodecyl sulfide. For example, the dispersion can include about 0.15 wt % surfactant.

In some embodiments, the viscosity of the dispersion can be no more than about 100 cP at a shear rate of around 50 Hz, no more than about 90 cP at a shear rate of around 50 Hz, no more than about 80 cP at a shear rate of around 50 Hz, or no more than about 70 cP at a shear rate of around 50 Hz. In some embodiments, the viscosity of the dispersion can be at least about 10 cP at a shear rate of around 50 Hz, at least about 20 cP at a shear rate of around 50 Hz, or at least about 30 cP at a shear rate of around 50 Hz.

Combinations of the above referenced ranges for the viscosity of the dispersion are also possible (e.g., a viscosity of at least about 10 cP and no more than about 100 cP at a shear rate of around 50 Hz, at least about 20 cP and no more than about 90 cP at a shear rate of around 50 Hz).

To produce dispersions that can coat well onto the support substrates, the order of addition of reagents can be important. For example, prior to deposition, dispersions that undergo carbodiimide coupling conditions require adjustment of the pH to be greater than 8.0 prior to the addition of 1-Ethyl-3-(3-dimethylaminopropyl)carbodiimide (EDC) and N-hydroxysuccinimide (NHS).

Prior to the reaction with a crosslinking moiety, the graphene oxide sheets can be functionalized with one or more desirable chemical groups. For example, the graphene oxide sheets can be functionalized with amines. See Navaee, A. & Salimi, A., "Efficient amine functionalization of graphene oxide through the Bucherer reaction: an extraordinary metal-free electrocatalyst for the oxygen reduction reaction," *RSC Adv.* 5, 59874-59880 (2015), the contents of which are incorporated by reference.

The graphene oxide sheets can also be functionalized with carboxylic groups. See Sydlík, S. A. & Swager, T. M., "Functional Graphenic Materials Via a Johnson-Claissen Rearrangement," *Adv. Funct. Mater.* 23, 1873-1882 (2012); Collins, W. R., et al., "Rearrangement of Graphite Oxide: A Route to Covalently Functionalized Graphenes," *Angew. Chem., Int. Ed.* 50, 8848-8852 (2011), the contents of each of which are incorporated by reference.

In some embodiments, the graphene oxide sheets can be functionalized with hydroxyl groups. For example, a graphene oxide sheet can react with an epoxide so that the graphene oxide sheet is functionalized with hydroxyl groups. Examples of epoxides include, but are not limited to, 1,2-epoxypropane, styrene oxide, ethylene oxide, epichlorohydrine, 1,2-epoxybutane, bisphenol A diglycidyl ether, 1,3-butadiene diepoxide, and 1,2,7,8-diepoxyoctane.

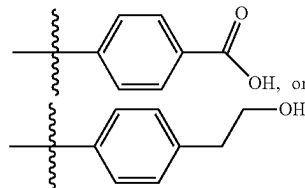
In some embodiments, the graphene oxide sheets can be functionalized with a functionalizing moiety having the structure in accordance with Formula II:



wherein: R_3 is an aryl or heteroaryl, which can be optionally substituted with one or more substituents. In some embodiments, the substituents can be selected from

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C_1 - C_6 alkyl and $\text{C}(=\text{O})\text{OR}_{3a}$, wherein the alkyl can be optionally substituted with halogen, OH, NH_2 , and SH; and R_{3a} can be H or C_1 - C_6 alkyl. In some embodiments, the aryl is phenyl. In some embodiments, R_3 is



where  denotes the point of coupling with $-\text{NH}$ in Formula II.

In some embodiments, the functionalizing moiety can include 4-aminophenylacetic acid, 2-(4-aminophenyl) ethanol, 2-(4-aminophenyl) propanol, 2-(4-aminophenyl) butanol, or any combination thereof.

In some embodiments, the graphene oxide sheets can be functionalized with a functionalizing moiety having the structure in accordance with Formula III:



wherein: R_4 is a C_1 - C_6 alkyl or a C_2 - C_6 alkenyl, each of which can be optionally substituted with one or more substituents. In some embodiments, non-limiting examples of R_4 can include methyl, ethyl, n-propyl, isopropyl, n-butyl, isobutyl, sec-butyl, tert-butyl, ethenyl, propenyl, and butenyl.

In some embodiments, the functionalizing moiety can include acrylamide, propionamide, isobutyramide, pivalamide, or any combination thereof.

Once the graphene oxide sheets have the desirable chemical groups suitable for crosslinking, they can be in contact with a crosslinking moiety to initiate crosslinking between the graphene oxide sheets. The reaction conditions for crosslinking can vary, depending on the crosslinking moieties used. As compared to existing processes, some embodiments of the processes of the present disclosure can be performed under ambient environments (i.e., in the presence of oxygen and humidity). In some embodiments, the reaction between the functionalizing moiety and the crosslinking moiety produces the chemical linker.

In some embodiments, the crosslinking moiety can be $\text{H}_2\text{N}-\text{C}_4-\text{C}_{10}$ alkyl- NH_2 , $\text{HS}-\text{C}_4-\text{C}_{10}$ alkyl-SH, alkyl-I, p-phenylenediamine, or 1,4-bis(bromomethyl)benzene. In some embodiments, the crosslinking moiety is $\text{H}_2\text{N}-\text{C}_6$ alkyl- NH_2 . In some embodiments, the crosslinking moiety is $\text{HS}-\text{C}_6$ alkyl-SH. In some embodiments, the crosslinking moiety is 1- C_5 alkyl-I. In some embodiments, the crosslinking moiety is p-phenylenediamine. In some embodiments, the crosslinking moiety is 1,4-bis(bromomethyl)benzene.

In some embodiments, triethylamine (TEA), 1,4-diazabicyclo[2.2.2]octane (DABCO), potassium hydroxide (KOH), sodium hydroxide (NaOH), cesium hydroxide (CsOH), lithium hydroxide (LiOH), or other bases can be used to deprotonate the relevant alcohol groups and accelerate the reaction in many cases. In some embodiments, the base is triethylamine. In some embodiments, the base is CsOH. In some embodiments, the pH of the reaction is in the range of about 9 to 13, e.g., about 9, about 9.5, about 10, about 10.5, about 11, about 11.5, about 12, about 12.5, or about 13.

In some embodiments, the reaction temperature is about 50° C. to 70° C., e.g., about 50° C., about 55° C., about 60° C., about 65° C., or about 70° C.

In some embodiments, molecules useful for initiating crosslinking between graphene oxide sheets can include, but are not restricted to, ester groups, sulfonated esters, ether groups, amines, carboxyl groups, carboxylic acids, carbonyl groups, amides, halides, thiols, alkanes, fluoroalkanes, alkyl groups, methyl groups, ethyl groups, propyl groups, butyl groups, pentyl groups, heptyl groups, octyl groups, nonyl groups, decyl groups, isopropyl, cyclopropyl, isobutyl, t-butyl, cyclobutyl, cyclohexyl, chloromethyl, bromoethyl, trifluoromethyl, methylamine, dimethylamine, ethylamine, diethylamine, methylethylamine, iso-propylamine, piperidine, trimethylamine, propylamine, hydroxy groups, hydroxyl groups, thio groups, 1,3,5-benzenetricarbonyl trichloride, aromatic dichlorides, aromatic trichlorides, terephthaloyl chloride, adipoyl chloride, propanediol, pentanediol, hexanediol, heptanediol, naphthyl, biphenyl, benzyl, hexyldiamine, 1,6-diiodohexane, 1,6-dibromohexane, 1,6-dichlorohexane, α,α' -dichloro-p-xylene, α,α' -diiodo-p-xylene, α,α' -dibromo-p-xylene, dichloromethylnaphthalene, trichloromethylbenzene, dichloromethylbiphenyl, dibromomethylnaphthalene, tribromomethylbenzene, dibromomethylbiphenyl, diiodomethylnaphthalene, triiodomethylbenzene, diiodomethylbiphenyl, any other suitable crosslinking moieties, or combinations thereof.

In some embodiments, the crosslinking moiety can be coupled to at least one graphene oxide sheet through esterification under appropriate reaction conditions. For example, FIG. 5 shows an embodiment of esterification.

In some embodiments, the crosslinking moiety can be coupled to at least one graphene oxide sheet through amidation under appropriate reaction conditions, where the carboxylic group on graphene oxide reacts with the amine group in the crosslinking moiety. For example, FIG. 6 shows an embodiment of amidation.

In some embodiments, the crosslinking moiety can be coupled to at least one graphene oxide sheet through etherification under appropriate reaction conditions. For example, FIGS. 8, 10, and 11 show embodiments of etherification.

Other reaction schemes can be found at FIGS. 7 and 9, according to some embodiments.

Additionally, FIGS. 14A-14G present several examples of reactions involving graphene oxide sheets and chemical linkers.

In FIGS. 14A and 14B, the graphene oxide sheets are reacted with 4-aminobenzoic acid in a solution of CsOH and water and treated with NHS, EDC, TEA, and water at 60° C. (FIG. 14A) or 80° C. (FIG. 14B) to form a functionalized graphene oxide sheet. In FIG. 14A, a thiol-based alkane is reacted with the functionalized graphene oxide sheets and rinsed with dimethylformamide (DMF) at room temperature to form crosslinked graphene oxide sheets. In FIG. 14B, an amine-based alkane is reacted with the functionalized graphene oxide sheets and rinsed with DMF and then water at room temperature to form crosslinked graphene oxide sheets.

In FIG. 14C, the graphene oxide sheets are reacted with 2-(4-aminophenyl)ethan-1-ol in an aqueous solution of CsOH, and then treated with NHS, EDC, and TEA, and water at 60° C. to form a functionalized graphene oxide sheet. This functionalized graphene oxide sheets are then reacted with a xylene derivative in CsOH, 18-Crown-6 (18 C6) in DMF at room temperature or 65° C. to form cross-linked graphene oxide sheets.

In FIG. 14D, the graphene oxide sheets are reacted with 4-aminobenzoic acid in an aqueous solution of CsOH and then treated with NHS, EDC, TEA, and water at 80° C. to form a functionalized graphene oxide sheet. The functionalized graphene oxide sheets are then reacted with an amine-based chemical linker to form crosslinked graphene oxide sheets.

In FIG. 14E, the graphene oxide sheets are reacted with acrylamide in a solution of NHS, EDC, TEA, and water at 70° C. to form a functionalized graphene oxide sheet. The functionalized graphene oxide sheets are then reacted with a thiol-based chemical linker and rinsed with DMF at room temperature to form crosslinked graphene oxide sheets.

In FIG. 14F, the graphene oxide sheets are reacted with acrylamide in water at 70° C. to form functionalized graphene oxide sheets. The functionalized graphene oxide sheets are then reacted with a thiol-based chemical linker in DMF at room temperature to form crosslinked graphene oxide sheets.

In general, the graphene oxide sheets and/or the membrane may undergo any suitable number or type of chemical treatments. In some embodiments, the type of chemical treatment is selected based on the intended application for the membrane. For instance, a membrane for separation of neutral species may not undergo a chemical treatment that imparts charge neutrality. In some embodiments, a membrane intended for use in high pH environments may undergo one or more chemical treatments to produce a membrane having chemical linkages comprising an ether. A membrane that may be exposed to oxidizing agents, such as hypochlorite or peroxide, may undergo one or more chemical treatments to produce a membrane having chemical linkages comprising an ester. As another example, a membrane intended to separate salts may undergo one or more chemical treatments to produce charge neutrality. A membrane that will be used to separate low molecular weight species (e.g., about 250 Da) may undergo one or more chemical treatments to produce a membrane having a relatively small solvent saturated d-spacing.

In some embodiments, the density of the chemical linker can be tuned by varying the weight ratio of the chemical linker versus the graphene oxide sheet. Generally, the lower the weight ratio, the lower the density.

In some embodiments, double fabrication methods—graphene deposition and crosslinking followed by a second layer of graphene deposition and crosslinking—can improve the sharpness of the molecular weight cutoff. In other words, adding additional membrane layers with the same d-spacing (size of the space between graphene layers) can lead to more absolute exclusion of particles above or below a particular molecular weight (size) and/or can lead to better removal of salts or other components from a solution. Without wishing to be bound by any particular theory, the combination of deposition and functionalization as well as the specific chemistry used to link the graphene oxide sheets can be used to engineer the technical specifications of the membrane.

For both single and double layer graphene oxide membranes, vacuum filtration can be used to deposit the graphene oxide sheets from an aqueous solution. In some embodiments, the graphene oxide sheets are allowed to dry before being crosslinked. In some embodiments, the deposition kinetics and drying process as well as the method of deposition can impact the alignment of the graphene oxide sheets, completeness of the film, and filtration performance. Gravure coating or slot-die coating can also be used to deposit the graphene oxide sheets.

Manufacture of the Filtration Apparatus

In some embodiments, the filtration apparatus can be manufactured by depositing the graphene oxide membrane onto a suitable support substrate.

In some embodiments, the filtration apparatus can be manufactured by depositing a plurality of graphene oxide sheets on a suitable support substrate and then forming a graphene oxide membrane in situ.

In some embodiments, the filtration apparatus can be manufactured according to the following steps: (a) depositing a plurality of graphene oxide sheets onto a suitable support substrate; (b) immersing the graphene oxide sheets and support substrate in a solution containing a crosslinking moiety; (c) reacting the graphene oxide with the crosslinking moiety under appropriate conditions; and (d) washing the graphene oxide sheets and support substrate. The graphene oxide sheets can be functionalized by one or more chemical groups prior to the deposition. If the substrate comprises PES, then DMF is not used to wash the substrate.

In some embodiments, the filtration apparatus can be manufactured according to the following steps: (a) mixing a plurality of graphene oxide sheets and a crosslinking moiety in a solvent; (b) depositing the graphene oxide sheets and the crosslinking moiety onto a suitable support substrate; (c) immersing or coating the support substrate with an activating agent; and (d) washing the graphene oxide sheets and support substrate. The graphene oxide sheets can be functionalized by one or more chemical groups prior to the deposition.

Applications

The graphene oxide membrane or filtration apparatus disclosed herein can be used for a wide range of nanofiltration or microfiltration applications, including but not limited to, concentration of molecules (e.g., whey, lactose), desalting (e.g., lactose, dye, chemicals, pharmaceuticals), fractionation (e.g., sugars), extraction (e.g., nutraceuticals, plant oils), recovery (e.g., catalyst, solvent), and purification (e.g., pharmaceutical, chemical, fuel). For example, a fluid comprising a plurality of species (e.g., plurality of retentate species) may be placed in contact with a first side of the graphene oxide membrane. The graphene oxide membrane may have interlayer spacing and/or intralayer spacing that are sized to prevent at least a portion of the species from traversing the membrane through the interlayer spacing and/or intralayer spacing, i.e., flowing from the first side of the graphene oxide membrane and to a second, opposing side of the graphene oxide membrane. In some embodiments, the fluid may include one or more types of species (e.g., a retentate species or a permeate species). In some embodiments, the graphene oxide membrane may have an average interlayer spacing and/or intralayer spacing that is sized to prevent at least a portion of the retentate species from traversing the graphene oxide membrane, while allowing at least a portion (e.g., substantially all) of the permeate species to traverse the graphene oxide membrane.

The graphene oxide membrane or filtration apparatus disclosed herein can be used in reverse osmosis to remove ions, molecules, and larger particles from a fluid, e.g., drinking water.

In some embodiments, the graphene oxide membrane or filtration apparatus disclosed herein can be used in methods for filtering raw milk, cheese whey, whey protein concentrate, mixtures comprising lactose, and whey protein isolate. The methods can include flowing the raw milk through the graphene oxide membrane.

The graphene oxide membrane or filtration apparatus disclosed herein can also be used for the removal of lignin

from black liquor. Weak black liquor (WBL) from pulp digestion is generally produced at 80° C. to 90° C. Cooling the WBL prior to filtration would be very expensive and energy intensive. Without the need for cooling, the WBL can pass through the graphene oxide membrane described herein at a high temperature, e.g., 80° C. to 90° C. or 75° C. to 85° C. In some embodiments, WBL can be flowed through the filtration apparatus described herein, wherein the WBL comprises lignin, sodium sulfate, sodium carbonate, sodium hydrosulfide, sodium thiosulfate, and/or sodium hydroxide.

The performance of the membrane for WBL filtration can be assessed by the rejection rate on a total solids basis. In some embodiments, the rejection rate is between about 75% and about 95% on a total solids basis, e.g., between about 75% and about 90%, between about 75% and about 85%, or between 80% and about 95% on a total solids basis.

In some embodiments, the graphene oxide membrane can reject at least a portion of the lignin. In some embodiments, the graphene oxide membrane can reject at least about 50%, at least about 60%, at least about 70%, at least about 80%, at least about 90%, at least about 95%, at least about 96%, at least about 97%, at least about 98%, at least about 99%, or at least about 99.5% of the lignin.

In some embodiments, the graphene oxide membrane can reject at least a portion of the sodium sulfate. In some embodiments, the graphene oxide membrane can reject at least about 50%, at least about 60%, at least about 70%, at least about 80%, at least about 90%, at least about 95%, at least about 96%, at least about 97%, at least about 98%, at least about 99%, or at least about 99.5% of the sodium sulfate.

In some embodiments, the graphene oxide membrane can reject at least a portion of the sodium carbonate. In some embodiments, the graphene oxide membrane can reject at least about 50%, at least about 60%, at least about 70%, at least about 80%, at least about 90%, at least about 95%, at least about 96%, at least about 97%, at least about 98%, at least about 99%, or at least about 99.5% of the sodium carbonate.

In some embodiments, the graphene oxide membrane can reject at least a portion of the sodium hydrosulfide. In some embodiments, the graphene oxide membrane can reject at least about 50%, at least about 60%, at least about 70%, at least about 80%, at least about 90%, at least about 95%, at least about 96%, at least about 97%, at least about 98%, at least about 99%, or at least about 99.5% of the sodium hydrosulfide.

In some embodiments, the graphene oxide membrane can reject at least a portion of the sodium thiosulfate. In some embodiments, the graphene oxide membrane can reject at least about 50%, at least about 60%, at least about 70%, at least about 80%, at least about 90%, at least about 95%, at least about 96%, at least about 97%, at least about 98%, at least about 99%, or at least about 99.5% of the sodium thiosulfate.

In some embodiments, the graphene oxide membrane can reject at least a portion of the sodium hydroxide. In some embodiments, the graphene oxide membrane can reject at least about 50%, at least about 60%, at least about 70%, at least about 80%, at least about 90%, at least about 95%, at least about 96%, at least about 97%, at least about 98%, at least about 99%, or at least about 99.5% of the sodium hydroxide.

The graphene oxide membrane or filtration apparatus disclosed herein can also be used in: (1) point-of-use water purification for military operation missions and for humanitarian relief to disaster-ridden and impoverished areas; (2)

on-site treatment of hydrofracking flowback water; (3) renewable energy production; and (4) desalination of water.

In some embodiments, the graphene oxide membrane can be designed using simulations. FIG. 16 is modeling diagram that models arbitrarily sized membrane system using simple picture of permeability and rejection from dead end or crossflow. Based on the results of these simulations, a high-rejection graphene oxide membrane can process dilute black liquor at high operating pressures. In some embodiments, a lower (<80%) rejection rates of specific salts can also accomplish the desired rejection percentage by passing permeate through a second, small system. In some embodiments, membranes designed have specific rejection targets for hydroxide, monovalent salts, divalent salts, WBL, and/or different organic constituents of WBL. In some embodiments, the composition of permeates can be estimated based on the selected membrane systems.

In some embodiments, models of arbitrarily sized membrane system can be developed based on permeability data and rejection from dead end or crossflow. In some embodiments, every constituent species can be tracked. Counter ions (e.g., sodium) can be given the same rejection rate as pair ion (e.g., sodium paired with lignin, sodium paired with Cl^- , sodium paired with sulfates). In some embodiments, a total or species flow rate can be calculated based on the formula presented in FIG. 16.

In some embodiments, a system can be designed with a first pass only ("single-pass system"). In some embodiments, a feed can be split over sets of 5 spirals in series. In some embodiments, the system can be designed as a multi-pass system (e.g., a 2-pass system). In some embodiments, permeate from a first system can be passed to a second simulation as feed. In some embodiments, concentrates from both systems can be collected and combined. In some embodiments, system design is intended to lower operating pressure and/or provide cleaner permeate.

In some embodiments, a model of a filtration system can include fixed parameters and/or variable parameters. In some embodiments, the fixed parameters can include feed spacing, permeability, temperature, viscosity, molar volume, density, rejection of large organics, minimum driving pressure (feed pressure-feed osmotic pressure), relative amounts of each WBL component (percent of total dissolved solids for each non-hydroxide solute component), molecular weights, and/or methanol level. In some embodiments, the temperature can be about 80° C. In some embodiments, the minimum driving pressure can be about 1 MPa. In some embodiments, the molecular weight of lignin can be about 5,000 Da. In some embodiments, the molecular weight for dissolved carbohydrates and hemicelluloses can be about 500. In some embodiments, methanol level can be about 1 wt %.

In some embodiments, the variable parameters can include the number of modules in each pass, total dissolved solids (TDS), pressure, divalent rejection, difference between divalent and monovalent rejection, and/or hydroxide rejection. In some embodiments, TDS can be about 9 wt % to about 18 wt %. In some embodiments, TDS can be about 9 wt %, about 10 wt %, about 11 wt %, about 12 wt %, about 13 wt %, about 14 wt %, about 15 wt %, about 16 wt %, about 17 wt %, about 18 wt %, inclusive of all values and ranges therebetween. In some embodiments, TDS can be about 12.5 wt % to about 16.5 wt %. In some embodiments, pressure can be about 4 MPa to about 10 MPa. In some embodiments, the organic rejection can be about 1. In some embodiments, divalent rejection can be about 0.4 to about 1. In some embodiments, monovalent rejection can be

about 0.05 to about 0.2 less than the divalent rejection. In some embodiments, hydroxide rejection can be about 0 to about equal to the monovalent rejection.

The high temperature capability of these membranes enables their use in reverse osmosis like concentration of WBL in the Kraft and sulfite pulping industries. This can be illustrated in FIGS. 21-26E. The concentration, production rate, and composition of WBL varies widely from mill to mill.

The concentration of liquors by membrane process was modeled. A discretized fluid flowing between two parallel sheets of membrane (FIG. 16) at high pressure is considered. An amount of fluid passes through the membrane sheets on either side according to the permeability of the membrane and the pressure of the fluid. The rejection properties of the membrane determine the composition of the permeate. The fluid then passes to the next discretized fluid element according to mass conservation. When the fluid reaches the end of the membrane system, the solution is now considered to be the concentrate. The sum of all the discretized permeate elements is the total permeate. The recovery ratio or percentage of the feed which becomes the permeate thus relates to the strength of the concentrate and the rate of permeate production by the system. Different feed pressures, membrane rejections and liquor strengths are illustrated.

FIG. 21 shows how the composition of this permeate depends on the total strength (% solids) of the feed weak black liquor when the membrane rejects 100% of dissolved organics and divalent anions. The membrane rejects between 80 and 95% of monovalent anions. The lower the permeate TDS, the cleaner and easier it is to reuse the permeate. All membrane systems show low permeate solids for even a broad range of feed dissolved solids.

FIGS. 22A-22D illustrate the effect of different membrane rejections on the permeate quality. Divalent ion rejection varies between 70% and 100%. Within each figure, the monovalent anion rejection is 5 to 20 percentage points lower than the divalent. High rejection membrane designs are very effective at producing clean (<1%, <0.5%, or even <0.1% TDS) permeates. Lower rejecting systems produce even higher recovery percentages due to their reduced osmotic pressure and as can be shown in later figures, further purified by a second pass with the same or higher rejecting membranes.

FIGS. 23A-23D illustrate the range of operating pressures which can be used for the reverse osmosis type weak black liquor concentration using the assumption of an ideal solution (non-dilute). Low pressure (e.g., 800 psi) is not sufficient to produce high recovery ratios unless the liquor is weaker. High pressure (e.g., 1400 psi) can produce very high recovery ratios even with these high rejecting membranes and clean permeates (<0.1% or even 0.05% TDS).

FIGS. 24A-24D illustrate that for a fixed system feed rate, the size of the membranes system (more or less modules) changes the recovery ratio. The recovery ratio relates to the amount and the strength of the concentrate.

Additionally, the solution can pass through two membranes. The permeate from previous FIGS. 21-24D can then be passed between additional membranes to produce an even cleaner permeate. FIGS. 25A-25D shows that very clean permeates can thus be achieved with membranes rejecting only 70% or more of the divalent anions such as sulfate. With very high rejections of divalent (90% to 100%) and monovalent anions (80% to 95%), nearly perfectly clean permeates can be produced from this second membrane system.

FIGS. 25A-25D consider only fixed sizes of membranes systems for this second pass, while FIGS. 26A-26E show that for a fixed first system size (and thus amount of permeate passed to the second system), the size of the second system can be varied to produce high recovery ratios and clean permeates (<1%, <0.5%, or <0.1% TDS). The maximum recovery ratio observed in the bottom panel is set by the size of the first system (750 modules at 1000 psi). These durable membranes can produce high recovery ratios (approaching 50%) and clean permeates.

While the present teachings have been described in conjunction with various embodiments and examples, it is not intended that the present teachings be limited to such embodiments or examples. On the contrary, the present teachings encompass various alternatives, modifications, and equivalents, as will be appreciated by those of skill in the art.

While various inventive embodiments have been described and illustrated herein, those of ordinary skill in the art will readily envision a variety of other means and/or structures for performing the function and/or obtaining the results and/or one or more of the advantages described herein, and each of such variations and/or modifications is deemed to be within the scope of the inventive embodiments described herein. More generally, those skilled in the art will readily appreciate that all parameters, dimensions, materials, and configurations described herein are meant to be exemplary and that the actual parameters, dimensions, materials, and/or configurations will depend upon the specific application or applications for which the inventive teachings is/are used. Those skilled in the art will recognize many equivalents to the specific inventive embodiments described herein. It is, therefore, to be understood that the foregoing embodiments are presented by way of example only and that, within the scope of the appended claims and equivalents thereto, inventive embodiments may be practiced otherwise than as specifically described and claimed. Inventive embodiments of the present disclosure are directed to each individual feature, system, article, material, kit, and/or method described herein. In addition, any combination of two or more such features, systems, articles, materials, kits, and/or methods, if such features, systems, articles, materials, kits, and/or methods are not mutually inconsistent, is included within the inventive scope of the present disclosure.

All definitions, as defined and used herein, should be understood to control over dictionary definitions, definitions in documents incorporated by reference, and/or ordinary meanings of the defined terms.

The indefinite articles “a” and “an,” as used herein in the specification and in the claims, unless clearly indicated to the contrary, should be understood to mean “at least one.” Any ranges cited herein are inclusive.

The terms “substantially,” “approximately,” and “about” used throughout this Specification and the claims generally mean plus or minus 10% of the value stated, e.g., about 100 would include 90 to 110.

The phrase “and/or,” as used herein in the specification and in the claims, should be understood to mean “either or both” of the elements so conjoined, i.e., elements that are conjunctively present in some cases and disjunctively present in other cases. Multiple elements listed with “and/or” should be construed in the same fashion, i.e., “one or more” of the elements so conjoined. Other elements may optionally be present other than the elements specifically identified by the “and/or” clause, whether related or unrelated to those elements specifically identified. Thus, as a non-limiting

example, a reference to “A and/or B”, when used in conjunction with open-ended language such as “comprising” may refer, in one embodiment, to A only (optionally including elements other than B); in another embodiment, to B only (optionally including elements other than A); in yet another embodiment, to both A and B (optionally including other elements); etc.

As used herein in the specification and in the claims, “or” should be understood to have the same meaning as “and/or” as defined above. For example, when separating items in a list, “or” or “and/or” shall be interpreted as being inclusive, i.e., the inclusion of at least one, but also including more than one, of a number or list of elements, and, optionally, additional unlisted items. Only terms clearly indicated to the contrary, such as “only one of” or “exactly one of,” or, when used in the claims, “consisting of,” will refer to the inclusion of exactly one element of a number or list of elements. In general, the term “or” as used herein shall only be interpreted as indicating exclusive alternatives (i.e. “one or the other but not both”) when preceded by terms of exclusivity, such as “either,” “one of” “only one of,” or “exactly one of.” “Consisting essentially of,” when used in the claims, shall have its ordinary meaning as used in the field of patent law.

As used herein in the specification and in the claims, the phrase “at least one,” in reference to a list of one or more elements, should be understood to mean at least one element selected from any one or more of the elements in the list of elements, but not necessarily including at least one of each and every element specifically listed within the list of elements and not excluding any combinations of elements in the list of elements. This definition also allows that elements may optionally be present other than the elements specifically identified within the list of elements to which the phrase “at least one” refers, whether related or unrelated to those elements specifically identified. Thus, as a non-limiting example, “at least one of A and B” (or, equivalently, “at least one of A or B,” or, equivalently “at least one of A and/or B”) may refer, in one embodiment, to at least one, optionally including more than one, A, with no B present (and optionally including elements other than B); in another embodiment, to at least one, optionally including more than one, B, with no A present (and optionally including elements other than A); in yet another embodiment, to at least one, optionally including more than one, A, and at least one, optionally including more than one, B (and optionally including other elements); etc.

As used herein, the term “basic” means pH greater than 7.

As used herein, “wt %” refers to weight percent.

In the claims, as well as in the specification above, all transitional phrases such as “comprising,” “including,” “carrying,” “having,” “containing,” “involving,” “holding,” “composed of,” and the like are to be understood to be open-ended, i.e., to mean including but not limited to. Only the transitional phrases “consisting of” and “consisting essentially of” shall be closed or semi-closed transitional phrases, respectively, as set forth in the United States Patent Office Manual of Patent Examining Procedures, Section 2111.03.

As used herein, the term “graphene oxide sheet” means a single atomic graphene oxide layer or a plurality of atomic graphene oxide layers. Each atomic graphene oxide layer may include out-of-plane chemical moieties attached to one or more carbon atoms on the layer. In some embodiments, the term “graphene oxide sheet” means 1 to about 20 atomic graphene oxide layers, e.g., 1 to about 18, 1 to about 16, 1 to about 14, 1 to about 12, 1 to about 10, 1 to about 8, 1 to about 6, 1 to about 4, or 1 to about 3 atomic graphene oxide

layers. In some embodiments, the term “graphene oxide sheet” means 1, 2, or 3 atomic graphene oxide layers.

As used herein in the specification and in the claims, the phrase “chemical linker” refers to a molecule that can couple two adjacent graphene oxide sheets through a covalent bond, a noncovalent interaction, or a combination thereof. Non-limiting examples of noncovalent interactions include ionic interactions, hydrogen bonding, halogen bonding, Van der Waals forces (e.g., dipole-dipole interactions, dipole-induced dipole interactions, or London dispersion forces), π -effects (e.g., π - π interactions, cation- π interactions, anion- π interactions, or polar- π interactions), and the hydrophobic effect.

As used herein, the term “flux” means flow rate. It describes the permeability of a membrane.

As used herein, the term “crosslink” refers to the process of coupling two adjacent graphene oxide sheets through one or more chemical linkers.

As used herein, the term “optionally substituted” is understood to mean that a given chemical moiety (e.g., an alkyl group) can (but is not required to) be bonded other substituents (e.g., heteroatoms). For instance, an alkyl group that is optionally substituted can be a fully saturated alkyl chain (i.e., a pure hydrocarbon). Alternatively, the same optionally substituted alkyl group can have substituents different from hydrogen. For instance, it can, at any point along the chain be bounded to a halogen atom, a hydroxyl group, or any other substituent described herein. Thus the term “optionally substituted” means that a given chemical moiety has the potential to contain other functional groups, but does not necessarily have any further functional groups. Suitable substituents used in the optional substitution of the described groups include, without limitation, halogen, oxo, —OH, —CN, —COOH, —CH₂CN, —O—(C₁-C₆) alkyl, (C₁-C₆) alkyl, C₁-C₆ alkoxy, (C₁-C₆) haloalkyl, C₁-C₆ haloalkoxy, —O—(C₂-C₆) alkenyl, —O—(C₂-C₆) alkynyl, (C₂-C₆) alkenyl, (C₂-C₆) alkynyl, —OH, —OP(O)(OH)₂, —OC(O)(C₁-C₆) alkyl, —C(O)(C₁-C₆) alkyl, —OC(O)O(C₁-C₆) alkyl, —NH₂, —NH((C₁-C₆) alkyl), —N((C₁-C₆) alkyl)₂, —NHC(O)(C₁-C₆) alkyl, —C(O)NH(C₁-C₆) alkyl, —S(O)₂(C₁-C₆) alkyl, —S(O)NH(C₁-C₆) alkyl, and —S(O)N((C₁-C₆) alkyl)₂. The substituents can themselves be optionally substituted.

As used herein, the term “hydroxy” or “hydroxyl” refers to the group —OH or —O⁻.

As used herein, “halo” or “halogen” refers to fluoro, chloro, bromo, and iodo.

The term “carbonyl” includes compounds and moieties which contain a carbon connected with a double bond to an oxygen atom. Examples of moieties containing a carbonyl include, but are not limited to, aldehydes, ketones, carboxylic acids, amides, esters, anhydrides, etc.

The term “carboxyl” refers to —COOH or its C₁-C₆ alkyl ester.

“Acyl” includes moieties that contain the acyl radical (R—C(O)—) or a carbonyl group. “Substituted acyl” includes acyl groups where one or more of the hydrogen atoms are replaced by, for example, alkyl groups, alkynyl groups, halogen, hydroxyl, alkylcarbonyloxy, arylcarbonyloxy, alkoxycarbonyloxy, aryloxy carbonyloxy, carboxylate, alkylcarbonyl, arylcarbonyl, alkoxycarbonyl, aminocarbonyl, alkylaminocarbonyl, dialkylaminocarbonyl, alkylthiocarbonyl, alkoxy, phosphate, phosphonate, phosphinato, amino (including alkylamino, dialkylamino, arylamino, diarylamino and alkylarylamino), acylamino (including alkylcarbonylamino, arylcarbonylamino, carbamoyl and ureido), amidino, imino, sulfhydryl, alkylthio, arylthio, thiocarboxylate, sulfates, alkylsulfinyl, sulfonate, sulfamoyl, sulfona-

mido, nitro, trifluoromethyl, cyano, azido, heterocycl, alkylaryl, or an aromatic or heteroaromatic moiety.

The term “alkoxy” or “alkoxyl” includes substituted and unsubstituted alkyl, alkenyl and alkynyl groups covalently linked to an oxygen atom. Examples of alkoxy groups or alkoxyl radicals include, but are not limited to, methoxy, ethoxy, isopropoxy, propoxy, butoxy and pentoxy groups. Examples of substituted alkoxy groups include halogenated alkoxy groups. The alkoxy groups can be substituted with groups such as alkenyl, alkynyl, halogen, hydroxyl, alkylcarbonyloxy, arylcarbonyloxy, alkoxycarbonyloxy, aryloxy carbonyloxy, carboxylate, alkylcarbonyl, arylcarbonyl, alkoxycarbonyl, aminocarbonyl, alkylaminocarbonyl, dialkylaminocarbonyl, alkylthiocarbonyl, alkoxy, phosphate, phosphonate, phosphinato, amino (including alkylamino, dialkylamino, arylamino, diarylamino, and alkylarylamino), acylamino (including alkylcarbonylamino, arylcarbonylamino, carbamoyl and ureido), amidino, imino, sulfhydryl, alkylthio, arylthio, thiocarboxylate, sulfates, alkylsulfinyl, sulfonate, sulfamoyl, sulfonamido, nitro, trifluoromethyl, cyano, azido, heterocycl, alkylaryl, or an aromatic or heteroaromatic moieties. Examples of halogen substituted alkoxy groups include, but are not limited to, fluoromethoxy, difluoromethoxy, trifluoromethoxy, chloromethoxy, dichloromethoxy and trichloromethoxy.

The term “ether” or “alkoxy” includes compounds or moieties which contain an oxygen bonded to two carbon atoms or heteroatoms. For example, the term includes “alkoxyalkyl,” which refers to an alkyl, alkenyl, or alkynyl group covalently bonded to an oxygen atom which is covalently bonded to an alkyl group.

The term “ester” includes compounds or moieties which contain a carbon or a heteroatom bound to an oxygen atom which is bonded to the carbon of a carbonyl group. The term “ester” includes alkoxycarboxy groups such as methoxycarbonyl, ethoxycarbonyl, propoxycarbonyl, butoxycarbonyl, pentoxycarbonyl, etc.

The term “thioalkyl” includes compounds or moieties which contain an alkyl group connected with a sulfur atom. The thioalkyl groups can be substituted with groups such as alkyl, alkenyl, alkynyl, halogen, hydroxyl, alkylcarbonyloxy, arylcarbonyloxy, alkoxycarbonyloxy, aryloxy carbonyloxy, carboxylate, carboxylic acid, alkylcarbonyl, arylcarbonyl, alkoxycarbonyl, aminocarbonyl, alkylaminocarbonyl, dialkylaminocarbonyl, alkylthiocarbonyl, alkoxy, amino (including alkylamino, dialkylamino, arylamino, diarylamino and alkylarylamino), acylamino (including alkylcarbonylamino, arylcarbonylamino, carbamoyl and ureido), amidino, imino, sulfhydryl, alkylthio, arylthio, thiocarboxylate, sulfates, alkylsulfinyl, sulfonate, sulfamoyl, sulfonamido, nitro, trifluoromethyl, cyano, azido, heterocycl, alkylaryl, or an aromatic or heteroaromatic moieties.

The term “thiocarbonyl” or “thiocarboxy” includes compounds and moieties which contain a carbon connected with a double bond to a sulfur atom.

The term “thioether” includes moieties which contain a sulfur atom bonded to two carbon atoms or heteroatoms. Examples of thioethers include, but are not limited to alkthioalkyls, alkthioalkenyls, and alkthioalkynyls. The term “alkthioalkyls” include moieties with an alkyl, alkenyl, or alkynyl group bonded to a sulfur atom which is bonded to an alkyl group. Similarly, the term “alkthioalkenyls” refers to moieties wherein an alkyl, alkenyl or alkynyl group is bonded to a sulfur atom which is covalently bonded to an alkenyl group; and “alkthioalkynyls” refers to moieties wherein an alkyl, alkenyl or alkynyl group is bonded to a sulfur atom which is covalently bonded to an alkynyl group.

As used herein, “amino” or “amine,” as used herein, refers to a primary ($-\text{NH}_2$), secondary ($-\text{NHR}_x$), tertiary ($-\text{NR}_x\text{R}_y$), or quaternary amine ($-\text{N}^+\text{R}_x\text{R}_y\text{R}_z$), where R_x , R_y , and R_z are independently an aliphatic, alicyclic, heteroaliphatic, heterocyclic, aryl, or heteroaryl moiety, as defined herein. Examples of amine groups include, but are not limited to, methylamine, dimethylamine, ethylamine, diethylamine, methylethylamine, iso-propylamine, piperidine, trimethylamine, and propylamine. “Alkylamino” includes groups of compounds wherein the nitrogen of $-\text{NH}_2$ is bound to at least one alkyl group. Examples of alkylamino groups include benzylamino, methylamino, ethylamino, phenethylamino, etc. “Dialkylamino” includes groups wherein the nitrogen of $-\text{NH}_2$ is bound to two alkyl groups. Examples of dialkylamino groups include, but are not limited to, dimethylamino and diethylamino. “Arylamino” and “diarylamino” include groups wherein the nitrogen is bound to at least one or two aryl groups, respectively. “Aminoaryl” and “aminoaryloxy” refer to aryl and aryloxy substituted with amino. “Alkylarylamino,” “alkylaminoaryl” or “arylaminoalkyl” refers to an amino group which is bound to at least one alkyl group and at least one aryl group. “Alkaminalkyl” refers to an alkyl, alkenyl, or alkynyl group bound to a nitrogen atom which is also bound to an alkyl group. “Acylamino” includes groups wherein nitrogen is bound to an acyl group. Examples of acylamino include, but are not limited to, alkylcarbonylamino, arylcarbonylamino, carbamoyl and ureido groups.

The term “amide” or “aminocarboxy” includes compounds or moieties that contain a nitrogen atom that is bound to the carbon of a carbonyl or a thiocarbonyl group. The term includes “alkaminocarboxy” groups that include alkyl, alkenyl or alkynyl groups bound to an amino group which is bound to the carbon of a carbonyl or thiocarbonyl group. It also includes “arylamino-carboxy” groups that include aryl or heteroaryl moieties bound to an amino group that is bound to the carbon of a carbonyl or thiocarbonyl group. The terms “alkylaminocarboxy,” “alkenylaminocarboxy,” “alkynylaminocarboxy” and “arylamino-carboxy” include moieties wherein alkyl, alkenyl, alkynyl and aryl moieties, respectively, are bound to a nitrogen atom which is in turn bound to the carbon of a carbonyl group. Amides can be substituted with substituents such as straight chain alkyl, branched alkyl, cycloalkyl, aryl, heteroaryl or heterocycle. Substituents on amide groups may be further substituted.

Unless otherwise specifically defined, the term “aryl” refers to cyclic, aromatic hydrocarbon groups that have 1 to 3 aromatic rings, including monocyclic or bicyclic groups such as phenyl, biphenyl or naphthyl. Where containing two aromatic rings (bicyclic, etc.), the aromatic rings of the aryl group may be joined at a single point (e.g., biphenyl), or fused (e.g., naphthyl). The aryl group may be optionally substituted by one or more substituents, e.g., 1 to 5 substituents, at any point of attachment. Exemplary substituents include, but are not limited to, $-\text{H}$, $-\text{halogen}$, $-\text{O}-(\text{C}_1-\text{C}_6)$ alkyl, (C_1-C_6) alkyl, $-\text{O}-(\text{C}_2-\text{C}_6)$ alkenyl, $-\text{O}-(\text{C}_2-\text{C}_6)$ alkynyl, (C_2-C_6) alkenyl, (C_2-C_6) alkynyl, $-\text{OH}$, $-\text{OP}(\text{O})(\text{OH})_2$, $-\text{OC}(\text{O})(\text{C}_1-\text{C}_6)$ alkyl, $-\text{C}(\text{O})(\text{C}_1-\text{C}_6)$ alkyl, $-\text{OC}(\text{O})(\text{C}_1-\text{C}_6)$ alkyl, NH_2 , $\text{NH}((\text{C}_1-\text{C}_6)$ alkyl), $\text{N}((\text{C}_1-\text{C}_6)$ alkyl) $_2$, $-\text{S}(\text{O})_2-(\text{C}_1-\text{C}_6)$ alkyl, $-\text{S}(\text{O})\text{NH}(\text{C}_1-\text{C}_6)$ alkyl, and $-\text{S}(\text{O})\text{N}((\text{C}_1-\text{C}_6)$ alkyl) $_2$. The substituents can themselves be optionally substituted. Furthermore, when containing two fused rings the aryl groups herein defined may have an unsaturated or partially saturated ring fused with a fully saturated ring. Exemplary ring systems of these aryl groups include, but are not limited to, phenyl, biphenyl, naphthyl,

anthracenyl, phenalenyl, phenanthrenyl, indanyl, indenyl, tetrahydronaphthalenyl, tetrahydrobenzoannulenyl, and the like.

Unless otherwise specifically defined, “heteroaryl” means a monocyclic aromatic radical of 5 to 24 ring atoms or a polycyclic aromatic radical, containing one or more ring heteroatoms selected from N, O, or S, the remaining ring atoms being C. Heteroaryl as herein defined also means a bicyclic heteroaromatic group wherein the heteroatom is selected from N, O, or S. The aromatic radical is optionally substituted independently with one or more substituents described herein. Examples include, but are not limited to, furyl, thienyl, pyrrolyl, pyridyl, pyrazolyl, pyrimidinyl, imidazolyl, isoxazolyl, oxazolyl, oxadiazolyl, pyrazinyl, indolyl, thiophen-2-yl, quinolyl, benzopyran, isothiazolyl, thiazolyl, thiadiazole, indazole, benzimidazolyl, thieno[3,2-b]thiophene, triazolyl, triazinyl, imidazo[1,2-b]pyrazolyl, furo[2,3-c]pyridinyl, imidazo[1,2-a]pyridinyl, indazolyl, pyrrolo[2,3-c]pyridinyl, pyrrolo[3,2-c]pyridinyl, pyrazolo[3,4-c]pyridinyl, thieno[3,2-c]pyridinyl, thieno[2,3-c]pyridinyl, thieno[2,3-b]pyridinyl, benzothiazolyl, indolyl, indolinyl, indolinonyl, dihydrobenzothiophenyl, dihydrobenzofuran, benzofuran, chromanyl, thiochromanyl, tetrahydroquinolyl, dihydrobenzothiazine, dihydrobenzoxanyl, quinolyl, isoquinolyl, 1,6-naphthyridinyl, benzo[de]isoquinolyl, pyrido[4,3-b][1,6]naphthyridinyl, thieno[2,3-b]pyrazinyl, quinazolinyl, tetrazolo[1,5-a]pyridinyl, [1,2,4]triazolo[4,3-a]pyridinyl, isoindolyl, pyrrolo[2,3-b]pyridinyl, pyrrolo[3,4-b]pyridinyl, pyrrolo[3,2-b]pyridinyl, imidazo[5,4-b]pyridinyl, pyrrolo[1,2-a]pyrimidinyl, tetrahydro pyrrolo[1,2-a]pyrimidinyl, 3,4-dihydro-2H-1 λ^2 -pyrrolo[2,1-b]pyrimidine, dibenzo[b,d] thiophene, pyridin-2-one, furo[3,2-c]pyridinyl, furo[2,3-c]pyridinyl, 1H-pyrido[3,4-b][1,4]thiazinyl, benzoisoxazolyl, benzoisoxazolyl, furo[2,3-b]pyridinyl, benzothiophenyl, 1,5-naphthyridinyl, furo[3,2-b]pyridine, [1,2,4]triazolo[1,5-a]pyridinyl, benzo[1,2,3]triazolyl, imidazo[1,2-a]pyrimidinyl, [1,2,4]triazolo[4,3-b]pyridazinyl, benzo[c][1,2,5]thiadiazolyl, benzo[c][1,2,5]oxadiazole, 1,3-dihydro-2H-benzo[d]imidazol-2-one, 3,4-dihydro-2H-pyrazolo[1,5-b][1,2]oxazinyl, 4,5,6,7-tetrahydropyrazolo[1,5-a]pyridinyl, thiazolo[5,4-d]thiazolyl, imidazo[2,1-b][1,3,4]thiadiazolyl, thieno[2,3-b]pyrrolyl, 3H-indolyl, and derivatives thereof. Furthermore, when containing two fused rings the aryl groups herein defined may have an unsaturated or partially saturated ring fused with a fully saturated ring. Exemplary ring systems of these heteroaryl groups include indolinyl, indolinonyl, dihydrobenzothiophenyl, dihydrobenzofuran, chromanyl, thiochromanyl, tetrahydroquinolyl, dihydrobenzothiazine, 3,4-dihydro-1H-isoquinolyl, 2,3-dihydrobenzofuran, indolinyl, indolyl, and dihydrobenzoxanyl.

Furthermore, the terms “aryl” and “heteroaryl” include multicyclic aryl and heteroaryl groups, e.g., tricyclic, bicyclic, e.g., naphthalene, benzoxazole, benzodioxazole, benzothiazole, benzoimidazole, benzothiophene, quinoline, isoquinoline, naphthyridine, indole, benzofuran, purine, benzofuran, deazapurine, indolizine.

“cycloalkyl” refers to a saturated or partially saturated ring structure having about 3 to about 8 ring members that has only carbon atoms as ring atoms and can include divalent radicals. Examples of cycloalkyl groups include but are not limited to, cyclopropyl, cyclobutyl, cyclopentyl, cyclohexyl, cyclohexene, cyclopentenyl, cyclohexenyl.

“Heterocycloalkyl” refers to a saturated or partially unsaturated 3-8 membered monocyclic, 7-12 membered bicyclic (fused, bridged, or spiro rings), or 11-14 membered tricyclic ring system (fused, bridged, or spiro rings) having

one or more heteroatoms (such as O, N, S, P, or Se), e.g., 1 or 1-2 or 1-3 or 1-4 or 1-5 or 1-6 heteroatoms, or e.g. 1, 2, 3, 4, 5, or 6 heteroatoms, independently selected from the group consisting of nitrogen, oxygen and sulfur, unless specified otherwise. Examples of heterocycloalkyl groups include, but are not limited to, piperidinyl, piperazinyl, pyrrolidinyl, dioxanyl, tetrahydrofuranyl, isoindolinyl, indolinyl, imidazolidinyl, pyrazolidinyl, oxazolidinyl, isoxazolidinyl, triazolidinyl, oxiranyl, azetidiny, oxetanyl, thietanyl, 1,2,3,6-tetrahydropyridinyl, tetrahydropyran, dihydropyran, pyran, morpholinyl, tetrahydrothiopyran, 1,4-diazepanyl, 1,4-oxazepanyl, 2-oxa-5-azabicyclo[2.2.1]heptanyl, 2,5-diazabicyclo[2.2.1]heptanyl, 2-oxa-6-azaspiro[3.3]heptanyl, 2,6-diazaspiro[3.3]heptanyl, 1,4-dioxo-8-azaspiro[4.5]decanyl, 1,4-dioxaspiro[4.5]decanyl, 1-oxaspiro[4.5]decanyl, 1-azaspiro[4.5]decanyl, 3'H-spiro[cyclohexane-1,1'-isobenzofuran]-yl, 7'H-spiro[cyclohexane-1,5'-furo[3,4-b]pyridin]-yl, 3'H-spiro[cyclohexane-1,1'-furo[3,4-c]pyridin]-yl, 3-azabicyclo[3.1.0]hexanyl, 3-azabicyclo[3.1.0]hexan-3-yl, 1,4,5,6-tetrahydropyrrolo[3,4-c]pyrazolyl, 3,4,5,6,7,8-hexahydropyrido[4,3-d]pyrimidinyl, 4,5,6,7-tetrahydro-1H-pyrazolo[3,4-c]pyridinyl, 5,6,7,8-tetrahydropyrido[4,3-d]pyrimidinyl, 2-azaspiro[3.3]heptanyl, 2-methyl-2-azaspiro[3.3]heptanyl, 2-azaspiro[3.5]nonanyl, 2-methyl-2-azaspiro[3.5]nonanyl, 2-azaspiro[4.5]decanyl, 2-methyl-2-azaspiro[4.5]decanyl, 2-oxa-azaspiro[3.4]octanyl, 2-oxa-azaspiro[3.4]octan-6-yl, and the like. In the case of multicyclic heterocycloalkyl, only one of the rings in the heterocycloalkyl needs to be non-aromatic (e.g., 4,5,6,7-tetrahydrobenzo[c]isoxazolyl).

"Alkyl" refers to a straight or branched chain saturated hydrocarbon. C₁-C₆ alkyl groups contain 1 to 6 carbon atoms. Examples of a C₁-C₆ alkyl group include, but are not limited to, methyl, ethyl, propyl, butyl, pentyl, isopropyl, isobutyl, sec-butyl and tert-butyl, isopentyl and neopentyl.

An optionally substituted alkyl refers to unsubstituted alkyl or alkyl having designated substituents replacing one or more hydrogen atoms on one or more carbons of the hydrocarbon backbone. Such substituents can include, for example, alkyl, alkenyl, alkynyl, halogen, hydroxyl, alkyl-carbonyloxy, arylcarbonyloxy, alkoxy-carbonyloxy, aryloxy-carbonyloxy, carboxylate, alkylcarbonyl, arylcarbonyl, alkoxy-carbonyl, aminocarbonyl, alkylaminocarbonyl, dialkylaminocarbonyl, alkylthiocarbonyl, alkoxy, phosphate, phosphonate, phosphinato, amino (including alkylamino, dialkylamino, arylamino, diarylamino and alkylarylamin), acylamino (including alkylcarbonylamino, arylcarbonylamino, carbamoyl and ureido), amidino, imino, sulfhydryl, alkylthio, arylthio, thiocarboxylate, sulfates, alkylsulfonate, sulfonate, sulfamoyl, sulfonamido, nitro, trifluoromethyl, cyano, azido, heterocyclyl, alkylaryl, or an aromatic or heteroaromatic moiety.

The term "hydroxyalkyl" means an alkyl group as defined above, where the alkyl group is substituted with one or more OH groups. Examples of hydroxyalkyl groups include HO-CH₂-, HO-CH₂-CH₂- and CH₃-CH(OH)-.

As used herein, "alkylene linker" is intended to include C₁, C₂, C₃, C₄, C₅, C₆, C₇, C₈, C₉, or C₁₀ straight chain (linear) saturated divalent aliphatic hydrocarbon groups and C₂, C₃, C₄, C₅ or C₆, C₇, C₈, C₉, or C₁₀ branched saturated aliphatic hydrocarbon groups. For example, C₁-C₆ alkylene linker is intended to include C₁, C₂, C₃, C₄, C₅ and C₆ alkylene linker groups. Examples of alkylene linker include, moieties having from one to six carbon atoms, such as, but not limited to, methyl (-CH₂-), ethyl (-CH₂CH₂-), n-propyl (-CH₂CH₂CH₂-), i-propyl (-CHCH₃CH₂-), n-butyl (-CH₂CH₂CH₂CH₂-), s-butyl

(-CHCH₃CH₂CH₂-), i-butyl (-C(CH₃)₂CH₂-), n-pentyl (-CH₂CH₂CH₂CH₂CH₂-), s-pentyl (-CHCH₃CH₂CH₂CH₂-) or n-hexyl (-CH₂CH₂CH₂CH₂CH₂CH₂-).

As used herein, "alkenyl" includes unsaturated aliphatic groups analogous in length and possible substitution to the alkyls described above, but that contain at least one double bond. For example, the term "alkenyl" includes straight chain alkenyl groups (e.g., ethenyl, propenyl, butenyl, pentenyl, hexenyl, heptenyl, octenyl, nonenyl, decenyl), and branched alkenyl groups.

An optionally substituted alkenyl refers to unsubstituted alkenyl or alkenyl having designated substituents replacing one or more hydrogen atoms on one or more hydrocarbon backbone carbon atoms. Such substituents can include, for example, alkyl, alkenyl, alkynyl, halogen, hydroxyl, alkyl-carbonyloxy, arylcarbonyloxy, alkoxy-carbonyloxy, aryloxy-carbonyloxy, carboxylate, alkylcarbonyl, arylcarbonyl, alkoxy-carbonyl, aminocarbonyl, alkylaminocarbonyl, dialkylaminocarbonyl, alkylthiocarbonyl, alkoxy, phosphate, phosphonate, phosphinato, amino (including alkylamino, dialkylamino, arylamino, diarylamino and alkylarylamin), acylamino (including alkylcarbonylamino, arylcarbonylamino, carbamoyl and ureido), amidino, imino, sulfhydryl, alkylthio, arylthio, thiocarboxylate, sulfates, alkylsulfonate, sulfonate, sulfamoyl, sulfonamido, nitro, trifluoromethyl, cyano, heterocyclyl, alkylaryl, or an aromatic or heteroaromatic moiety.

As used herein, "C₂-C₁₀ alkenylene linker" is intended to include C₂, C₃, C₄, C₅, C₆, C₇, C₈, C₉, or C₁₀ chain (linear or branched) divalent unsaturated aliphatic hydrocarbon groups. For example, C₂-C₁₀ alkenylene linker is intended to include C₂, C₃, C₄, C₅, C₆, C₇, C₈, C₉, and C₁₀ alkenylene linker groups.

"Alkynyl" includes unsaturated aliphatic groups analogous in length and possible substitution to the alkyls described above, but which contain at least one triple bond. For example, "alkynyl" includes straight chain alkynyl groups (e.g., ethynyl, propynyl, butynyl, pentynyl, hexynyl, heptyn, octynyl, nonynyl, decynyl), and branched alkynyl groups. In certain embodiments, a straight chain or branched alkynyl group has six or fewer carbon atoms in its backbone (e.g., C₂-C₆ for straight chain, C₃-C₆ for branched chain). The term "C₂-C₆" includes alkynyl groups containing two to six carbon atoms. The term "C₃-C₆" includes alkynyl groups containing three to six carbon atoms.

An optionally substituted alkynyl refers to unsubstituted alkynyl or alkynyl having designated substituents replacing one or more hydrogen atoms on one or more hydrocarbon backbone carbon atoms. Such substituents can include, for example, alkyl, alkenyl, alkynyl, halogen, hydroxyl, alkyl-carbonyloxy, arylcarbonyloxy, alkoxy-carbonyloxy, aryloxy-carbonyloxy, carboxylate, alkylcarbonyl, arylcarbonyl, alkoxy-carbonyl, aminocarbonyl, alkylaminocarbonyl, dialkylaminocarbonyl, alkylthiocarbonyl, alkoxy, phosphate, phosphonate, phosphinato, amino (including alkylamino, dialkylamino, arylamino, diarylamino and alkylarylamin), acylamino (including alkylcarbonylamino, arylcarbonylamino, carbamoyl and ureido), amidino, imino, sulfhydryl, alkylthio, arylthio, thiocarboxylate, sulfates, alkylsulfonate, sulfonate, sulfamoyl, sulfonamido, nitro, trifluoromethyl, cyano, azido, heterocyclyl, alkylaryl, or an aromatic or heteroaromatic moiety.

As used herein, the term "molecular weight cutoff" refers to at least 90% (e.g., at least 92%, at least 95%, or at least 98%) rejection rate for molecules with molecular weights greater than the cutoff value.

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As used herein, the term “room temperature” can refer to a temperature of about 15° C., about 16° C., about 17° C., about 18° C., about 19° C., about 20° C., about 21° C., about 22° C., about 23° C., about 24° C., or about 25° C. In some embodiments, the room temperature is about 20° C.

As used herein, the term “substantially the same” refers to a first value that is within 10% of a second value. For example, if A is substantially the same as B, and B is 100, A can have a value ranging from 90 to 110. If A is substantially the same as B, and B is 200, A can have a value ranging from 180 to 220.

As used herein, the term “grayscale mode value” refers to the mode value of an image recorded using the RGB color model, calculated with the aid of an image processing software (e.g., ImageJ) by first converting the image to grayscale, where each pixel is converted to grayscale using the formula: $\text{gray} = 0.299 \times \text{red} + 0.587 \times \text{green} + 0.114 \times \text{blue}$, and then quantifying the mode of the distribution of the intensity of the pixels.

The claims should not be read as limited to the described order or elements unless stated to that effect. It should be understood that various changes in form and detail may be made by one of ordinary skill in the art without departing from the spirit and scope of the appended claims. All embodiments that come within the spirit and scope of the following claims and equivalents thereto are claimed.

EXAMPLES

Example 1

The graphene oxide membrane can be produced in accordance with the schematic illustration of FIG. 5. The procedure can include the following steps: (1) mix 0.5 mL of 0.4 wt % aqueous graphene oxide sheets with 19.5 mL of MilliQ-water; (2) pour onto a 90-mm support which is mounted on fritted glass (active area ~78 mm); (3) pull vacuum for 3 hours; (4) air dry; (5) dissolve 66 mg α, α' -dibromo-p-xylene in 5 mL dimethylformamide (DMF) and 10 mg KOH in 1 mL H₂O, which are then added to the dry graphene oxide sheets in a glass Petri dish; (6) swirl and rest for 3 hours; and (7) wash with water.

The resulting membrane has a lactose rejection rate of 100%, a MgSO₄ rejection rate of 83%, and a flux of 5.25×10^{-4} GFD/psi.

Example 2

The graphene oxide membrane can be produced in accordance with the schematic illustration of FIG. 6. The procedure can include the following steps: (1) mix 0.5 mL of 0.4 wt % aqueous graphene oxide sheets with 19.5 mL of MilliQ-water; (2) pour onto a 90-mm support which is mounted on fritted glass (active area ~78 mm); (3) pull vacuum for 3 hours; (4) air dry; (5) soak membrane in a Petri dish for 30 min in 10 mL 0.1 2-(N-morpholino)ethanesulfonic acid (MES) at pH 6.3; (6) decant and add 10 mL 5 mM N-hydroxysuccinimide (NHS) and 10 mL 2 mM 1-ethyl-3-(3-dimethylaminopropyl)carbodiimide (EDC); (7) soak for one hour; (8) decant and add 30 mg p-phenylenediamine and 84 mg NaHCO₃ in 10 mL H₂O for one hour; and (9) wash with water.

The resulting membrane has a lactose rejection rate of 83%, a MgSO₄ rejection rate of 69%, and a flux of 1.40×10^{-3} GFD/psi.

Alternatively, the procedure can include the following steps: (1) mix 0.5 mL of 0.4 wt % aqueous graphene oxide

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sheets with 19.5 mL of MilliQ-water; (2) pour onto a 90-mm support which is mounted on fritted glass (active area ~78 mm); (3) pull vacuum for 3 hours; (4) air dry; (5) soak membrane in 5 mL acetonitrile (ACN) and 1 mL thionyl chloride in glass Petri dish for 1 hour; (6) decant; (7) add 80 mg p-phenylenediamine dissolved in 5 mL ACN and soak for 3 hours; and (8) wash with water.

The resulting membrane has a lactose rejection rate of 91%, a MgSO₄ rejection rate of 83%, and a flux of 3.94×10^{-4} GFD/psi.

Example 3

The graphene oxide membrane can be produced in accordance with the schematic illustration of FIG. 7. The procedure can include the following steps: (1) mix 0.5 mL of 0.4 wt % aqueous graphene oxide sheets with 19.5 mL of MilliQ-water; (2) pour onto a 90-mm support which is mounted on fritted glass (active area ~78 mm); (3) pull vacuum for 3 hours; (4) air dry; (5) mix 7 mg 1,4-diazabicyclo[2.2.2]octane (DABCO) with 0.5 mL DMF+1 mL isopropanol and cooled to 0° C., which is added to the graphene oxide sheets via pipette on surface; (6) add 0.05 mL ethyl propiolate; (7) over the dish to prevent solvent evaporation; (8) let cool to room temperature over 4 hours; (9) wash and air dry; (10) add 30 mg p-phenylenediamine and 84 mg NaHCO₃ in 10 mL DMF to membrane in Petri dish; (11) seal in metal enclosure and heat in oven at 65° C. for 5 hours; and (12) wash with water.

The resulting membrane has a lactose rejection rate of 94%, a MgSO₄ rejection rate of 85%, and a flux of 1.63×10^{-3} GFD/psi.

Example 4

The graphene oxide membrane can be produced in accordance with the schematic illustration of FIG. 8. The procedure can include the following steps: (1) mix 0.5 mL of 0.4 wt % aqueous graphene oxide sheets with 19.5 mL of MilliQ-water; (2) pour onto a 90 mm support which is mounted on fritted glass (active area ~78 mm); (3) pull vacuum for 3 hours; (4) air dry. (5) add 0.1 mL styrene oxide in 2 mL DMF to the graphene oxide sheets; (6) add cover to prevent solvent evaporation; (7) heat the graphene oxide sheets for 3 hours at 65° C.; (8) wash the graphene oxide sheets with DMF; (9) add 30 mg α, α' -dibromo-p-xylene in 2 mL DMF and 11 mg DABCO; (10) heat at 65° C. for 17 hours; and (11) wash with water.

The resulting membrane has a lactose rejection rate of 94%, a MgSO₄ rejection rate of 81%, and a flux of 2.06×10^{-3} GFD/psi.

An alternative procedure can include the following steps: (1) mix 0.5 mL of 0.4 wt % aqueous graphene oxide sheets with 19.5 mL of MilliQ-water; (2) pour onto a 90 mm support which is mounted on fritted glass (active area ~78 mm); (3) pull vacuum for 3 hours; (4) air dry; (5) add styrene oxide in 2 mL DMF and 5 mg CuBF₄ to the graphene oxide sheets; (6) add cover to prevent solvent evaporation at room temperature for 17 hours; (7) wash the graphene oxide sheets with DMF; (8) add 30 mg α, α' -dibromo-p-xylene dissolved in 2 mL DMF and 0.02 mL 1,8-diazabicyclo[5.4.0]undec-7-ene; (9) heat at 65° C. for 3 hours; (10) wash with water.

The resulting membrane has a lactose rejection rate of 85%, a MgSO₄ rejection rate of 78%, and a flux of 9.60×10^{-4} GFD/psi.

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Example 5

The graphene oxide membrane can be produced in accordance with the schematic illustration of FIG. 9. The procedure can include the following steps: (1) mix 0.5 mL of 0.4 wt % aqueous graphene oxide sheets with 19.5 mL of MilliQ-water; (2) pour onto a 90-mm support which is mounted on fritted glass (active area ~78 mm); (3) pull vacuum for 3 hours; (4) air dry; (5) soak the graphene oxide sheets in 5 mg KOH in 0.1 mL H₂O and 2.5 mL DMF and add 0.1 mL tert-Butyl acrylate to surface; (6) let rest for 6 hours; (7) rinse with DMF; (8) add 20 mg p-phenylenediamine in 2.5 mL DMF and heat at 65° C. for 6 hours; (9) wash with water.

The resulting membrane has a lactose rejection rate of 89%, a MgSO₄ rejection rate of 76%, and a flux of 2.59×10⁻³ GFD/psi.

Example 6

The graphene oxide membrane can be produced in accordance with the schematic illustration of FIG. 10. The procedure can include the following steps: (1) mix 0.5 mL of 0.4 wt % aqueous graphene oxide sheets with 19.5 mL of MilliQ-water; (2) pour onto a 90 mm support which is mounted on fritted glass (active area ~78 mm); (3) pull vacuum for 3 hours; (4) air dry; (5) soak membrane in 0.1 mL 1,2-epoxypropane dissolved in 2.5 mL DMF; (6) heat at 65° C. for 1 hour; (7) wash the graphene oxide sheets with DMF; (8) soak in 60 mg α,α'-dibromo-p-xylene and 7 mg DABCO in 2.5 mL DMF; (9) heat at 65° C. for 3 hours; and (10) wash with water.

The resulting membrane has a lactose rejection rate of 92%, a MgSO₄ rejection rate of 84%, and a flux of 1.10×10⁻³ GFD/psi.

Example 7

The graphene oxide membrane can be produced in accordance with the schematic illustration of FIG. 11. The graphene oxide membrane can be produced in accordance with the following procedure: (1) mix 0.5 mL of 0.4 wt % aqueous graphene oxide sheets with 19.5 mL of MilliQ-water; (2) pour onto a 90-mm support which is mounted on fritted glass (active area ~78 mm); (3) pull vacuum for 3 hours; (4) air dry; (5) mix 0.5 mL styrene oxide with 5 mL DMF, which is then added to the graphene oxide sheets; (6) add cover to prevent solvent evaporation; (7) put in metal enclosure and heat at 65° C. for 7 hours; (8) wash with DMF; (9) add 0.1 mL 1,5-diiodopentane in 5 mL DMF and 0.04 mL N,N-diisopropylethylamine (DIPEA); (10) heat for 16 hours at 65° C.; and (11) wash with water.

The resulting membrane has a lactose rejection rate of 92%, a MgSO₄ rejection rate of 85%, and a flux of 6.13×10⁻⁴ GFD/psi.

Example 8

A graphene oxide membrane was produced with the following procedure: (1) 15 mL of 4 mg/mL graphene oxide sheets was treated with 0.8 mL TEA, 22 mg NHS, and 72 mg EDC. (2) Mixture was incubated at room temperature for 1 hour. (3) A 5 mL aliquot of the mixture was treated with 106.6 mg acrylamide. (4) Resulting mixture was heated at 4 hours in a 80° C. oven. (5) Mixture was subsequently cast onto substrate, dried, and washed with DMF/Ethanol. (6)

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Mixture was then treated with 3 mL DMF, 0.07 mL 1,8-octanedithiol, and (7) incubated at room temperature for 3 hours.

The membrane was evaluated using the durability testing procedure outlined below: (1) cut a 47 to 50 mm disc from the graphene oxide membrane using a razor blade or laser cutter; (2) load the disc with the graphene oxide side up onto a porous stainless steel frit, which is then mounted into a Sterlitech HP4750 filtration cell; (3) add 60 to 100 mL of 1 wt % lactose solution; (4) place the setup on a stir plate at approximately 750 rpm; (5) close the feed chamber and pressurize it to 50 to 100 psi; (6) Approximately 15 to 30 mL of permeate is collected of a 1 wt % lactose and 0.1 wt % MgSO₄ solution at room temperature using 75 psi of driving pressure and ~500 rpm stir speed; and (6) Membrane is then exposed to partial phosphate buffer at exposure temperature for 4 hours and then tested for sugar and salt rejection again.

The partial phosphate buffer is made by mixing 500 mL of 0.025 M Na₂HPO₄ (1.77 g in 500 mL) with 111 mL of 0.1 M NaOH (0.444 g in 111 mL). Mixture is either sonicated to dissolve or allowed to rest overnight. Buffer is stored in the 4° C. fridge.

The membrane exhibited an initial lactose rejection rate of 86% and a lactose rejection rate of 84% after exposure to a temperature of 70° C.

Example 9

A graphene oxide membrane was produced with the following procedure: (1) 15 mL of 4 mg/mL graphene oxide sheets was treated with 0.8 mL TEA, 22 mg NHS, and 72 mg EDC. (2) Mixture was incubated at room temperature for 1 hour. (3) A 5 mL aliquot of the mixture was treated with 53.3 mg acrylamide. (4) Resulting mixture was heated at 4 hours in a 80° C. oven. (5) Mixture was subsequently cast onto substrate, dried, and washed with DMF/Ethanol. (6) Mixture was then treated with 3 mL DMF, 0.07 mL 1,8-octanedithiol, and (7) incubated at room temperature for 3 hours.

The membrane was evaluated using the durability testing procedure outlined in Example 8. The membrane exhibited an initial lactose rejection rate of 91% and a lactose rejection rate of 86% after exposure to a temperature of 70° C.

Example 10

A graphene oxide membrane was produced with the following procedure (See reaction scheme in FIG. 14E for similar procedure): (1) 15 mL of 4 mg/mL graphene oxide sheets was treated with 0.08 mL TEA, 22 mg NHS, and 72 mg EDC. (2) Mixture was incubated at room temperature for 1 hour. (3) A 5 mL aliquot of the mixture was treated with 10.6 mg acrylamide. (4) Resulting mixture was heated at 4 hours in a 65° C. oven. (5) Mixture was then cast onto substrate, dried, and washed with DMF/Ethanol. (6) Mixture was then treated with 3 mL of DMF and 0.07 mL of 1,8-octanedithiol. (7) Mixture was then incubated for 3 hours at room temperature, and (8) washed with DMF/Water.

The membrane was evaluated using the durability testing procedure outlined in Example 8. The membrane exhibited an initial lactose rejection rate of 90% and a lactose rejection rate of 87% after exposure to a temperature of 70° C.

Example 11

A graphene oxide membrane was produced with the following procedure (See reaction scheme in FIG. 14F for

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similar procedure): (1) 15 mL of 4 mg/mL graphene oxide sheets was treated with 53.3 mg acrylamide. (2) Mixture was heated at 80° C. for 3 hours, 20 minutes and then cast onto a substrate. (3) Resulting mixture was dried and washed with DMF/Water. (4) Mixture was subsequently treated with 3 mL DMF and 0.14 mL 1,8-octanedithiol. (5) Mixture was then incubated for 3 hours at room temperature.

The membrane was evaluated using the durability testing procedure outlined in Example 8. The membrane exhibited an initial lactose rejection rate of 96% and a lactose rejection rate of 75% after exposure to a temperature of 70° C.

Example 12

A graphene oxide membrane was produced with the following procedure (See FIG. 15A): (1) Vortex was applied to 5 mg CsOH and 10 mg 4-aminophenylacetic acid in 0.2 mL of water for 70 seconds (not all dissolved-all dispersed). (2) Mixture was added to 3 mL of 4 mg/mL graphene oxide sheet. (3) Mixture was then heated for 19 hours at 60° C. while stirring and then cooled to room temperature while stirring. (4) Mixture was then treated with 0.012 mL TEA, 5 mg NHS, 16 mg EDC, and (5) incubated 2 hours at room temperature. (6) Rod cast was allowed to dry in air. (7) A piece was cut to fit in 90 mm petri dish. (8) 3 mL DMF and 0.07 mL 1,8-octanedithiol were added. (9) Material was incubated 3 hours at room temperature. (10) Material was washed with DMF then water.

The membrane was evaluated using the durability testing procedure outlined in Example 8. The membrane exhibited an initial lactose rejection rate of 89% and a lactose rejection rate of 68% after exposure to a temperature of 70° C.

Example 13

A graphene oxide membrane was produced with the following procedure (See FIG. 15B): (1) Vortex was applied to 5 mg CsOH and 10 mg 4-aminophenylacetic acid in 0.2 mL of water for 70 seconds (not all dissolved-all dispersed). (2) Non solids were syringed and (3) mixture was added to 3 mL of 4 mg/mL graphene oxide sheet. (4) Mixture was then heated for 1 hour at 80° C. while stirring and then cooled to room temperature. (5) Mixture was then treated with 0.012 mL TEA, 5 mg NHS, 16 mg EDC, and (6) incubated 2 hours at room temperature. (7) Rod cast was allowed to dry in air. (8) 3 mL DMF and 0.10 mL 1,8-octanedithiol were added. (9) Material was incubated 3 hrs at room temperature. (10) Material was washed with DMF then water.

The membrane was evaluated using the durability testing procedure outlined in Example 8. The membrane exhibited an initial lactose rejection rate of 90% and a lactose rejection rate of 55% after exposure to a temperature of 70° C.

Example 14

A graphene oxide membrane was produced with the following procedure (See FIG. 15C): (1) A Rod (40 g) Vortex was applied to 5 mg CsOH and 30 mg 2-(4-aminophenyl) ethanol in 0.5 mL of water for 70 seconds (not all dissolved). (2) Non solids were syringed and (3) mixture was added to 3 mL of 4 mg/mL graphene oxide sheet. (4) Mixture was then heated for 30 minutes at 65° C. without stirring and then cooled to room temperature. (5) Rod cast was allowed to dry in air. (6) A piece was cut to fit in 90 mm petri dish. (7) Material was soaked in 10 mg CsOH, 3 mg 18-crown-6-ether and 5 mL DMF for 1 hour at room

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temperature. (8) 20 mg of solid Bis-Br-Bn was then added. (9) Mixture was swirled and then allowed to sit for 6 hours at room temperature. (10) Material was washed with DMF then water.

The membrane was evaluated using the durability testing procedure outlined in Example 8. The membrane exhibited an initial lactose rejection rate of 92% and a lactose rejection rate of 59% after exposure at a temperature of 70° C.

Example 15

A graphene oxide membrane was produced with the following procedure (See FIG. 15D): (1) Vortex was applied to 5 mg CsOH and 10 mg 4-aminophenylacetic acid in 0.2 mL of water for 70 seconds (not all dissolved-all dispersed). (2) Mixture was added to 3 mL of 4 mg/mL graphene oxide sheet. (3) Mixture was then heated for 19 hours at 60° C. while stirring and then cooled to room temperature while stirring. (4) Mixture was then treated with 0.012 mL TEA, 5 mg NHS, 16 mg EDC, and (5) incubated 2 hours at room temperature. (6) Rod cast was allowed to dry in air. (7) A piece was cut to fit in 90 mm petri dish. (8) A solution of 1 mL Water and 0.1 mL poly(ethylenimine) were added to surface and mixture was swirled. (9) Material was incubated 3 hours at room temperature. (10) Material was washed with water.

The membrane was evaluated using the durability testing procedure outlined in Example 8. The membrane exhibited an initial lactose rejection rate of 61% and a lactose rejection rate of 67% after exposure to a temperature of 70° C.

Example 16. Effects of Surface Roughness of the Support Substrate

A Veeco Dektak 6M profilometer was used to measure the 2D Rsq roughness of two support materials. Addition of graphene oxide films of the thickness in these examples did not seem to impact the roughness with statistical significance. The 0.22 μ m porous polytetrafluoroethylene (PTFE) on a rough polypropylene (PP) nonwoven had a Rsq of 3.8+/-1.5 μ m and the 0.45 μ m porous PTFE on a PP nonwoven had a Rsq of 2.4+/-0.5 μ m (95% confidence intervals based on measurements in 6 random locations at different angles).

Filtration results based on a rough support substrate. 38.8 mg of graphene oxide as a 0.4 wt % aqueous graphene oxide solution mixed with 300 mL of water. Cast by vacuum filtration through a 250 mm diameter ultrafiltration membrane—0.22 μ m porous PTFE on a rough PP nonwoven and allowed to dry. Soaked in a solution of 639 mg 1,4-bis(bromomethyl)benzene, 484 mg KOH, and 50 mL DMF for 2 hours at room temperature. A 47 mm disk was cut from the membrane for dead end filtration testing with a 1 wt % lactose at 50 psi. Rejection rate was 66.7% for lactose. Rejection rate was 49% for MgSO₄. The flux was 1.40 $\times$ 10⁻³ GFD/psi.

Filtration results based on a smooth support substrate. 38.8 mg of graphene oxide as a 0.4 wt % aqueous graphene oxide solution mixed with 300 mL of water. Cast by vacuum filtration through a 250 mm diameter ultrafiltration membrane—0.45 μ m porous PTFE on a PP nonwoven and allowed to dry. Dissolve 96.8 mg KOH in 3.5 mL H₂O and ~10 mL DMF. Let cool. Dissolve 639 mg 1,4-bis(bromomethyl)benzene in ~15 mL DMF. Pour KOH solution and 25 mL DMF on membrane first. Swirl. Add 1,4-bis(bromomethyl)benzene solution and swirl. Allowed to soak at room temperature for 3 hours. A 47 mm disk was cut from the

membrane for dead end filtration testing with a 1 wt % lactose at 50 psi. Rejection rate was 95% for lactose for the first 22 mL of permeate and held steady at ~100% for lactose over the next 3 days. Rejection rate was 89% for MgSO_4 . The flux was 1.90×10^{-3} GFD/psi.

Example 17. X-ray Diffraction Measurements

A 0.4 wt % graphene oxide solution rod cast with a 40-gauge Meyer rod onto hydrophilic treated 0.45 μm porous PTFE on a PP nonwoven to yield a complete film. Graphene oxide was allowed to dry for several days before grazing incidence X-ray diffraction (GIXRD) measurement to determine average graphene oxide interplane spacing (d-spacing) using a Rigaku SmartLab. Two-theta scans from 10° to 90° with a 1° angle of incidence from the source were collected. Spectra were normalized and background subtracted using the spectra of the microporous support without graphene oxide coating. The d-spacing peak occurred at 11.1 degrees 2 theta or 8.0 angstroms.

Saturate a 5 mL H_2O solution with 150 mg p-phenylenediamine (not all dissolved, allow to settle). Take 1 mL of p-phenylenediamine solution and add to 5 mL of 0.4 wt % graphene oxide and immediately cast by Meyer rod draw down (40-gauge wire wound rod). Allow to dry ~2 hours in air. Add 1 mL of tert-butyl nitrite to 2 mL acetonitrile and add to surface of membrane. Let rest for two hours. Wash with DMF and water. Two pieces of the material were allowed to dry, one at room temperature for several days, and one overnight at 65° C. before GIXRD measurement to determine average graphene oxide interplane spacing (d-spacing) using a Rigaku SmartLab. 2-theta scans from 10° to 90° with a 1° angle of incidence from the source were collected. Spectra were normalized and background subtracted using the spectra of the microporous support without graphene oxide coating. The d-spacing peak occurred at 8.5 degrees 2 theta or 10.4 angstroms for the room temperature dried and 9.1 degrees 2 theta or 9.7 angstroms d-spacing for the 65° C. dried.

0.5 mL of a 0.4 wt % aqueous graphene oxide solution mixed with 20 mL of water. Cast by vacuum filtration through a 90 mm diameter ultrafiltration membrane—0.45 μm porous PTFE on a PP nonwoven and allowed to dry. Add 0.1 mL 1,2-epoxypropane to 2.5 mL DMF at room temperature and soaked membrane in solution for 24 hours. Wash with DMF. Soak in 40 mg Bis-Br-Bn and 15 mg DABCO in 2.5 mL DMF and heat at 65° C. for 3 hours. Wash with water. Membrane was allowed to dry for several days before GIXRD measurement to determine average graphene oxide interplane spacing (d-spacing) using a Rigaku SmartLab. 2-theta scans from 10° to 90° with a 1° angle of incidence from the source were collected. Spectra were normalized and background subtracted using the spectra of the microporous support without graphene oxide coating. The d-spacing peak occurred at 5.8 degrees 2 theta or 15.2 angstroms.

0.5 mL of a 0.4 wt % aqueous graphene oxide solution mixed with 20 mL of water. Cast by vacuum filtration through a 90 mm diameter ultrafiltration membrane—0.45 μm porous PTFE on a PP nonwoven and allowed to dry. Membrane soaked in 5 mL acetonitrile and 1 mL thionyl chloride for 1 hour. Decant liquid from surface of membrane. Add 80 mg p-phenylenediamine in 5 mL acetonitrile for 3 hours. Solids form upon addition. Wash with DMF, then H_2O . Dry. Soak in 40 mg 1,4-bis(bromomethyl)benzene and 7 mg DABCO in 2.5 mL DMF and heat at 65° C. for 3 hours. Wash with H_2O and air dry. A 47 mm disk was cut from the center of the membrane and tested at 75 psi in

a dead-end filtration cell. Flux of 0.09 GFD (at 75 psi) and lactose rejection of 98% were determined after 1 week of filtration testing with a 1 wt % lactose. Two pieces of the material were allowed to dry—one from the region outside the 47 mm disk which had not been tested and one from the region tested. GIXRD measurement was used to determine average graphene oxide interplane spacing (d-spacing) using a Rigaku SmartLab. 2-theta scans from 10° to 90° with a 1° angle of incidence from the source were collected. Spectra were normalized and background subtracted using the spectra of the microporous support without graphene oxide coating. The d-spacing peak occurred at 9.3 degrees 2 theta or 9.6 angstroms for the untested and 9.2 degrees 2 theta or 9.6 angstroms d-spacing for the tested piece.

0.5 mL of a 0.4 wt % aqueous graphene oxide solution mixed with 20 mL of water. Cast by vacuum filtration through a 90 mm diameter ultrafiltration membrane—0.45 μm porous PTFE on a PP nonwoven and allowed to dry. Dissolve 65 mg of 2-aminoethanethiol in 5 mL DMF. Solution poured over membrane surface in glass petri dish. Add 10 mg solid DABCO, swirl and soak for 3 hours. Wash with DMF then water. Air dry. Membrane was allowed to dry for several days before GIXRD measurement to determine average graphene oxide interplane spacing (d-spacing) using a Rigaku SmartLab. 2-theta scans from 10° to 90° with a 1° angle of incidence from the source were collected. Spectra were normalized and background subtracted using the spectra of the microporous support without graphene oxide coating. The d-spacing peak occurred at 9.8 degrees 2 theta or 9.0 angstroms.

Example 18. Pretreatment Flux Improvement

Flux is measured during filtration tests with 1 wt % lactose and 0.1 wt % MgSO_4 before and after exposure to elevated temperature and pH. On average, fluxes range between 0.3-0.6 gallons per square foot per day (GFD) with a feed pressure of 75 pounds per square inch (psi).

Table 1 summarizes the results of the fluxes before and after exposure to various conditions. All the membranes discussed in this example and referenced in Table 1 meet a minimum lactose rejection of 50%.

TABLE 1

	Exposure Conditions	Initial Flux (GFD)	Final Flux (GFD)	Flux Improvement	Chemistry type
Entry 1	50° C., pH 11.5, 4 hours	0.16 (25° C.)	0.74 (25° C.)	4.5X	p-phenylenediamine
Entry 2	50° C., pH 11.5, 4 hours	0.13 (25° C.)	0.48 (25° C.)	3.8X	Thionyl chloride, p-phenylenediamine, 1,4-bis(bromomethyl)benzene

Example 19. Quantification of Membrane Color

The color of a graphene oxide membrane was characterized by recording images of the graphene oxide membrane and quantifying its grayscale mode value. Images of the graphene oxide membrane were collected in a lightbox with dimensions 9.4×9.1×8.7" and two rows of 20 white LEDs on the top front and rear edge of the lightbox. Samples of the graphene oxide membrane being photographed were placed in the center of the lightbox, and images were captured at a roughly 90° angle between camera lens, object, and LED

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light source. Data presented was averaged over multiple samples. Color was be quantified using image processing software (e.g., ImageJ) by converting the relative intensity of each pixel in the images from the additive color model RGB to gray scale, and comparing the intensity of the resulting signals plotted as a pixel histogram. Each gray scale pixel histogram displayed the distribution of gray values for each image with the x-axis representing possible gray values (0-255) and the y-axis representing the number of pixels found at each gray value. The grayscale mode value was calculated from the histogram, with gray values closer to zero corresponding to darker images, and gray values closer to 255 correspond to lighter images.

FIGS. 19A-20H show the quantification results using the method described above.

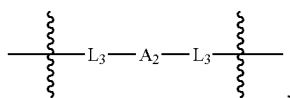
What is claimed is:

1. A filtration apparatus, comprising:

a support substrate; and

a graphene oxide membrane disposed on the support substrate, the graphene oxide membrane including a plurality of graphene oxide sheets, each of the graphene oxide sheets coupled to an adjacent graphene oxide sheet via a chemical linker,

wherein the chemical linker has the Formula Ib:



(Ib)

wherein:

L_3 is selected from $\text{---C(=O)---NH---(CH}_2\text{)}_m\text{---}$, $\text{---C(=O)---NH---C(=O)---(CH}_2\text{)}_2\text{---S---(CH}_2\text{)}_m\text{---}$, or $\text{---NH---C(=O)---(CH}_2\text{)}_2\text{---S---(CH}_2\text{)}_m\text{---}$;

A_2 is selected from aryl, heteroaryl, $\text{C}_4\text{--C}_{10}$ heterocycloalkyl, $\text{C}_4\text{--C}_{10}$ cycloalkyl, or $\text{C}_4\text{--C}_{10}$ alkyl, wherein the aryl, heteroaryl, heterocycloalkyl, cycloalkyl, and alkyl can each be optionally substituted by one or more substituents selected from halo, $\text{C}_1\text{--C}_4$ alkoxy, or $\text{C}_1\text{--C}_4$ alkyl;

m is 0-4; and

denotes the point of coupling with a carbon atom on a graphene oxide;

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the graphene oxide membrane has a flux of at least about 2.5×10^{-3} gallons per square foot per day per psi (GFD/psi) with a 1 wt. % lactose solution at room temperature; and

the graphene oxide membrane includes about 0.1 mg to 6 mg of the graphene oxide membrane per 5000 mm^2 membrane area.

2. The filtration apparatus of claim 1, wherein the support substrate is a microporous polymer substrate.

3. The filtration apparatus of claim 1, wherein the support substrate has a root mean squared surface roughness of less than about 3 μm .

4. The filtration apparatus of claim 1, wherein the support substrate has an average pore size of 0.1 μm to 5 μm .

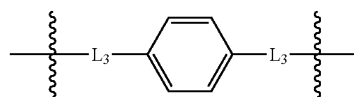
5. The filtration apparatus of claim 1, wherein the support substrate includes a material selected from polypropylene, polystyrene, polyethylene, polyethylene oxide, polyether-sulfone, polytetrafluoroethylene, polyvinylidene fluoride, polymethylmethacrylate, polydimethylsiloxane, polyester, cellulose, cellulose acetate, cellulose nitrate, polyacrylonitrile, glass fiber, quartz, alumina, silver, polycarbonate, nylon, aramid, or polyether ether ketone.

6. The filtration apparatus of claim 1, wherein the graphene oxide membrane has a lactose rejection rate of at least 50% with a 1 wt % lactose solution.

7. The filtration apparatus of claim 1, wherein the graphene oxide membrane has a MgSO_4 rejection rate of at least 80% with a 0.1 wt % MgSO_4 solution.

8. The filtration apparatus of claim 1, wherein A_2 is phenyl or linear $\text{C}_5\text{--C}_6$ alkyl.

9. The filtration apparatus of claim 8, wherein the chemical linker has Formula Ib-1:



(Ib-1)

10. The filtration apparatus of claim 1, wherein m is 0 or 1.

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